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Prodi : D3 - Teknologi Informasi

In [22]:
```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import tensorflow as tf
from sklearn.model_selection import train_test_split
from sklearn.preprocessing import StandardScaler
```

kode ini memanggil library yang akan dipakai untuk mengolah data, membuat model jaringan saraf, membagi dataset, dan menampilkan grafik.

- pandas dipakai untuk membaca dan mengelola dataset berbentuk tabel.
- numpy membantu operasi numerik dan array.
- matplotlib.pyplot dipakai untuk membuat grafik.
- tensorflow menjadi framework untuk membangun dan melatih jaringan saraf.
- train_test_split membagi data menjadi data training dan testing.
- standardscaler disiapkan untuk standardisasi fitur, tetapi pada notebook ini belum digunakan.

In [23]:
```
from google.colab import drive
drive.mount('/content/drive')
```

```
Drive already mounted at /content/drive; to attempt to forcibly remount, call drive.
mount("/content/drive", force_remount=True).
```

drive.mount('/content/drive') menghubungkan google colab dengan google drive agar file yang disimpan di drive dapat diakses dari notebook.

In [24]:
```
!pip install tensorflow
```

```
Requirement already satisfied: tensorflow in /usr/local/lib/python3.13/dist-packages
(2.21.0)
Requirement already satisfied: absl-py>=1.0.0 in /usr/local/lib/python3.13/dist-pack
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st-packages (from rich->keras>=3.12.0->tensorflow) (4.2.0)
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(from markdown-it-py>=2.2.0->rich->keras>=3.12.0->tensorflow) (0.1.2)
```

menginstall tennsorflow agar dapat digunakan di tab ini

In [25]:
```
!pip install pandas matplotlib scikit-learn
```

```
Requirement already satisfied: pandas in /usr/local/lib/python3.13/dist-packages (2.
2.3)
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ges (from pandas) (2.4.6)
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Requirement already satisfied: six>=1.5 in /usr/local/lib/python3.13/dist-packages
(from python-dateutil>=2.8.2->pandas) (1.17.0)
```

cell ini memasang pandas, matplotlib, dan scikit-learn. ketiganya mendukung tahapan sebelum dan sesudah training: pandas untuk membaca dataset, matplotlib untuk visualisasi, sedangkan scikit-learn menyediakan fungsi pembagian data.

In [26]:
```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import tensorflow as tf
from sklearn.model_selection import train_test_split
from sklearn.preprocessing import StandardScaler
```

library dipanggil kembali agar nama modul dan fungsi tersedia pada sesi eksekusi notebook. pemanggilan ulang tidak membuat model baru dan tidak mengubah dataset.

pada praktik ini, fungsi-fungsi tersebut akan digunakan untuk membaca data advertising, membagi data, dan membangun model mlp menggunakan tensorflow keras.

In [27]:
```
# Load dataset
dataset = pd.read_csv('/content/advertising.csv')

# Lihat beberapa baris dataset
dataset.head()
```

Out[27]:

	TV	Radio	Newspaper	Sales
0	230.1	37.8	69.2	22.1
1	44.5	39.3	45.1	10.4
2	17.2	45.9	69.3	9.3
3	151.5	41.3	58.5	18.5
4	180.8	10.8	58.4	12.9

pd.read_csv() membaca file advertising.csv dan menyimpannya ke variabel dataset. perintah dataset.head() menampilkan lima baris pertama agar isi dan nama kolom dapat diperiksa. dataset memiliki tiga fitur masukan, yaitu tv, radio, dan newspaper, serta satu target sales. tujuan model adalah mempelajari hubungan anggaran iklan pada ketiga media tersebut dengan nilai penjualan. jadi ini termasuk tugas regresi karena output yang diprediksi berupa angka, bukan label kelas. dan lima baris pertama yang ditampilkan adalah head nya.

In [28]:
```
dataset.describe()
```

Out[28]:

	TV	Radio	Newspaper	Sales
count	200.000000	200.000000	200.000000	200.000000
mean	147.042500	23.264000	30.554000	14.022500
std	85.854236	14.846809	21.778621	5.217457
min	0.700000	0.000000	0.300000	1.600000
25%	74.375000	9.975000	12.750000	10.375000
50%	149.750000	22.900000	25.750000	12.900000
75%	218.825000	36.525000	45.100000	17.400000
max	296.400000	49.600000	114.000000	27.000000

dataset.describe() menghasilkan ringkasan statistik untuk setiap kolom numerik. count menunjukkan jumlah data, mean nilai rata-rata, std simpangan baku, min nilai terkecil, kuartil 25%, 50%, 75%, dan max nilai terbesar. pada data untuk setiap kolom, sehingga tidak terlihat kolom numerik yang kekurangan jumlah observasi. rata-rata sales sekitar 15.13, sedangkan rata-rata anggaran tv sekitar 147.04, radio 23.264, dan newspaper 30.554. ringkasan ini membantu memahami skala dan sebaran data sebelum model dilatih.

In [29]:
```
# Cek nilai null
dataset.isnull().sum()
```

Out[29]:

	0
TV	0
Radio	0
Newspaper	0
Sales	0

dtype: int64

dataset.isnull() memeriksa setiap sel apakah nilainya kosong, lalu .sum() menghitung jumlah nilai kosong pada setiap kolom.

hasil menunjukkan tv, radio, newspaper, dan sales masing-masing memiliki nilai null sebanyak 0. artinya, tidak ditemukan missing value pada kolom-kolom tersebut, sehingga tidak diperlukan penghapusan atau pengisian nilai kosong pada tahap ini.

In [30]:
```
# Split data menjadi training dan testing set
from sklearn.model_selection import train_test_split

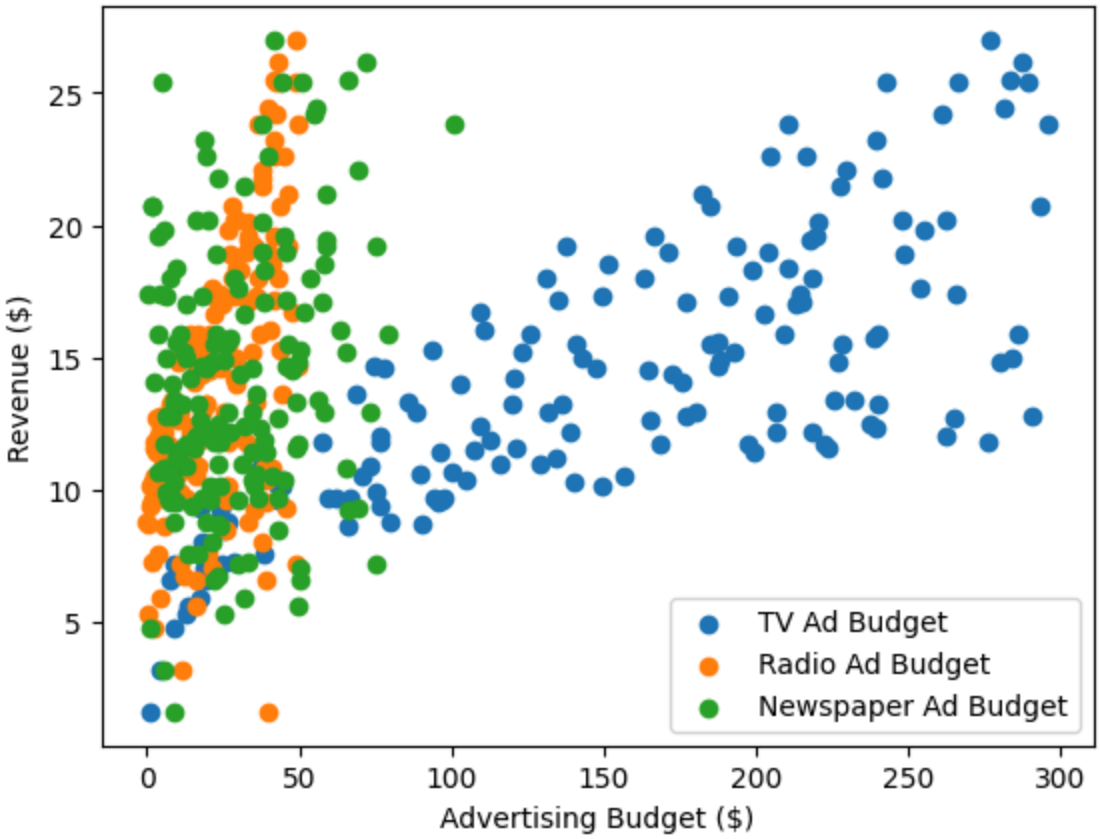
X = dataset.drop(['Sales'], axis=1)
y = dataset['Sales']
```

```
X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.2, random_sta
```

x berisi fitur masukan dengan menghapus kolom sales, sedangkan y berisi target yang akan diprediksi, yaitu sales. fungsi train_test_split() membagi dataset menjadi data training dan data testing.

test_size=0.2 berarti 20% data digunakan untuk testing dan 80% untuk training. dari total 200 baris, pembagiannya menjadi 160 data training dan 40 data testing. random_state=42 menjaga pembagian tetap sama ketika kode dijalankan kembali.

In [31]:
```
# Membuat plot dari dataset
plt.scatter(X_train['TV'], y_train, label='TV Ad Budget') # Changed column name to
plt.scatter(X_train['Radio'], y_train, label='Radio Ad Budget') # Changed column na
plt.scatter(X_train['Newspaper'], y_train, label='Newspaper Ad Budget') # Changed c
plt.xlabel('Advertising Budget ($)')
plt.ylabel('Revenue ($)')
plt.legend()
plt.show()
```



setiap plt.scatter() membuat scatter plot antara satu fitur anggaran iklan dengan nilai sales. sumbu x menunjukkan anggaran iklan dan sumbu y menunjukkan penjualan. tiga kelompok titik memperlihatkan hubungan tv, radio, dan newspaper dengan sales.

pada gambar ini pemeriksaan awal untuk melihat pola, sebaran, dan kemungkinan hubungan antarvariabel sebelum model dibangun. pola titik tidak harus membentuk garis

lurus karena mlp dapat mempelajari hubungan non-linear melalui hidden layer dan fungsi aktivasi.

hasilnya berupa kumpulan titik pada grafik. grafik ini belum merupakan hasil prediksi mlp, data yang ditampilkan berasal dari data training.

In [32]:
```
from tensorflow.keras import Sequential
from tensorflow.keras.layers import Dense

# Membangun model MLP
model = Sequential([
    Dense(512, input_shape=(3,), activation='relu'),  # Input layer
    Dense(512, activation='relu'),  # Hidden layer 1
    Dense(256, activation='relu'),  # Hidden layer 2
    Dense(1)  # Output layer (tidak perlu fungsi aktivasi karena ini masalah regres
])
model.compile(optimizer='adam', loss='mse', metrics=['mae'])
model.summary()
```

```
/usr/local/lib/python3.13/dist-packages/keras/src/layers/core/dense.py:106: UserWarn
ing: Do not pass an `input_shape`/`input_dim` argument to a layer. When using Sequen
tial models, prefer using an `Input(shape)` object as the first layer in the model i
nstead.
  super().__init__(activity_regularizer=activity_regularizer, **kwargs)
```

Model: "sequential_1"

Layer (type)	Output Shape	Param #
dense_4 (Dense)	(None, 512)	2,048
dense_5 (Dense)	(None, 512)	262,656
dense_6 (Dense)	(None, 256)	131,328
dense_7 (Dense)	(None, 1)	257

Total params: 396,289 (1.51 MB)

Trainable params: 396,289 (1.51 MB)

Non-trainable params: 0 (0.00 B)

cell ini membangun jaringan saraf multilayer perceptron (mlp) dengan sequential, yaitu susunan layer yang diproses secara berurutan. menurut modul, mlp memiliki input layer, hidden layer, dan output layer.

- layer pertama dense(512, input_shape=(3,), activation='relu') menerima tiga fitur: tv, radio, dan newspaper. 512 menunjukkan jumlah neuron pada layer tersebut. relu menerapkan fungsi max(0, x), yaitu nilai negatif menjadi nol dan nilai positif diteruskan.
- hidden layer berikutnya memiliki 512 neuron dan 256 neuron. layer-layer ini menggabungkan informasi dari fitur sebelumnya untuk mempelajari pola yang lebih kompleks.

- output layer dense(1) menghasilkan satu angka prediksi sales. aktivasi linear bawaan digunakan karena masalahnya regresi, bukan klasifikasi kelas. pada proses forward propagation, nilai input diteruskan melewati layer, dikalikan dengan bobot, ditambah bias, lalu diproses oleh aktivasi sampai menghasilkan prediksi. model.compile() menentukan adam sebagai optimizer, mse sebagai loss, dan mae sebagai metrik. mse mengukur rata-rata kuadrat selisih prediksi dan nilai sebenarnya, sedangkan mae mengukur rata-rata selisih absolut.

ketika model.summary() dijalankan, output menampilkan susunan layer, bentuk output, dan jumlah parameter. parameter adalah bobot dan bias yang akan dipelajari.

In [33]:
```
early_stop = tf.keras.callbacks.EarlyStopping(monitor='val_loss', patience=5)
history = model.fit(X_train, y_train, epochs=1400, validation_data=(X_test, y_test)
```

```
Epoch 1/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 1s 44ms/step - loss: 207.5248 - mae: 11.5186 - val_loss: 15
6.4665 - val_mae: 11.3837
Epoch 2/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 18ms/step - loss: 59.9790 - mae: 6.2667 - val_loss: 39.1
409 - val_mae: 4.8049
Epoch 3/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 16ms/step - loss: 20.8493 - mae: 3.2646 - val_loss: 18.2
375 - val_mae: 3.7724
Epoch 4/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 17ms/step - loss: 12.5046 - mae: 3.0049 - val_loss: 11.7
611 - val_mae: 2.6323
Epoch 5/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 18ms/step - loss: 7.8771 - mae: 2.0876 - val_loss: 6.541
4 - val_mae: 2.0966
Epoch 6/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 17ms/step - loss: 4.2929 - mae: 1.7490 - val_loss: 6.041
8 - val_mae: 1.9868
Epoch 7/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 17ms/step - loss: 3.3966 - mae: 1.4688 - val_loss: 4.914
0 - val_mae: 1.8337
Epoch 8/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 17ms/step - loss: 2.8811 - mae: 1.3678 - val_loss: 4.547
3 - val_mae: 1.7062
Epoch 9/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 17ms/step - loss: 2.6077 - mae: 1.2477 - val_loss: 4.239
4 - val_mae: 1.5357
Epoch 10/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 16ms/step - loss: 2.5737 - mae: 1.1941 - val_loss: 4.777
5 - val_mae: 1.7459
Epoch 11/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 16ms/step - loss: 2.6062 - mae: 1.2018 - val_loss: 4.774
2 - val_mae: 1.6416
Epoch 12/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 16ms/step - loss: 2.4268 - mae: 1.1700 - val_loss: 4.998
5 - val_mae: 1.5453
Epoch 13/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 17ms/step - loss: 2.3154 - mae: 1.1176 - val_loss: 5.124
8 - val_mae: 1.6409
Epoch 14/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 16ms/step - loss: 2.3909 - mae: 1.1870 - val_loss: 4.624
1 - val_mae: 1.5076
Epoch 15/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 17ms/step - loss: 2.3964 - mae: 1.1889 - val_loss: 5.179
4 - val_mae: 1.5266
Epoch 16/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 17ms/step - loss: 2.3068 - mae: 1.1191 - val_loss: 5.618
0 - val_mae: 1.8273
Epoch 17/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 17ms/step - loss: 2.5598 - mae: 1.2412 - val_loss: 4.617
3 - val_mae: 1.4736
Epoch 18/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 17ms/step - loss: 2.4231 - mae: 1.1844 - val_loss: 5.128
3 - val_mae: 1.5522
Epoch 19/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 21ms/step - loss: 2.7954 - mae: 1.2978 - val_loss: 5.221
```

```
1 - val_mae: 1.5901
Epoch 20/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 17ms/step - loss: 3.0311 - mae: 1.3581 - val_loss: 5.712
9 - val_mae: 1.8300
Epoch 21/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 18ms/step - loss: 2.6302 - mae: 1.2802 - val_loss: 4.844
4 - val_mae: 1.6259
Epoch 22/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 17ms/step - loss: 2.2075 - mae: 1.1703 - val_loss: 5.348
4 - val_mae: 1.6729
Epoch 23/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 17ms/step - loss: 2.3533 - mae: 1.1334 - val_loss: 5.000
5 - val_mae: 1.5398
Epoch 24/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 17ms/step - loss: 2.1666 - mae: 1.0938 - val_loss: 5.148
6 - val_mae: 1.4886
Epoch 25/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 17ms/step - loss: 2.1441 - mae: 1.0727 - val_loss: 5.389
3 - val_mae: 1.5238
Epoch 26/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 17ms/step - loss: 2.0299 - mae: 1.0165 - val_loss: 4.997
2 - val_mae: 1.4926
Epoch 27/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 18ms/step - loss: 2.0880 - mae: 1.0474 - val_loss: 5.051
0 - val_mae: 1.5475
Epoch 28/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 2.3470 - mae: 1.1560 - val_loss: 5.161
5 - val_mae: 1.7020
Epoch 29/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 18ms/step - loss: 2.1724 - mae: 1.0779 - val_loss: 4.902
0 - val_mae: 1.4888
Epoch 30/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 16ms/step - loss: 2.0818 - mae: 1.0991 - val_loss: 5.106
8 - val_mae: 1.6302
Epoch 31/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 17ms/step - loss: 2.3611 - mae: 1.1813 - val_loss: 5.066
8 - val_mae: 1.5083
Epoch 32/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 17ms/step - loss: 2.0357 - mae: 1.0355 - val_loss: 4.876
7 - val_mae: 1.4877
Epoch 33/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 16ms/step - loss: 1.9535 - mae: 1.0339 - val_loss: 4.721
4 - val_mae: 1.4878
Epoch 34/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 17ms/step - loss: 2.0672 - mae: 1.0741 - val_loss: 4.942
7 - val_mae: 1.4706
Epoch 35/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 18ms/step - loss: 1.9219 - mae: 0.9994 - val_loss: 5.366
7 - val_mae: 1.5966
Epoch 36/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 16ms/step - loss: 2.0713 - mae: 1.0584 - val_loss: 4.782
2 - val_mae: 1.4715
Epoch 37/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 16ms/step - loss: 1.9539 - mae: 1.0419 - val_loss: 4.898
7 - val_mae: 1.4207
Epoch 38/1400
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5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 17ms/step - loss: 1.9267 - mae: 0.9792 - val_loss: 5.162
4 - val_mae: 1.4772
Epoch 39/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 17ms/step - loss: 1.8795 - mae: 1.0145 - val_loss: 4.585
5 - val_mae: 1.4246
Epoch 40/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 1.9291 - mae: 1.0216 - val_loss: 5.126
7 - val_mae: 1.5098
Epoch 41/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 18ms/step - loss: 1.9217 - mae: 1.0138 - val_loss: 4.978
1 - val_mae: 1.4670
Epoch 42/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 37ms/step - loss: 1.8668 - mae: 0.9968 - val_loss: 5.255
3 - val_mae: 1.5048
Epoch 43/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 33ms/step - loss: 1.8674 - mae: 0.9728 - val_loss: 5.402
8 - val_mae: 1.5707
Epoch 44/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 34ms/step - loss: 2.1019 - mae: 1.1089 - val_loss: 4.821
6 - val_mae: 1.4312
Epoch 45/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 26ms/step - loss: 1.9731 - mae: 1.0744 - val_loss: 4.996
6 - val_mae: 1.5419
Epoch 46/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 28ms/step - loss: 2.0479 - mae: 1.0776 - val_loss: 4.787
4 - val_mae: 1.4052
Epoch 47/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 29ms/step - loss: 1.8939 - mae: 1.0020 - val_loss: 5.703
6 - val_mae: 1.5421
Epoch 48/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 34ms/step - loss: 1.8638 - mae: 1.0056 - val_loss: 5.410
6 - val_mae: 1.5587
Epoch 49/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 23ms/step - loss: 1.7123 - mae: 0.9776 - val_loss: 4.583
5 - val_mae: 1.3679
Epoch 50/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 34ms/step - loss: 1.6349 - mae: 0.9159 - val_loss: 4.940
4 - val_mae: 1.3951
Epoch 51/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 26ms/step - loss: 1.7013 - mae: 0.9253 - val_loss: 5.433
6 - val_mae: 1.6193
Epoch 52/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 27ms/step - loss: 2.1907 - mae: 1.1407 - val_loss: 5.609
8 - val_mae: 1.5213
Epoch 53/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 27ms/step - loss: 2.3636 - mae: 1.2383 - val_loss: 4.954
2 - val_mae: 1.6435
Epoch 54/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 30ms/step - loss: 1.7948 - mae: 1.0519 - val_loss: 4.764
3 - val_mae: 1.3537
Epoch 55/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 26ms/step - loss: 1.6581 - mae: 0.9407 - val_loss: 5.226
4 - val_mae: 1.3472
Epoch 56/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 17ms/step - loss: 1.6234 - mae: 0.9251 - val_loss: 4.575
6 - val_mae: 1.3556
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Epoch 57/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 17ms/step - loss: 1.6418 - mae: 0.9451 - val_loss: 4.483
7 - val_mae: 1.2872
Epoch 58/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 16ms/step - loss: 1.6178 - mae: 0.9268 - val_loss: 4.598
6 - val_mae: 1.4104
Epoch 59/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 16ms/step - loss: 1.4475 - mae: 0.8613 - val_loss: 4.644
3 - val_mae: 1.3330
Epoch 60/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 16ms/step - loss: 1.5767 - mae: 0.9082 - val_loss: 4.707
8 - val_mae: 1.2752
Epoch 61/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 16ms/step - loss: 1.4615 - mae: 0.8613 - val_loss: 4.195
3 - val_mae: 1.3553
Epoch 62/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 17ms/step - loss: 1.4332 - mae: 0.8491 - val_loss: 4.624
9 - val_mae: 1.3002
Epoch 63/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 17ms/step - loss: 1.5077 - mae: 0.8978 - val_loss: 5.192
3 - val_mae: 1.4643
Epoch 64/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 1.5858 - mae: 0.9484 - val_loss: 4.611
0 - val_mae: 1.4241
Epoch 65/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 16ms/step - loss: 1.4412 - mae: 0.8698 - val_loss: 5.526
6 - val_mae: 1.4540
Epoch 66/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 16ms/step - loss: 1.6114 - mae: 0.9750 - val_loss: 4.725
6 - val_mae: 1.4841
Epoch 67/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 17ms/step - loss: 1.4935 - mae: 0.9023 - val_loss: 5.093
9 - val_mae: 1.3813
Epoch 68/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 17ms/step - loss: 1.4633 - mae: 0.8945 - val_loss: 5.155
1 - val_mae: 1.4483
Epoch 69/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 17ms/step - loss: 1.4485 - mae: 0.8746 - val_loss: 4.634
6 - val_mae: 1.4791
Epoch 70/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 17ms/step - loss: 1.4370 - mae: 0.8866 - val_loss: 5.194
1 - val_mae: 1.3919
Epoch 71/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 21ms/step - loss: 1.4046 - mae: 0.9278 - val_loss: 4.388
1 - val_mae: 1.2167
Epoch 72/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 17ms/step - loss: 1.3235 - mae: 0.8338 - val_loss: 4.078
0 - val_mae: 1.2237
Epoch 73/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 17ms/step - loss: 1.3894 - mae: 0.8953 - val_loss: 4.200
3 - val_mae: 1.3749
Epoch 74/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 18ms/step - loss: 1.3627 - mae: 0.8659 - val_loss: 4.353
8 - val_mae: 1.2390
Epoch 75/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 21ms/step - loss: 1.2226 - mae: 0.7777 - val_loss: 4.196
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5 - val_mae: 1.1930
Epoch 76/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 17ms/step - loss: 1.1699 - mae: 0.7616 - val_loss: 4.025
6 - val_mae: 1.2421
Epoch 77/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 17ms/step - loss: 1.1833 - mae: 0.7740 - val_loss: 4.296
5 - val_mae: 1.2095
Epoch 78/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 17ms/step - loss: 1.2184 - mae: 0.8135 - val_loss: 4.236
6 - val_mae: 1.2258
Epoch 79/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 18ms/step - loss: 1.1988 - mae: 0.8011 - val_loss: 4.435
4 - val_mae: 1.1710
Epoch 80/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 17ms/step - loss: 1.2355 - mae: 0.8129 - val_loss: 4.666
0 - val_mae: 1.2808
Epoch 81/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 27ms/step - loss: 1.2346 - mae: 0.8204 - val_loss: 3.846
8 - val_mae: 1.1889
Epoch 82/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 1.1120 - mae: 0.7582 - val_loss: 4.862
3 - val_mae: 1.3839
Epoch 83/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 16ms/step - loss: 1.2435 - mae: 0.8280 - val_loss: 4.220
3 - val_mae: 1.2489
Epoch 84/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 16ms/step - loss: 1.2065 - mae: 0.8239 - val_loss: 4.002
1 - val_mae: 1.1958
Epoch 85/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 1.0045 - mae: 0.7091 - val_loss: 4.186
1 - val_mae: 1.1236
Epoch 86/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 17ms/step - loss: 0.9818 - mae: 0.6975 - val_loss: 3.604
6 - val_mae: 1.0999
Epoch 87/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 18ms/step - loss: 0.9421 - mae: 0.7033 - val_loss: 3.942
9 - val_mae: 1.2723
Epoch 88/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 17ms/step - loss: 1.1351 - mae: 0.8240 - val_loss: 3.697
2 - val_mae: 1.0650
Epoch 89/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 18ms/step - loss: 0.9985 - mae: 0.7529 - val_loss: 4.557
9 - val_mae: 1.3182
Epoch 90/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 16ms/step - loss: 1.3239 - mae: 0.9018 - val_loss: 4.464
2 - val_mae: 1.1643
Epoch 91/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 16ms/step - loss: 1.0077 - mae: 0.7401 - val_loss: 3.525
4 - val_mae: 1.0751
Epoch 92/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 21ms/step - loss: 0.8883 - mae: 0.6840 - val_loss: 3.151
2 - val_mae: 1.0214
Epoch 93/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 18ms/step - loss: 0.8691 - mae: 0.6819 - val_loss: 3.538
6 - val_mae: 1.0657
Epoch 94/1400
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5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 18ms/step - loss: 0.7969 - mae: 0.6374 - val_loss: 3.634
5 - val_mae: 1.1475
Epoch 95/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.8499 - mae: 0.6781 - val_loss: 3.926
5 - val_mae: 1.0925
Epoch 96/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 21ms/step - loss: 1.0174 - mae: 0.7843 - val_loss: 3.033
9 - val_mae: 1.0404
Epoch 97/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 1.0330 - mae: 0.7638 - val_loss: 3.507
2 - val_mae: 1.4008
Epoch 98/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 1.1123 - mae: 0.8144 - val_loss: 3.270
8 - val_mae: 0.9895
Epoch 99/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 18ms/step - loss: 1.0604 - mae: 0.7936 - val_loss: 3.824
0 - val_mae: 1.2807
Epoch 100/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.9820 - mae: 0.7767 - val_loss: 4.296
8 - val_mae: 1.3528
Epoch 101/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 18ms/step - loss: 1.0178 - mae: 0.7607 - val_loss: 3.217
5 - val_mae: 1.0232
Epoch 102/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 18ms/step - loss: 0.7812 - mae: 0.6483 - val_loss: 3.150
2 - val_mae: 1.0526
Epoch 103/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 18ms/step - loss: 0.7268 - mae: 0.5978 - val_loss: 2.954
7 - val_mae: 0.9442
Epoch 104/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 16ms/step - loss: 0.6413 - mae: 0.5792 - val_loss: 2.755
9 - val_mae: 1.0188
Epoch 105/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 17ms/step - loss: 0.6739 - mae: 0.6180 - val_loss: 2.872
5 - val_mae: 1.1787
Epoch 106/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 21ms/step - loss: 0.8624 - mae: 0.7002 - val_loss: 2.690
0 - val_mae: 0.9615
Epoch 107/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 18ms/step - loss: 0.6948 - mae: 0.6153 - val_loss: 3.097
7 - val_mae: 0.9936
Epoch 108/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 17ms/step - loss: 0.6031 - mae: 0.5632 - val_loss: 3.294
2 - val_mae: 1.0933
Epoch 109/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 17ms/step - loss: 0.7714 - mae: 0.6675 - val_loss: 2.671
1 - val_mae: 0.9314
Epoch 110/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 17ms/step - loss: 0.6438 - mae: 0.5884 - val_loss: 2.895
8 - val_mae: 1.1257
Epoch 111/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 17ms/step - loss: 0.7032 - mae: 0.6281 - val_loss: 3.047
6 - val_mae: 0.9678
Epoch 112/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 17ms/step - loss: 1.0102 - mae: 0.7953 - val_loss: 3.358
8 - val_mae: 1.1531
```

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Epoch 113/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 17ms/step - loss: 0.9745 - mae: 0.7857 - val_loss: 2.155
6 - val_mae: 0.8933
Epoch 114/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 17ms/step - loss: 0.8016 - mae: 0.6556 - val_loss: 2.736
2 - val_mae: 1.1309
Epoch 115/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 17ms/step - loss: 0.8503 - mae: 0.7051 - val_loss: 3.246
5 - val_mae: 1.0738
Epoch 116/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 17ms/step - loss: 0.7352 - mae: 0.6553 - val_loss: 3.385
7 - val_mae: 1.1140
Epoch 117/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 1.1199 - mae: 0.8442 - val_loss: 3.295
5 - val_mae: 1.4668
Epoch 118/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 17ms/step - loss: 1.3209 - mae: 0.9352 - val_loss: 2.706
4 - val_mae: 1.3787
Epoch 119/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 17ms/step - loss: 1.5507 - mae: 0.9897 - val_loss: 4.236
0 - val_mae: 1.3602
Epoch 120/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 17ms/step - loss: 2.2057 - mae: 1.1428 - val_loss: 6.665
5 - val_mae: 1.8164
Epoch 121/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 16ms/step - loss: 1.9032 - mae: 1.0970 - val_loss: 5.134
9 - val_mae: 1.4342
Epoch 122/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 16ms/step - loss: 1.6806 - mae: 1.0215 - val_loss: 3.255
0 - val_mae: 1.2003
Epoch 123/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 16ms/step - loss: 1.4281 - mae: 0.9529 - val_loss: 2.899
4 - val_mae: 1.2290
Epoch 124/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 17ms/step - loss: 0.9890 - mae: 0.7856 - val_loss: 3.085
2 - val_mae: 0.9985
Epoch 125/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 16ms/step - loss: 0.7917 - mae: 0.6872 - val_loss: 3.699
0 - val_mae: 1.1649
Epoch 126/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 18ms/step - loss: 0.9025 - mae: 0.7521 - val_loss: 2.313
7 - val_mae: 1.0558
Epoch 127/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 17ms/step - loss: 0.6908 - mae: 0.6308 - val_loss: 2.237
4 - val_mae: 0.9709
Epoch 128/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.6223 - mae: 0.5924 - val_loss: 2.524
3 - val_mae: 0.9250
Epoch 129/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 17ms/step - loss: 0.6107 - mae: 0.5988 - val_loss: 2.758
7 - val_mae: 0.9050
Epoch 130/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 18ms/step - loss: 0.5784 - mae: 0.5741 - val_loss: 2.777
1 - val_mae: 0.9596
Epoch 131/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 17ms/step - loss: 0.5227 - mae: 0.5515 - val_loss: 2.787
```

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2 - val_mae: 1.0258
Epoch 132/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 17ms/step - loss: 0.5747 - mae: 0.5750 - val_loss: 2.099
4 - val_mae: 0.8425
Epoch 133/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 17ms/step - loss: 0.4129 - mae: 0.4740 - val_loss: 2.235
8 - val_mae: 0.8888
Epoch 134/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 17ms/step - loss: 0.4169 - mae: 0.4745 - val_loss: 2.184
0 - val_mae: 0.8956
Epoch 135/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 18ms/step - loss: 0.4628 - mae: 0.4982 - val_loss: 2.292
8 - val_mae: 0.8754
Epoch 136/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 18ms/step - loss: 0.5355 - mae: 0.5676 - val_loss: 2.881
3 - val_mae: 1.0618
Epoch 137/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 18ms/step - loss: 0.9328 - mae: 0.7667 - val_loss: 2.524
9 - val_mae: 1.1080
Epoch 138/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 18ms/step - loss: 1.0138 - mae: 0.7796 - val_loss: 2.473
0 - val_mae: 1.1946
Epoch 139/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 1.3705 - mae: 0.9461 - val_loss: 1.978
1 - val_mae: 0.8902
Epoch 140/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 17ms/step - loss: 1.2167 - mae: 0.8507 - val_loss: 3.150
9 - val_mae: 1.1757
Epoch 141/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 18ms/step - loss: 1.1655 - mae: 0.8519 - val_loss: 2.617
0 - val_mae: 1.0471
Epoch 142/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 17ms/step - loss: 0.6565 - mae: 0.6253 - val_loss: 1.802
9 - val_mae: 0.8062
Epoch 143/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 17ms/step - loss: 0.5643 - mae: 0.5651 - val_loss: 1.802
3 - val_mae: 0.8054
Epoch 144/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 17ms/step - loss: 0.4194 - mae: 0.4874 - val_loss: 1.966
8 - val_mae: 0.7868
Epoch 145/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 17ms/step - loss: 0.3426 - mae: 0.4321 - val_loss: 2.101
8 - val_mae: 0.7914
Epoch 146/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 17ms/step - loss: 0.3450 - mae: 0.4331 - val_loss: 1.988
7 - val_mae: 0.8280
Epoch 147/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 17ms/step - loss: 0.3636 - mae: 0.4560 - val_loss: 1.896
5 - val_mae: 0.8618
Epoch 148/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 16ms/step - loss: 0.4462 - mae: 0.5066 - val_loss: 2.072
8 - val_mae: 0.9770
Epoch 149/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 18ms/step - loss: 0.4436 - mae: 0.4840 - val_loss: 2.041
7 - val_mae: 0.7749
Epoch 150/1400
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5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 16ms/step - loss: 0.3060 - mae: 0.4112 - val_loss: 1.936
6 - val_mae: 0.7658
Epoch 151/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 17ms/step - loss: 0.2922 - mae: 0.4060 - val_loss: 1.904
4 - val_mae: 0.7707
Epoch 152/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 18ms/step - loss: 0.3001 - mae: 0.3991 - val_loss: 1.933
7 - val_mae: 0.7588
Epoch 153/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 16ms/step - loss: 0.2936 - mae: 0.3957 - val_loss: 1.962
5 - val_mae: 0.7837
Epoch 154/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 18ms/step - loss: 0.3933 - mae: 0.4841 - val_loss: 2.472
4 - val_mae: 0.9463
Epoch 155/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 17ms/step - loss: 0.6533 - mae: 0.6238 - val_loss: 2.120
0 - val_mae: 0.9822
Epoch 156/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 17ms/step - loss: 0.6780 - mae: 0.6490 - val_loss: 1.984
5 - val_mae: 0.9757
Epoch 157/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 17ms/step - loss: 0.5307 - mae: 0.5614 - val_loss: 1.746
2 - val_mae: 0.7472
Epoch 158/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 17ms/step - loss: 0.3483 - mae: 0.4337 - val_loss: 2.284
8 - val_mae: 0.8694
Epoch 159/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 29ms/step - loss: 0.3567 - mae: 0.4625 - val_loss: 1.981
7 - val_mae: 0.7793
Epoch 160/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 34ms/step - loss: 0.3356 - mae: 0.4351 - val_loss: 1.711
6 - val_mae: 0.7404
Epoch 161/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 41ms/step - loss: 0.3003 - mae: 0.4187 - val_loss: 1.834
6 - val_mae: 0.7637
Epoch 162/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 49ms/step - loss: 0.3446 - mae: 0.4479 - val_loss: 1.946
3 - val_mae: 0.7396
Epoch 163/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 49ms/step - loss: 0.2921 - mae: 0.4048 - val_loss: 1.789
3 - val_mae: 0.7451
Epoch 164/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 57ms/step - loss: 0.2527 - mae: 0.3850 - val_loss: 1.586
9 - val_mae: 0.7117
Epoch 165/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 81ms/step - loss: 0.2781 - mae: 0.3901 - val_loss: 1.772
9 - val_mae: 0.7234
Epoch 166/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 70ms/step - loss: 0.3296 - mae: 0.4408 - val_loss: 2.083
1 - val_mae: 0.7837
Epoch 167/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 1s 55ms/step - loss: 0.3229 - mae: 0.4244 - val_loss: 1.993
7 - val_mae: 0.7835
Epoch 168/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 42ms/step - loss: 0.7047 - mae: 0.6952 - val_loss: 2.524
7 - val_mae: 1.3061
```

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Epoch 169/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 53ms/step - loss: 1.5024 - mae: 1.0262 - val_loss: 2.275
2 - val_mae: 1.0714
Epoch 170/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 40ms/step - loss: 0.8476 - mae: 0.7213 - val_loss: 1.936
5 - val_mae: 0.7858
Epoch 171/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 31ms/step - loss: 0.8473 - mae: 0.6827 - val_loss: 2.334
5 - val_mae: 0.9108
Epoch 172/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 52ms/step - loss: 0.4887 - mae: 0.5406 - val_loss: 2.013
9 - val_mae: 0.8523
Epoch 173/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 62ms/step - loss: 0.4267 - mae: 0.5234 - val_loss: 2.300
4 - val_mae: 0.8820
Epoch 174/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 1s 58ms/step - loss: 0.4869 - mae: 0.5550 - val_loss: 2.102
2 - val_mae: 0.7942
Epoch 175/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 46ms/step - loss: 0.5771 - mae: 0.5885 - val_loss: 1.864
4 - val_mae: 0.8034
Epoch 176/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 46ms/step - loss: 0.3517 - mae: 0.4452 - val_loss: 1.721
6 - val_mae: 0.7548
Epoch 177/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 59ms/step - loss: 0.2841 - mae: 0.4173 - val_loss: 1.783
9 - val_mae: 0.7523
Epoch 178/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 55ms/step - loss: 0.2838 - mae: 0.4180 - val_loss: 2.140
1 - val_mae: 0.8868
Epoch 179/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 56ms/step - loss: 0.3498 - mae: 0.4616 - val_loss: 1.964
5 - val_mae: 0.7978
Epoch 180/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 42ms/step - loss: 0.4205 - mae: 0.4958 - val_loss: 1.741
0 - val_mae: 0.9105
Epoch 181/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 18ms/step - loss: 0.3636 - mae: 0.4551 - val_loss: 1.787
4 - val_mae: 0.8598
Epoch 182/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 18ms/step - loss: 0.3225 - mae: 0.4445 - val_loss: 1.721
6 - val_mae: 0.7195
Epoch 183/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.2487 - mae: 0.3722 - val_loss: 1.593
1 - val_mae: 0.7013
Epoch 184/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 18ms/step - loss: 0.2798 - mae: 0.3990 - val_loss: 1.658
2 - val_mae: 0.7971
Epoch 185/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 17ms/step - loss: 0.3702 - mae: 0.4536 - val_loss: 1.800
9 - val_mae: 0.8582
Epoch 186/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.3956 - mae: 0.4956 - val_loss: 1.708
2 - val_mae: 0.7223
Epoch 187/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 18ms/step - loss: 0.2309 - mae: 0.3770 - val_loss: 1.618
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8 - val_mae: 0.7225
Epoch 188/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 21ms/step - loss: 0.2033 - mae: 0.3466 - val_loss: 1.632
2 - val_mae: 0.7273
Epoch 189/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 18ms/step - loss: 0.2271 - mae: 0.3615 - val_loss: 1.761
4 - val_mae: 0.7281
Epoch 190/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 18ms/step - loss: 0.1901 - mae: 0.3311 - val_loss: 1.536
4 - val_mae: 0.7072
Epoch 191/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 18ms/step - loss: 0.2217 - mae: 0.3563 - val_loss: 1.510
1 - val_mae: 0.7106
Epoch 192/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 18ms/step - loss: 0.2349 - mae: 0.3753 - val_loss: 1.730
9 - val_mae: 0.7245
Epoch 193/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.2073 - mae: 0.3475 - val_loss: 1.798
0 - val_mae: 0.7726
Epoch 194/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 18ms/step - loss: 0.2378 - mae: 0.3746 - val_loss: 1.625
5 - val_mae: 0.7410
Epoch 195/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 21ms/step - loss: 0.2661 - mae: 0.3805 - val_loss: 1.567
1 - val_mae: 0.7120
Epoch 196/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.2592 - mae: 0.3955 - val_loss: 1.746
4 - val_mae: 0.9070
Epoch 197/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.5422 - mae: 0.5630 - val_loss: 1.931
9 - val_mae: 1.0807
Epoch 198/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 21ms/step - loss: 0.4224 - mae: 0.5209 - val_loss: 1.674
2 - val_mae: 0.7866
Epoch 199/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.3042 - mae: 0.4249 - val_loss: 1.771
7 - val_mae: 0.7992
Epoch 200/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.3681 - mae: 0.4770 - val_loss: 1.625
2 - val_mae: 0.7446
Epoch 201/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 18ms/step - loss: 0.3108 - mae: 0.4279 - val_loss: 1.560
0 - val_mae: 0.7957
Epoch 202/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.2885 - mae: 0.4080 - val_loss: 1.590
0 - val_mae: 0.7269
Epoch 203/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 18ms/step - loss: 0.2775 - mae: 0.4019 - val_loss: 2.038
7 - val_mae: 0.8320
Epoch 204/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 18ms/step - loss: 0.3703 - mae: 0.4695 - val_loss: 1.751
1 - val_mae: 0.7467
Epoch 205/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.2834 - mae: 0.4052 - val_loss: 1.549
2 - val_mae: 0.8138
Epoch 206/1400
```

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5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 18ms/step - loss: 0.2841 - mae: 0.4234 - val_loss: 1.516
3 - val_mae: 0.8068
Epoch 207/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 18ms/step - loss: 0.3918 - mae: 0.4788 - val_loss: 1.801
0 - val_mae: 0.7575
Epoch 208/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 22ms/step - loss: 0.2830 - mae: 0.3898 - val_loss: 2.117
6 - val_mae: 0.8491
Epoch 209/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.2801 - mae: 0.4065 - val_loss: 1.673
6 - val_mae: 0.7183
Epoch 210/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.1716 - mae: 0.3120 - val_loss: 1.524
3 - val_mae: 0.7110
Epoch 211/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.1844 - mae: 0.3223 - val_loss: 1.667
7 - val_mae: 0.7278
Epoch 212/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.1633 - mae: 0.2994 - val_loss: 1.703
8 - val_mae: 0.7357
Epoch 213/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 18ms/step - loss: 0.1730 - mae: 0.3149 - val_loss: 1.586
8 - val_mae: 0.7155
Epoch 214/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 18ms/step - loss: 0.1509 - mae: 0.2959 - val_loss: 1.609
4 - val_mae: 0.7162
Epoch 215/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 18ms/step - loss: 0.1700 - mae: 0.3158 - val_loss: 1.521
5 - val_mae: 0.6974
Epoch 216/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 18ms/step - loss: 0.1712 - mae: 0.3164 - val_loss: 1.554
3 - val_mae: 0.7614
Epoch 217/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.3214 - mae: 0.4302 - val_loss: 1.612
0 - val_mae: 0.7280
Epoch 218/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 18ms/step - loss: 0.1984 - mae: 0.3401 - val_loss: 1.525
1 - val_mae: 0.7196
Epoch 219/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 17ms/step - loss: 0.1805 - mae: 0.3200 - val_loss: 1.587
0 - val_mae: 0.7495
Epoch 220/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 18ms/step - loss: 0.1731 - mae: 0.3132 - val_loss: 1.561
9 - val_mae: 0.7388
Epoch 221/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 18ms/step - loss: 0.1605 - mae: 0.3161 - val_loss: 1.530
9 - val_mae: 0.7247
Epoch 222/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.1464 - mae: 0.2881 - val_loss: 1.590
5 - val_mae: 0.7433
Epoch 223/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 17ms/step - loss: 0.2189 - mae: 0.3398 - val_loss: 1.512
4 - val_mae: 0.7247
Epoch 224/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.6107 - mae: 0.6058 - val_loss: 1.591
7 - val_mae: 0.8819
```

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Epoch 225/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 18ms/step - loss: 0.3885 - mae: 0.4862 - val_loss: 1.772
2 - val_mae: 0.8234
Epoch 226/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 18ms/step - loss: 0.2493 - mae: 0.3841 - val_loss: 1.568
9 - val_mae: 0.7442
Epoch 227/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 17ms/step - loss: 0.1809 - mae: 0.3210 - val_loss: 1.490
9 - val_mae: 0.7420
Epoch 228/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 21ms/step - loss: 0.1810 - mae: 0.3297 - val_loss: 1.533
0 - val_mae: 0.7183
Epoch 229/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 18ms/step - loss: 0.1463 - mae: 0.2789 - val_loss: 1.703
1 - val_mae: 0.7653
Epoch 230/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.1629 - mae: 0.3078 - val_loss: 1.513
2 - val_mae: 0.7350
Epoch 231/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 17ms/step - loss: 0.1545 - mae: 0.2904 - val_loss: 1.562
1 - val_mae: 0.7580
Epoch 232/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 17ms/step - loss: 0.1850 - mae: 0.3290 - val_loss: 1.448
6 - val_mae: 0.7522
Epoch 233/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 17ms/step - loss: 0.2372 - mae: 0.3710 - val_loss: 1.427
5 - val_mae: 0.7164
Epoch 234/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 17ms/step - loss: 0.2321 - mae: 0.3592 - val_loss: 2.356
4 - val_mae: 1.0041
Epoch 235/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.6250 - mae: 0.6242 - val_loss: 1.427
3 - val_mae: 0.7479
Epoch 236/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 18ms/step - loss: 0.2887 - mae: 0.4367 - val_loss: 1.557
6 - val_mae: 0.7628
Epoch 237/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 18ms/step - loss: 0.2390 - mae: 0.3953 - val_loss: 1.557
7 - val_mae: 0.7673
Epoch 238/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 21ms/step - loss: 0.2255 - mae: 0.3618 - val_loss: 1.610
6 - val_mae: 0.7767
Epoch 239/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.3031 - mae: 0.4338 - val_loss: 1.769
2 - val_mae: 0.8069
Epoch 240/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 29ms/step - loss: 0.2879 - mae: 0.4344 - val_loss: 1.398
6 - val_mae: 0.6926
Epoch 241/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 25ms/step - loss: 0.2877 - mae: 0.4180 - val_loss: 1.526
5 - val_mae: 0.8410
Epoch 242/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 34ms/step - loss: 0.3262 - mae: 0.4445 - val_loss: 1.398
8 - val_mae: 0.7018
Epoch 243/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 31ms/step - loss: 0.2649 - mae: 0.4022 - val_loss: 1.994
```

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2 - val_mae: 0.9144
Epoch 244/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 36ms/step - loss: 0.6906 - mae: 0.6604 - val_loss: 1.655
6 - val_mae: 0.7909
Epoch 245/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 41ms/step - loss: 0.5342 - mae: 0.5699 - val_loss: 1.491
4 - val_mae: 0.8619
Epoch 246/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 25ms/step - loss: 0.3995 - mae: 0.5054 - val_loss: 1.598
5 - val_mae: 0.8813
Epoch 247/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 26ms/step - loss: 0.5727 - mae: 0.5856 - val_loss: 1.540
5 - val_mae: 0.7527
Epoch 248/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 37ms/step - loss: 0.3476 - mae: 0.4639 - val_loss: 2.387
7 - val_mae: 1.0369
Epoch 249/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 24ms/step - loss: 0.5901 - mae: 0.5992 - val_loss: 1.629
3 - val_mae: 0.7307
Epoch 250/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 27ms/step - loss: 0.2287 - mae: 0.3803 - val_loss: 1.525
5 - val_mae: 0.7786
Epoch 251/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 38ms/step - loss: 0.2337 - mae: 0.3736 - val_loss: 1.548
7 - val_mae: 0.7247
Epoch 252/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 18ms/step - loss: 0.1518 - mae: 0.2922 - val_loss: 1.688
1 - val_mae: 0.7284
Epoch 253/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 17ms/step - loss: 0.1637 - mae: 0.3197 - val_loss: 1.491
9 - val_mae: 0.7051
Epoch 254/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 18ms/step - loss: 0.1721 - mae: 0.3258 - val_loss: 1.461
2 - val_mae: 0.6994
Epoch 255/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 17ms/step - loss: 0.1423 - mae: 0.2990 - val_loss: 1.567
0 - val_mae: 0.7244
Epoch 256/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 17ms/step - loss: 0.1740 - mae: 0.3191 - val_loss: 1.408
5 - val_mae: 0.7241
Epoch 257/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 18ms/step - loss: 0.1599 - mae: 0.3063 - val_loss: 1.355
0 - val_mae: 0.7686
Epoch 258/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 18ms/step - loss: 0.3358 - mae: 0.4223 - val_loss: 1.696
6 - val_mae: 0.7783
Epoch 259/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.3375 - mae: 0.4538 - val_loss: 1.736
5 - val_mae: 0.8453
Epoch 260/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 18ms/step - loss: 0.2522 - mae: 0.3901 - val_loss: 1.331
5 - val_mae: 0.7184
Epoch 261/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 21ms/step - loss: 0.1978 - mae: 0.3325 - val_loss: 1.494
1 - val_mae: 0.7299
Epoch 262/1400
```

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5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 17ms/step - loss: 0.1752 - mae: 0.3259 - val_loss: 1.485
3 - val_mae: 0.7260
Epoch 263/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 18ms/step - loss: 0.1262 - mae: 0.2647 - val_loss: 1.476
9 - val_mae: 0.7145
Epoch 264/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 18ms/step - loss: 0.1678 - mae: 0.3119 - val_loss: 1.428
1 - val_mae: 0.6941
Epoch 265/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 18ms/step - loss: 0.1306 - mae: 0.2640 - val_loss: 1.440
8 - val_mae: 0.7100
Epoch 266/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.1280 - mae: 0.2685 - val_loss: 1.415
2 - val_mae: 0.6981
Epoch 267/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.1779 - mae: 0.3282 - val_loss: 1.434
0 - val_mae: 0.7102
Epoch 268/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 18ms/step - loss: 0.1157 - mae: 0.2564 - val_loss: 1.458
2 - val_mae: 0.7112
Epoch 269/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.1297 - mae: 0.2741 - val_loss: 1.323
6 - val_mae: 0.6915
Epoch 270/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.1076 - mae: 0.2487 - val_loss: 1.463
1 - val_mae: 0.7143
Epoch 271/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 23ms/step - loss: 0.1213 - mae: 0.2533 - val_loss: 1.388
2 - val_mae: 0.7059
Epoch 272/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 18ms/step - loss: 0.1543 - mae: 0.2891 - val_loss: 1.412
4 - val_mae: 0.7201
Epoch 273/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 18ms/step - loss: 0.4296 - mae: 0.4908 - val_loss: 1.264
9 - val_mae: 0.6878
Epoch 274/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.3965 - mae: 0.4764 - val_loss: 1.353
1 - val_mae: 0.8680
Epoch 275/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 21ms/step - loss: 0.3498 - mae: 0.4426 - val_loss: 1.732
4 - val_mae: 0.9730
Epoch 276/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.3817 - mae: 0.4868 - val_loss: 1.480
6 - val_mae: 0.7342
Epoch 277/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.2891 - mae: 0.4038 - val_loss: 2.181
6 - val_mae: 0.9551
Epoch 278/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 18ms/step - loss: 0.5233 - mae: 0.5733 - val_loss: 1.599
6 - val_mae: 0.7452
Epoch 279/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.4380 - mae: 0.4996 - val_loss: 1.589
3 - val_mae: 0.8765
Epoch 280/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 18ms/step - loss: 0.3707 - mae: 0.4711 - val_loss: 1.580
2 - val_mae: 0.7894
```

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Epoch 281/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 22ms/step - loss: 0.6074 - mae: 0.6226 - val_loss: 1.757
6 - val_mae: 0.8306
Epoch 282/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 17ms/step - loss: 0.3454 - mae: 0.4539 - val_loss: 1.958
8 - val_mae: 0.9000
Epoch 283/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 17ms/step - loss: 0.3612 - mae: 0.4611 - val_loss: 1.848
3 - val_mae: 0.8255
Epoch 284/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 17ms/step - loss: 0.3012 - mae: 0.4133 - val_loss: 1.304
7 - val_mae: 0.6913
Epoch 285/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.2130 - mae: 0.3529 - val_loss: 1.606
4 - val_mae: 0.7434
Epoch 286/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 18ms/step - loss: 0.2193 - mae: 0.3711 - val_loss: 1.510
7 - val_mae: 0.7037
Epoch 287/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 18ms/step - loss: 0.2071 - mae: 0.3497 - val_loss: 1.460
9 - val_mae: 0.7794
Epoch 288/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 18ms/step - loss: 0.2578 - mae: 0.3853 - val_loss: 1.376
9 - val_mae: 0.7058
Epoch 289/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 18ms/step - loss: 0.1464 - mae: 0.2985 - val_loss: 1.451
2 - val_mae: 0.7120
Epoch 290/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 17ms/step - loss: 0.1609 - mae: 0.3082 - val_loss: 1.381
0 - val_mae: 0.7502
Epoch 291/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 21ms/step - loss: 0.3132 - mae: 0.4366 - val_loss: 1.373
2 - val_mae: 0.7264
Epoch 292/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 17ms/step - loss: 0.2317 - mae: 0.3638 - val_loss: 1.727
6 - val_mae: 0.7798
Epoch 293/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 18ms/step - loss: 0.2296 - mae: 0.3812 - val_loss: 1.679
2 - val_mae: 0.8168
Epoch 294/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 18ms/step - loss: 0.3381 - mae: 0.4411 - val_loss: 1.354
8 - val_mae: 0.7261
Epoch 295/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.2016 - mae: 0.3461 - val_loss: 1.374
7 - val_mae: 0.6992
Epoch 296/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 17ms/step - loss: 0.2800 - mae: 0.4001 - val_loss: 1.756
8 - val_mae: 0.8217
Epoch 297/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 17ms/step - loss: 0.2972 - mae: 0.4327 - val_loss: 1.339
8 - val_mae: 0.7036
Epoch 298/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 16ms/step - loss: 0.2193 - mae: 0.3768 - val_loss: 1.242
1 - val_mae: 0.6867
Epoch 299/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 17ms/step - loss: 0.1544 - mae: 0.3074 - val_loss: 1.249
```

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1 - val_mae: 0.7065
Epoch 300/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 17ms/step - loss: 0.1580 - mae: 0.3123 - val_loss: 1.341
6 - val_mae: 0.7438
Epoch 301/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 18ms/step - loss: 0.2871 - mae: 0.4293 - val_loss: 1.568
8 - val_mae: 0.7803
Epoch 302/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 21ms/step - loss: 0.2175 - mae: 0.3628 - val_loss: 1.313
2 - val_mae: 0.6909
Epoch 303/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 17ms/step - loss: 0.1700 - mae: 0.3181 - val_loss: 1.363
9 - val_mae: 0.7380
Epoch 304/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 18ms/step - loss: 0.1833 - mae: 0.3328 - val_loss: 1.430
7 - val_mae: 0.7166
Epoch 305/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 17ms/step - loss: 0.1758 - mae: 0.3328 - val_loss: 1.924
0 - val_mae: 0.9066
Epoch 306/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 18ms/step - loss: 0.4234 - mae: 0.4885 - val_loss: 1.393
4 - val_mae: 0.7856
Epoch 307/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.2558 - mae: 0.3881 - val_loss: 1.431
5 - val_mae: 0.7634
Epoch 308/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 17ms/step - loss: 0.2162 - mae: 0.3381 - val_loss: 1.744
5 - val_mae: 0.7835
Epoch 309/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 18ms/step - loss: 0.2004 - mae: 0.3386 - val_loss: 1.419
3 - val_mae: 0.7179
Epoch 310/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 18ms/step - loss: 0.1222 - mae: 0.2658 - val_loss: 1.474
3 - val_mae: 0.7208
Epoch 311/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.1445 - mae: 0.2890 - val_loss: 1.621
0 - val_mae: 0.7618
Epoch 312/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 18ms/step - loss: 0.1936 - mae: 0.3484 - val_loss: 1.518
7 - val_mae: 0.7352
Epoch 313/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 18ms/step - loss: 0.1390 - mae: 0.2931 - val_loss: 1.382
1 - val_mae: 0.7444
Epoch 314/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 18ms/step - loss: 0.2727 - mae: 0.4156 - val_loss: 1.358
4 - val_mae: 0.6996
Epoch 315/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 18ms/step - loss: 0.3221 - mae: 0.4268 - val_loss: 1.988
7 - val_mae: 0.9526
Epoch 316/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 18ms/step - loss: 0.4094 - mae: 0.5007 - val_loss: 1.399
3 - val_mae: 0.6948
Epoch 317/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 18ms/step - loss: 0.2185 - mae: 0.3774 - val_loss: 1.228
7 - val_mae: 0.7279
Epoch 318/1400
```

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5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 18ms/step - loss: 0.2993 - mae: 0.4356 - val_loss: 1.343
5 - val_mae: 0.6919
Epoch 319/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 18ms/step - loss: 0.3631 - mae: 0.4646 - val_loss: 2.121
3 - val_mae: 0.9643
Epoch 320/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 18ms/step - loss: 0.2913 - mae: 0.4149 - val_loss: 1.433
7 - val_mae: 0.7143
Epoch 321/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 17ms/step - loss: 0.2069 - mae: 0.3552 - val_loss: 1.393
5 - val_mae: 0.8241
Epoch 322/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.4642 - mae: 0.5552 - val_loss: 1.501
9 - val_mae: 0.7894
Epoch 323/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 17ms/step - loss: 0.3183 - mae: 0.4480 - val_loss: 1.663
7 - val_mae: 0.7700
Epoch 324/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 17ms/step - loss: 0.2313 - mae: 0.3948 - val_loss: 1.596
3 - val_mae: 0.7875
Epoch 325/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.3407 - mae: 0.4616 - val_loss: 1.569
6 - val_mae: 0.7630
Epoch 326/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 18ms/step - loss: 0.2761 - mae: 0.4214 - val_loss: 1.439
5 - val_mae: 0.7327
Epoch 327/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 17ms/step - loss: 0.2446 - mae: 0.3730 - val_loss: 1.389
6 - val_mae: 0.7785
Epoch 328/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 18ms/step - loss: 0.2207 - mae: 0.3759 - val_loss: 1.734
8 - val_mae: 0.7566
Epoch 329/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 18ms/step - loss: 0.1586 - mae: 0.2995 - val_loss: 1.429
7 - val_mae: 0.7101
Epoch 330/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.2324 - mae: 0.3765 - val_loss: 1.389
1 - val_mae: 0.7978
Epoch 331/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 17ms/step - loss: 0.3335 - mae: 0.4479 - val_loss: 1.861
0 - val_mae: 0.8385
Epoch 332/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 22ms/step - loss: 0.3897 - mae: 0.4775 - val_loss: 2.372
1 - val_mae: 1.0388
Epoch 333/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 17ms/step - loss: 0.4206 - mae: 0.5050 - val_loss: 1.876
7 - val_mae: 0.8974
Epoch 334/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 18ms/step - loss: 0.5858 - mae: 0.6135 - val_loss: 1.509
7 - val_mae: 0.7317
Epoch 335/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 18ms/step - loss: 0.3684 - mae: 0.4687 - val_loss: 1.344
8 - val_mae: 0.7376
Epoch 336/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 17ms/step - loss: 0.3804 - mae: 0.4897 - val_loss: 1.648
8 - val_mae: 0.9416
```

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Epoch 337/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 18ms/step - loss: 0.4486 - mae: 0.4967 - val_loss: 1.627
8 - val_mae: 0.8468
Epoch 338/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 18ms/step - loss: 0.2154 - mae: 0.3670 - val_loss: 1.368
2 - val_mae: 0.7056
Epoch 339/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 16ms/step - loss: 0.1447 - mae: 0.2999 - val_loss: 1.469
9 - val_mae: 0.7132
Epoch 340/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 16ms/step - loss: 0.1189 - mae: 0.2668 - val_loss: 1.386
7 - val_mae: 0.6938
Epoch 341/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 17ms/step - loss: 0.1409 - mae: 0.2828 - val_loss: 1.294
7 - val_mae: 0.6960
Epoch 342/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 18ms/step - loss: 0.1534 - mae: 0.2995 - val_loss: 1.508
0 - val_mae: 0.7265
Epoch 343/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 23ms/step - loss: 0.1213 - mae: 0.2680 - val_loss: 1.571
4 - val_mae: 0.7426
Epoch 344/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 18ms/step - loss: 0.1521 - mae: 0.2960 - val_loss: 1.256
5 - val_mae: 0.6777
Epoch 345/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.1259 - mae: 0.2733 - val_loss: 1.400
0 - val_mae: 0.7422
Epoch 346/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 17ms/step - loss: 0.2049 - mae: 0.3471 - val_loss: 1.464
9 - val_mae: 0.7226
Epoch 347/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 16ms/step - loss: 0.1582 - mae: 0.2977 - val_loss: 1.739
8 - val_mae: 0.8229
Epoch 348/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 17ms/step - loss: 0.5530 - mae: 0.5836 - val_loss: 2.330
5 - val_mae: 1.0498
Epoch 349/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.6312 - mae: 0.6331 - val_loss: 1.620
8 - val_mae: 0.7441
Epoch 350/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 17ms/step - loss: 0.4787 - mae: 0.5428 - val_loss: 1.278
8 - val_mae: 0.7221
Epoch 351/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 33ms/step - loss: 0.3305 - mae: 0.4326 - val_loss: 1.458
1 - val_mae: 0.7215
Epoch 352/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 29ms/step - loss: 0.1647 - mae: 0.3215 - val_loss: 1.440
5 - val_mae: 0.6911
Epoch 353/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 36ms/step - loss: 0.1490 - mae: 0.3078 - val_loss: 1.510
4 - val_mae: 0.7107
Epoch 354/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 26ms/step - loss: 0.1809 - mae: 0.3241 - val_loss: 1.558
4 - val_mae: 0.7252
Epoch 355/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 37ms/step - loss: 0.1271 - mae: 0.2769 - val_loss: 1.448
```

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5 - val_mae: 0.7392
Epoch 356/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 26ms/step - loss: 0.1812 - mae: 0.3227 - val_loss: 1.479
1 - val_mae: 0.7050
Epoch 357/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 33ms/step - loss: 0.1776 - mae: 0.3285 - val_loss: 1.385
2 - val_mae: 0.7458
Epoch 358/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 37ms/step - loss: 0.2729 - mae: 0.4180 - val_loss: 1.325
8 - val_mae: 0.7731
Epoch 359/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 22ms/step - loss: 0.4497 - mae: 0.5421 - val_loss: 1.305
5 - val_mae: 0.6694
Epoch 360/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 28ms/step - loss: 0.3930 - mae: 0.4811 - val_loss: 1.726
8 - val_mae: 0.8076
Epoch 361/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 29ms/step - loss: 0.5643 - mae: 0.5711 - val_loss: 2.509
5 - val_mae: 1.1179
Epoch 362/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 18ms/step - loss: 0.5624 - mae: 0.6076 - val_loss: 2.563
4 - val_mae: 1.0866
Epoch 363/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 18ms/step - loss: 0.7034 - mae: 0.6693 - val_loss: 1.994
2 - val_mae: 0.9397
Epoch 364/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 18ms/step - loss: 0.3751 - mae: 0.4705 - val_loss: 1.361
9 - val_mae: 0.7540
Epoch 365/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.5512 - mae: 0.6303 - val_loss: 1.752
9 - val_mae: 0.8411
Epoch 366/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 17ms/step - loss: 0.4359 - mae: 0.5152 - val_loss: 1.702
8 - val_mae: 0.7811
Epoch 367/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 18ms/step - loss: 0.2291 - mae: 0.3792 - val_loss: 1.337
1 - val_mae: 0.7800
Epoch 368/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 18ms/step - loss: 0.3160 - mae: 0.4470 - val_loss: 1.388
1 - val_mae: 0.7894
Epoch 369/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 18ms/step - loss: 0.2727 - mae: 0.3943 - val_loss: 1.582
3 - val_mae: 0.7221
Epoch 370/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 18ms/step - loss: 0.1368 - mae: 0.2902 - val_loss: 1.601
0 - val_mae: 0.7322
Epoch 371/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 18ms/step - loss: 0.1566 - mae: 0.3038 - val_loss: 1.522
4 - val_mae: 0.7247
Epoch 372/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 18ms/step - loss: 0.1402 - mae: 0.2897 - val_loss: 1.408
6 - val_mae: 0.7032
Epoch 373/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 18ms/step - loss: 0.1502 - mae: 0.2925 - val_loss: 1.572
3 - val_mae: 0.7610
Epoch 374/1400
```

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5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 18ms/step - loss: 0.1725 - mae: 0.3082 - val_loss: 1.512
7 - val_mae: 0.7148
Epoch 375/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 17ms/step - loss: 0.1363 - mae: 0.2876 - val_loss: 1.318
5 - val_mae: 0.6921
Epoch 376/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 17ms/step - loss: 0.1250 - mae: 0.2662 - val_loss: 1.271
8 - val_mae: 0.6762
Epoch 377/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 18ms/step - loss: 0.1113 - mae: 0.2537 - val_loss: 1.381
1 - val_mae: 0.6973
Epoch 378/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 17ms/step - loss: 0.1141 - mae: 0.2574 - val_loss: 1.312
2 - val_mae: 0.6800
Epoch 379/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 17ms/step - loss: 0.1120 - mae: 0.2469 - val_loss: 1.359
6 - val_mae: 0.6903
Epoch 380/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 18ms/step - loss: 0.1109 - mae: 0.2458 - val_loss: 1.240
9 - val_mae: 0.6673
Epoch 381/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.1040 - mae: 0.2360 - val_loss: 1.173
4 - val_mae: 0.6669
Epoch 382/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 17ms/step - loss: 0.0994 - mae: 0.2356 - val_loss: 1.304
9 - val_mae: 0.6841
Epoch 383/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 17ms/step - loss: 0.0964 - mae: 0.2356 - val_loss: 1.290
1 - val_mae: 0.6794
Epoch 384/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 18ms/step - loss: 0.0909 - mae: 0.2152 - val_loss: 1.262
1 - val_mae: 0.6733
Epoch 385/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 17ms/step - loss: 0.0876 - mae: 0.2171 - val_loss: 1.259
7 - val_mae: 0.6745
Epoch 386/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.1206 - mae: 0.2539 - val_loss: 1.326
3 - val_mae: 0.6891
Epoch 387/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 17ms/step - loss: 0.1407 - mae: 0.2830 - val_loss: 1.494
6 - val_mae: 0.7636
Epoch 388/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.1417 - mae: 0.2826 - val_loss: 1.402
7 - val_mae: 0.7275
Epoch 389/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 18ms/step - loss: 0.1318 - mae: 0.2798 - val_loss: 1.327
7 - val_mae: 0.6953
Epoch 390/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 18ms/step - loss: 0.1052 - mae: 0.2460 - val_loss: 1.227
6 - val_mae: 0.6749
Epoch 391/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 18ms/step - loss: 0.0991 - mae: 0.2272 - val_loss: 1.202
4 - val_mae: 0.6648
Epoch 392/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 17ms/step - loss: 0.0939 - mae: 0.2293 - val_loss: 1.234
7 - val_mae: 0.6669
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Epoch 393/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 16ms/step - loss: 0.1030 - mae: 0.2267 - val_loss: 1.401
0 - val_mae: 0.7169
Epoch 394/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 30ms/step - loss: 0.1207 - mae: 0.2584 - val_loss: 1.281
9 - val_mae: 0.6800
Epoch 395/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 22ms/step - loss: 0.1869 - mae: 0.3096 - val_loss: 1.188
0 - val_mae: 0.7240
Epoch 396/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 18ms/step - loss: 0.2090 - mae: 0.3569 - val_loss: 1.227
3 - val_mae: 0.7147
Epoch 397/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 17ms/step - loss: 0.1686 - mae: 0.3154 - val_loss: 1.336
4 - val_mae: 0.6926
Epoch 398/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 17ms/step - loss: 0.1501 - mae: 0.2991 - val_loss: 1.344
5 - val_mae: 0.6982
Epoch 399/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 17ms/step - loss: 0.1694 - mae: 0.2997 - val_loss: 1.371
6 - val_mae: 0.7211
Epoch 400/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 17ms/step - loss: 0.1610 - mae: 0.3065 - val_loss: 1.431
1 - val_mae: 0.7312
Epoch 401/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 18ms/step - loss: 0.1308 - mae: 0.2686 - val_loss: 1.635
8 - val_mae: 0.7958
Epoch 402/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 17ms/step - loss: 0.1559 - mae: 0.3119 - val_loss: 1.236
3 - val_mae: 0.6702
Epoch 403/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.1496 - mae: 0.2993 - val_loss: 1.146
6 - val_mae: 0.6812
Epoch 404/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 21ms/step - loss: 0.1369 - mae: 0.2871 - val_loss: 1.299
6 - val_mae: 0.7193
Epoch 405/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 18ms/step - loss: 0.1962 - mae: 0.3385 - val_loss: 1.215
0 - val_mae: 0.7023
Epoch 406/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 18ms/step - loss: 0.3057 - mae: 0.4349 - val_loss: 1.486
5 - val_mae: 0.7988
Epoch 407/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 18ms/step - loss: 0.3487 - mae: 0.4580 - val_loss: 2.054
8 - val_mae: 0.9220
Epoch 408/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 18ms/step - loss: 0.2961 - mae: 0.4105 - val_loss: 1.460
3 - val_mae: 0.7798
Epoch 409/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 17ms/step - loss: 0.1956 - mae: 0.3446 - val_loss: 1.455
6 - val_mae: 0.7922
Epoch 410/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 17ms/step - loss: 0.2789 - mae: 0.3954 - val_loss: 1.304
2 - val_mae: 0.6836
Epoch 411/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 17ms/step - loss: 0.1527 - mae: 0.3060 - val_loss: 1.385
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3 - val_mae: 0.7337
Epoch 412/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 18ms/step - loss: 0.2075 - mae: 0.3555 - val_loss: 1.358
2 - val_mae: 0.7151
Epoch 413/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 17ms/step - loss: 0.2081 - mae: 0.3436 - val_loss: 1.289
3 - val_mae: 0.6773
Epoch 414/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 28ms/step - loss: 0.2503 - mae: 0.3935 - val_loss: 1.451
1 - val_mae: 0.7371
Epoch 415/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 17ms/step - loss: 0.2446 - mae: 0.3803 - val_loss: 1.430
3 - val_mae: 0.7063
Epoch 416/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 16ms/step - loss: 0.1512 - mae: 0.3034 - val_loss: 1.311
0 - val_mae: 0.7211
Epoch 417/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 18ms/step - loss: 0.2210 - mae: 0.3734 - val_loss: 1.474
8 - val_mae: 0.7563
Epoch 418/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.2514 - mae: 0.3852 - val_loss: 1.764
1 - val_mae: 0.8828
Epoch 419/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 17ms/step - loss: 0.2818 - mae: 0.4197 - val_loss: 1.787
4 - val_mae: 0.8409
Epoch 420/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.1915 - mae: 0.3358 - val_loss: 1.614
8 - val_mae: 0.7823
Epoch 421/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 18ms/step - loss: 0.1891 - mae: 0.3325 - val_loss: 1.536
0 - val_mae: 0.7770
Epoch 422/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 18ms/step - loss: 0.2275 - mae: 0.3638 - val_loss: 1.181
7 - val_mae: 0.6430
Epoch 423/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.1916 - mae: 0.3342 - val_loss: 1.234
2 - val_mae: 0.7077
Epoch 424/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 21ms/step - loss: 0.2094 - mae: 0.3472 - val_loss: 1.266
9 - val_mae: 0.7228
Epoch 425/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.2180 - mae: 0.3635 - val_loss: 1.589
0 - val_mae: 0.7591
Epoch 426/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 17ms/step - loss: 0.2637 - mae: 0.4074 - val_loss: 1.489
8 - val_mae: 0.7661
Epoch 427/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.2556 - mae: 0.3908 - val_loss: 1.887
3 - val_mae: 0.8357
Epoch 428/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 17ms/step - loss: 0.4237 - mae: 0.5081 - val_loss: 1.176
2 - val_mae: 0.7092
Epoch 429/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 18ms/step - loss: 0.3744 - mae: 0.4617 - val_loss: 1.425
2 - val_mae: 0.8812
Epoch 430/1400
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5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 17ms/step - loss: 0.3168 - mae: 0.4457 - val_loss: 1.295
5 - val_mae: 0.8110
Epoch 431/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.2704 - mae: 0.4253 - val_loss: 1.218
0 - val_mae: 0.7473
Epoch 432/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 18ms/step - loss: 0.2222 - mae: 0.3611 - val_loss: 1.283
8 - val_mae: 0.6907
Epoch 433/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 18ms/step - loss: 0.2574 - mae: 0.3882 - val_loss: 1.458
0 - val_mae: 0.7095
Epoch 434/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.2647 - mae: 0.4104 - val_loss: 1.980
7 - val_mae: 0.9706
Epoch 435/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 17ms/step - loss: 0.6373 - mae: 0.6608 - val_loss: 2.173
0 - val_mae: 1.0793
Epoch 436/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 17ms/step - loss: 0.5797 - mae: 0.5797 - val_loss: 1.333
6 - val_mae: 0.7171
Epoch 437/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 17ms/step - loss: 0.3674 - mae: 0.4635 - val_loss: 1.290
7 - val_mae: 0.6700
Epoch 438/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 18ms/step - loss: 0.4150 - mae: 0.5029 - val_loss: 1.372
4 - val_mae: 0.7985
Epoch 439/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 18ms/step - loss: 0.2835 - mae: 0.4277 - val_loss: 1.493
5 - val_mae: 0.7719
Epoch 440/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 18ms/step - loss: 0.1791 - mae: 0.3352 - val_loss: 1.157
8 - val_mae: 0.6549
Epoch 441/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.1539 - mae: 0.3035 - val_loss: 1.264
9 - val_mae: 0.6576
Epoch 442/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 17ms/step - loss: 0.1233 - mae: 0.2664 - val_loss: 1.374
0 - val_mae: 0.6827
Epoch 443/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 17ms/step - loss: 0.1370 - mae: 0.2774 - val_loss: 1.446
5 - val_mae: 0.7137
Epoch 444/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.1780 - mae: 0.3287 - val_loss: 1.226
2 - val_mae: 0.6603
Epoch 445/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 17ms/step - loss: 0.1588 - mae: 0.2986 - val_loss: 1.098
7 - val_mae: 0.6527
Epoch 446/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 17ms/step - loss: 0.2242 - mae: 0.3616 - val_loss: 1.224
2 - val_mae: 0.7261
Epoch 447/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 18ms/step - loss: 0.1808 - mae: 0.3212 - val_loss: 1.250
1 - val_mae: 0.6894
Epoch 448/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.2540 - mae: 0.3946 - val_loss: 1.219
1 - val_mae: 0.6646
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Epoch 449/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 17ms/step - loss: 0.1345 - mae: 0.2791 - val_loss: 1.483
5 - val_mae: 0.7208
Epoch 450/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 17ms/step - loss: 0.1516 - mae: 0.2974 - val_loss: 1.511
8 - val_mae: 0.7101
Epoch 451/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.1803 - mae: 0.3172 - val_loss: 1.354
0 - val_mae: 0.6827
Epoch 452/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 18ms/step - loss: 0.1518 - mae: 0.2959 - val_loss: 1.231
0 - val_mae: 0.6596
Epoch 453/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 17ms/step - loss: 0.1654 - mae: 0.3162 - val_loss: 1.240
0 - val_mae: 0.6771
Epoch 454/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 17ms/step - loss: 0.1706 - mae: 0.3136 - val_loss: 1.168
0 - val_mae: 0.6616
Epoch 455/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 21ms/step - loss: 0.1335 - mae: 0.2884 - val_loss: 1.262
0 - val_mae: 0.6888
Epoch 456/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 17ms/step - loss: 0.1336 - mae: 0.2847 - val_loss: 1.265
3 - val_mae: 0.6781
Epoch 457/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 18ms/step - loss: 0.1287 - mae: 0.2712 - val_loss: 1.476
9 - val_mae: 0.7679
Epoch 458/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 17ms/step - loss: 0.1442 - mae: 0.2913 - val_loss: 1.152
7 - val_mae: 0.6464
Epoch 459/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 36ms/step - loss: 0.1172 - mae: 0.2625 - val_loss: 1.350
6 - val_mae: 0.6926
Epoch 460/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 25ms/step - loss: 0.1117 - mae: 0.2511 - val_loss: 1.271
4 - val_mae: 0.6736
Epoch 461/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 37ms/step - loss: 0.1021 - mae: 0.2430 - val_loss: 1.266
6 - val_mae: 0.6766
Epoch 462/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 26ms/step - loss: 0.1087 - mae: 0.2474 - val_loss: 1.220
5 - val_mae: 0.6665
Epoch 463/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.0965 - mae: 0.2291 - val_loss: 1.254
4 - val_mae: 0.6799
Epoch 464/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 33ms/step - loss: 0.0906 - mae: 0.2216 - val_loss: 1.271
1 - val_mae: 0.6753
Epoch 465/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 24ms/step - loss: 0.0848 - mae: 0.2110 - val_loss: 1.283
1 - val_mae: 0.6824
Epoch 466/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 24ms/step - loss: 0.0932 - mae: 0.2222 - val_loss: 1.198
7 - val_mae: 0.6669
Epoch 467/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 26ms/step - loss: 0.1348 - mae: 0.2848 - val_loss: 1.148
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4 - val_mae: 0.6569
Epoch 468/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 36ms/step - loss: 0.1620 - mae: 0.3012 - val_loss: 1.214
5 - val_mae: 0.6936
Epoch 469/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 23ms/step - loss: 0.1341 - mae: 0.2764 - val_loss: 1.247
1 - val_mae: 0.6626
Epoch 470/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 29ms/step - loss: 0.1285 - mae: 0.2594 - val_loss: 1.705
4 - val_mae: 0.8396
Epoch 471/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 18ms/step - loss: 0.2269 - mae: 0.3500 - val_loss: 1.184
5 - val_mae: 0.6681
Epoch 472/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 18ms/step - loss: 0.3465 - mae: 0.4448 - val_loss: 1.179
3 - val_mae: 0.6747
Epoch 473/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 17ms/step - loss: 0.4249 - mae: 0.5077 - val_loss: 1.422
1 - val_mae: 0.8228
Epoch 474/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 17ms/step - loss: 0.5163 - mae: 0.5706 - val_loss: 1.531
9 - val_mae: 0.9510
Epoch 475/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.5556 - mae: 0.5824 - val_loss: 1.151
5 - val_mae: 0.6906
Epoch 476/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.3953 - mae: 0.4770 - val_loss: 1.657
3 - val_mae: 0.7581
Epoch 477/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 18ms/step - loss: 0.2778 - mae: 0.4148 - val_loss: 1.382
4 - val_mae: 0.7100
Epoch 478/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 18ms/step - loss: 0.5513 - mae: 0.6294 - val_loss: 1.728
8 - val_mae: 0.8820
Epoch 479/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 17ms/step - loss: 0.5661 - mae: 0.6098 - val_loss: 2.016
2 - val_mae: 0.9189
Epoch 480/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 17ms/step - loss: 0.3557 - mae: 0.4664 - val_loss: 1.724
6 - val_mae: 0.8182
Epoch 481/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 17ms/step - loss: 0.2782 - mae: 0.4057 - val_loss: 1.561
4 - val_mae: 0.8174
Epoch 482/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 18ms/step - loss: 0.2700 - mae: 0.4092 - val_loss: 1.448
5 - val_mae: 0.6969
Epoch 483/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.2179 - mae: 0.3509 - val_loss: 1.273
3 - val_mae: 0.6965
Epoch 484/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 17ms/step - loss: 0.2005 - mae: 0.3462 - val_loss: 1.193
4 - val_mae: 0.6869
Epoch 485/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 18ms/step - loss: 0.1543 - mae: 0.3154 - val_loss: 1.528
2 - val_mae: 0.7028
Epoch 486/1400
```

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5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.1716 - mae: 0.3125 - val_loss: 1.428
1 - val_mae: 0.6884
Epoch 487/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 18ms/step - loss: 0.1219 - mae: 0.2687 - val_loss: 1.343
0 - val_mae: 0.6920
Epoch 488/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 18ms/step - loss: 0.1105 - mae: 0.2545 - val_loss: 1.542
9 - val_mae: 0.7523
Epoch 489/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 68ms/step - loss: 0.1327 - mae: 0.2810 - val_loss: 1.410
3 - val_mae: 0.6958
Epoch 490/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 18ms/step - loss: 0.1101 - mae: 0.2574 - val_loss: 1.127
7 - val_mae: 0.6337
Epoch 491/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0948 - mae: 0.2308 - val_loss: 1.174
1 - val_mae: 0.6516
Epoch 492/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 18ms/step - loss: 0.1069 - mae: 0.2429 - val_loss: 1.295
0 - val_mae: 0.6761
Epoch 493/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 17ms/step - loss: 0.1048 - mae: 0.2434 - val_loss: 1.411
8 - val_mae: 0.7119
Epoch 494/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 21ms/step - loss: 0.1214 - mae: 0.2675 - val_loss: 1.369
8 - val_mae: 0.7125
Epoch 495/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 36ms/step - loss: 0.1591 - mae: 0.3025 - val_loss: 1.121
4 - val_mae: 0.6422
Epoch 496/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 33ms/step - loss: 0.1445 - mae: 0.2826 - val_loss: 1.252
8 - val_mae: 0.6977
Epoch 497/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 22ms/step - loss: 0.1691 - mae: 0.3123 - val_loss: 1.264
3 - val_mae: 0.7038
Epoch 498/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.1528 - mae: 0.2922 - val_loss: 1.295
4 - val_mae: 0.6676
Epoch 499/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.1124 - mae: 0.2498 - val_loss: 1.192
4 - val_mae: 0.6533
Epoch 500/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.1225 - mae: 0.2607 - val_loss: 1.275
1 - val_mae: 0.6969
Epoch 501/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.1033 - mae: 0.2421 - val_loss: 1.409
3 - val_mae: 0.7208
Epoch 502/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.1210 - mae: 0.2547 - val_loss: 1.596
9 - val_mae: 0.8125
Epoch 503/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.2340 - mae: 0.3816 - val_loss: 1.361
3 - val_mae: 0.7460
Epoch 504/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.2907 - mae: 0.4200 - val_loss: 1.208
3 - val_mae: 0.6598
```

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Epoch 505/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.1779 - mae: 0.3284 - val_loss: 1.258
8 - val_mae: 0.7308
Epoch 506/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.1769 - mae: 0.3249 - val_loss: 1.280
5 - val_mae: 0.7244
Epoch 507/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.1780 - mae: 0.3187 - val_loss: 1.393
5 - val_mae: 0.6815
Epoch 508/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 18ms/step - loss: 0.1677 - mae: 0.3064 - val_loss: 1.403
5 - val_mae: 0.7253
Epoch 509/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.1328 - mae: 0.2820 - val_loss: 1.560
5 - val_mae: 0.8023
Epoch 510/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 18ms/step - loss: 0.3213 - mae: 0.4570 - val_loss: 1.279
2 - val_mae: 0.6655
Epoch 511/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 32ms/step - loss: 0.1201 - mae: 0.2606 - val_loss: 1.256
4 - val_mae: 0.7200
Epoch 512/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 17ms/step - loss: 0.1993 - mae: 0.3418 - val_loss: 1.187
6 - val_mae: 0.6448
Epoch 513/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 21ms/step - loss: 0.1586 - mae: 0.3094 - val_loss: 1.251
9 - val_mae: 0.6743
Epoch 514/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 18ms/step - loss: 0.1323 - mae: 0.2722 - val_loss: 1.338
1 - val_mae: 0.6923
Epoch 515/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.1057 - mae: 0.2571 - val_loss: 1.298
3 - val_mae: 0.6747
Epoch 516/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 18ms/step - loss: 0.1605 - mae: 0.3048 - val_loss: 1.128
5 - val_mae: 0.6378
Epoch 517/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.1293 - mae: 0.2826 - val_loss: 1.242
5 - val_mae: 0.6589
Epoch 518/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.1213 - mae: 0.2580 - val_loss: 1.193
8 - val_mae: 0.6718
Epoch 519/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 18ms/step - loss: 0.1673 - mae: 0.3149 - val_loss: 1.139
0 - val_mae: 0.6422
Epoch 520/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 18ms/step - loss: 0.1503 - mae: 0.2787 - val_loss: 1.564
9 - val_mae: 0.7926
Epoch 521/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 22ms/step - loss: 0.1495 - mae: 0.2885 - val_loss: 1.095
5 - val_mae: 0.6544
Epoch 522/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 18ms/step - loss: 0.1271 - mae: 0.2714 - val_loss: 1.294
4 - val_mae: 0.7070
Epoch 523/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 18ms/step - loss: 0.1312 - mae: 0.2742 - val_loss: 1.150
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6 - val_mae: 0.6421
Epoch 524/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 17ms/step - loss: 0.1088 - mae: 0.2483 - val_loss: 1.163
0 - val_mae: 0.6562
Epoch 525/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0959 - mae: 0.2306 - val_loss: 1.195
7 - val_mae: 0.6552
Epoch 526/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 18ms/step - loss: 0.1505 - mae: 0.2945 - val_loss: 1.345
2 - val_mae: 0.6915
Epoch 527/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 17ms/step - loss: 0.1919 - mae: 0.3372 - val_loss: 1.475
8 - val_mae: 0.7759
Epoch 528/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.1854 - mae: 0.3258 - val_loss: 1.370
2 - val_mae: 0.7162
Epoch 529/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 18ms/step - loss: 0.1482 - mae: 0.3023 - val_loss: 1.331
7 - val_mae: 0.6921
Epoch 530/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 18ms/step - loss: 0.1359 - mae: 0.2825 - val_loss: 1.282
3 - val_mae: 0.6831
Epoch 531/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 21ms/step - loss: 0.1188 - mae: 0.2550 - val_loss: 1.355
6 - val_mae: 0.7047
Epoch 532/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 18ms/step - loss: 0.1206 - mae: 0.2605 - val_loss: 1.102
8 - val_mae: 0.6336
Epoch 533/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 17ms/step - loss: 0.0982 - mae: 0.2295 - val_loss: 1.268
6 - val_mae: 0.6802
Epoch 534/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.2103 - mae: 0.3531 - val_loss: 1.202
1 - val_mae: 0.6862
Epoch 535/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.1811 - mae: 0.3391 - val_loss: 1.287
8 - val_mae: 0.6650
Epoch 536/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 18ms/step - loss: 0.1518 - mae: 0.2842 - val_loss: 1.482
4 - val_mae: 0.7479
Epoch 537/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 18ms/step - loss: 0.3490 - mae: 0.4349 - val_loss: 1.152
2 - val_mae: 0.6579
Epoch 538/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 18ms/step - loss: 0.3598 - mae: 0.4596 - val_loss: 1.359
4 - val_mae: 0.8832
Epoch 539/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 17ms/step - loss: 0.3856 - mae: 0.4943 - val_loss: 1.289
9 - val_mae: 0.7561
Epoch 540/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.4928 - mae: 0.5374 - val_loss: 1.186
4 - val_mae: 0.7039
Epoch 541/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.3610 - mae: 0.4829 - val_loss: 1.664
5 - val_mae: 0.7965
Epoch 542/1400
```

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5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.2231 - mae: 0.3610 - val_loss: 1.802
2 - val_mae: 0.7846
Epoch 543/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 18ms/step - loss: 0.3097 - mae: 0.4216 - val_loss: 1.171
9 - val_mae: 0.6343
Epoch 544/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.2544 - mae: 0.4009 - val_loss: 1.302
2 - val_mae: 0.7533
Epoch 545/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.2891 - mae: 0.4152 - val_loss: 1.453
2 - val_mae: 0.7526
Epoch 546/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.3814 - mae: 0.4725 - val_loss: 1.354
5 - val_mae: 0.8072
Epoch 547/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.3619 - mae: 0.4947 - val_loss: 1.297
3 - val_mae: 0.7278
Epoch 548/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 18ms/step - loss: 0.2806 - mae: 0.4155 - val_loss: 1.721
0 - val_mae: 0.7792
Epoch 549/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.3961 - mae: 0.4960 - val_loss: 1.970
7 - val_mae: 0.9854
Epoch 550/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 17ms/step - loss: 0.4118 - mae: 0.4868 - val_loss: 1.445
4 - val_mae: 0.7364
Epoch 551/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.2116 - mae: 0.3456 - val_loss: 1.105
5 - val_mae: 0.6336
Epoch 552/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 18ms/step - loss: 0.1432 - mae: 0.2902 - val_loss: 1.140
8 - val_mae: 0.6485
Epoch 553/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 17ms/step - loss: 0.1366 - mae: 0.2918 - val_loss: 1.289
0 - val_mae: 0.6748
Epoch 554/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.1199 - mae: 0.2679 - val_loss: 1.141
1 - val_mae: 0.6885
Epoch 555/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 17ms/step - loss: 0.1813 - mae: 0.3340 - val_loss: 1.266
1 - val_mae: 0.6747
Epoch 556/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 17ms/step - loss: 0.1342 - mae: 0.2798 - val_loss: 1.381
7 - val_mae: 0.7098
Epoch 557/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 18ms/step - loss: 0.1514 - mae: 0.2994 - val_loss: 1.242
8 - val_mae: 0.6669
Epoch 558/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 18ms/step - loss: 0.1197 - mae: 0.2692 - val_loss: 1.271
9 - val_mae: 0.6665
Epoch 559/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 18ms/step - loss: 0.1450 - mae: 0.3035 - val_loss: 1.245
7 - val_mae: 0.6828
Epoch 560/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 18ms/step - loss: 0.1631 - mae: 0.3177 - val_loss: 1.126
0 - val_mae: 0.6467
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Epoch 561/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 29ms/step - loss: 0.1233 - mae: 0.2790 - val_loss: 1.408
5 - val_mae: 0.6885
Epoch 562/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 28ms/step - loss: 0.1609 - mae: 0.3120 - val_loss: 1.194
7 - val_mae: 0.6718
Epoch 563/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 27ms/step - loss: 0.1244 - mae: 0.2602 - val_loss: 1.387
4 - val_mae: 0.7253
Epoch 564/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 28ms/step - loss: 0.1225 - mae: 0.2700 - val_loss: 1.090
5 - val_mae: 0.6388
Epoch 565/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1543 - mae: 0.2965 - val_loss: 1.188
4 - val_mae: 0.6622
Epoch 566/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 46ms/step - loss: 0.1197 - mae: 0.2583 - val_loss: 1.282
3 - val_mae: 0.6833
Epoch 567/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 33ms/step - loss: 0.1180 - mae: 0.2542 - val_loss: 1.203
1 - val_mae: 0.6676
Epoch 568/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 25ms/step - loss: 0.1848 - mae: 0.3311 - val_loss: 1.095
7 - val_mae: 0.6479
Epoch 569/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 26ms/step - loss: 0.1067 - mae: 0.2454 - val_loss: 1.218
3 - val_mae: 0.6683
Epoch 570/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 26ms/step - loss: 0.0980 - mae: 0.2226 - val_loss: 1.205
3 - val_mae: 0.6702
Epoch 571/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 37ms/step - loss: 0.0919 - mae: 0.2223 - val_loss: 1.243
6 - val_mae: 0.6898
Epoch 572/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 33ms/step - loss: 0.0913 - mae: 0.2234 - val_loss: 1.142
7 - val_mae: 0.6607
Epoch 573/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.1020 - mae: 0.2387 - val_loss: 1.302
6 - val_mae: 0.6936
Epoch 574/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 18ms/step - loss: 0.1201 - mae: 0.2611 - val_loss: 1.154
9 - val_mae: 0.6604
Epoch 575/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0852 - mae: 0.2119 - val_loss: 1.230
6 - val_mae: 0.6799
Epoch 576/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 18ms/step - loss: 0.1092 - mae: 0.2303 - val_loss: 1.415
1 - val_mae: 0.7489
Epoch 577/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.1929 - mae: 0.3339 - val_loss: 1.208
8 - val_mae: 0.6795
Epoch 578/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.1514 - mae: 0.3014 - val_loss: 1.335
8 - val_mae: 0.7041
Epoch 579/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 18ms/step - loss: 0.1297 - mae: 0.2712 - val_loss: 1.236
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3 - val_mae: 0.6798
Epoch 580/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 30ms/step - loss: 0.0926 - mae: 0.2246 - val_loss: 1.070
7 - val_mae: 0.6419
Epoch 581/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0934 - mae: 0.2298 - val_loss: 1.145
3 - val_mae: 0.6540
Epoch 582/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 17ms/step - loss: 0.0873 - mae: 0.2197 - val_loss: 1.195
0 - val_mae: 0.6721
Epoch 583/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0906 - mae: 0.2138 - val_loss: 1.206
4 - val_mae: 0.6786
Epoch 584/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0887 - mae: 0.2208 - val_loss: 1.216
7 - val_mae: 0.6688
Epoch 585/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 18ms/step - loss: 0.0801 - mae: 0.2118 - val_loss: 1.190
6 - val_mae: 0.6651
Epoch 586/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 18ms/step - loss: 0.0958 - mae: 0.2290 - val_loss: 1.150
6 - val_mae: 0.6579
Epoch 587/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0813 - mae: 0.2033 - val_loss: 1.139
6 - val_mae: 0.6617
Epoch 588/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0821 - mae: 0.2065 - val_loss: 1.150
7 - val_mae: 0.6607
Epoch 589/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 17ms/step - loss: 0.1083 - mae: 0.2345 - val_loss: 1.326
7 - val_mae: 0.7061
Epoch 590/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 23ms/step - loss: 0.1127 - mae: 0.2467 - val_loss: 1.171
9 - val_mae: 0.6605
Epoch 591/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 18ms/step - loss: 0.1233 - mae: 0.2530 - val_loss: 1.193
0 - val_mae: 0.6708
Epoch 592/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 21ms/step - loss: 0.1137 - mae: 0.2422 - val_loss: 1.387
4 - val_mae: 0.7408
Epoch 593/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.1517 - mae: 0.3096 - val_loss: 1.259
8 - val_mae: 0.6704
Epoch 594/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 18ms/step - loss: 0.1545 - mae: 0.2873 - val_loss: 1.275
9 - val_mae: 0.7597
Epoch 595/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 18ms/step - loss: 0.1888 - mae: 0.3261 - val_loss: 1.323
9 - val_mae: 0.6829
Epoch 596/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.1839 - mae: 0.3350 - val_loss: 1.474
8 - val_mae: 0.7560
Epoch 597/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 18ms/step - loss: 0.2886 - mae: 0.4132 - val_loss: 1.313
8 - val_mae: 0.7401
Epoch 598/1400
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5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 18ms/step - loss: 0.4287 - mae: 0.5350 - val_loss: 1.123
5 - val_mae: 0.6825
Epoch 599/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 18ms/step - loss: 0.3008 - mae: 0.4259 - val_loss: 1.710
6 - val_mae: 0.8997
Epoch 600/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 22ms/step - loss: 0.3259 - mae: 0.4530 - val_loss: 1.333
1 - val_mae: 0.7901
Epoch 601/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 18ms/step - loss: 0.5229 - mae: 0.5853 - val_loss: 1.377
6 - val_mae: 0.8509
Epoch 602/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.4437 - mae: 0.5223 - val_loss: 1.344
7 - val_mae: 0.7372
Epoch 603/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.2765 - mae: 0.4087 - val_loss: 1.411
7 - val_mae: 0.7331
Epoch 604/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 18ms/step - loss: 0.1799 - mae: 0.3307 - val_loss: 1.689
8 - val_mae: 0.8214
Epoch 605/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.2718 - mae: 0.4039 - val_loss: 1.162
1 - val_mae: 0.6951
Epoch 606/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.2255 - mae: 0.3652 - val_loss: 1.426
3 - val_mae: 0.7320
Epoch 607/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.2264 - mae: 0.3752 - val_loss: 1.143
6 - val_mae: 0.6818
Epoch 608/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 21ms/step - loss: 0.1764 - mae: 0.3265 - val_loss: 1.093
1 - val_mae: 0.6391
Epoch 609/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 23ms/step - loss: 0.1157 - mae: 0.2631 - val_loss: 1.442
1 - val_mae: 0.7191
Epoch 610/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.1033 - mae: 0.2511 - val_loss: 1.289
9 - val_mae: 0.7143
Epoch 611/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 18ms/step - loss: 0.1818 - mae: 0.3192 - val_loss: 1.124
4 - val_mae: 0.6470
Epoch 612/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 18ms/step - loss: 0.1080 - mae: 0.2483 - val_loss: 1.118
6 - val_mae: 0.6545
Epoch 613/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 18ms/step - loss: 0.1556 - mae: 0.3042 - val_loss: 1.500
8 - val_mae: 0.8169
Epoch 614/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.3182 - mae: 0.4274 - val_loss: 1.425
5 - val_mae: 0.7602
Epoch 615/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 18ms/step - loss: 0.3183 - mae: 0.4323 - val_loss: 1.663
3 - val_mae: 0.9944
Epoch 616/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 18ms/step - loss: 0.5549 - mae: 0.5954 - val_loss: 1.181
4 - val_mae: 0.7487
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Epoch 617/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 18ms/step - loss: 0.3403 - mae: 0.4647 - val_loss: 1.488
6 - val_mae: 0.8014
Epoch 618/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.2354 - mae: 0.3936 - val_loss: 1.638
6 - val_mae: 0.7925
Epoch 619/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 22ms/step - loss: 0.1801 - mae: 0.3370 - val_loss: 1.330
5 - val_mae: 0.7037
Epoch 620/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.1255 - mae: 0.2801 - val_loss: 1.359
6 - val_mae: 0.7081
Epoch 621/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 18ms/step - loss: 0.1288 - mae: 0.2760 - val_loss: 1.123
1 - val_mae: 0.6562
Epoch 622/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.1219 - mae: 0.2668 - val_loss: 1.384
3 - val_mae: 0.7288
Epoch 623/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.1277 - mae: 0.2811 - val_loss: 1.244
2 - val_mae: 0.6869
Epoch 624/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 18ms/step - loss: 0.1551 - mae: 0.3008 - val_loss: 1.060
9 - val_mae: 0.6791
Epoch 625/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 17ms/step - loss: 0.1312 - mae: 0.2816 - val_loss: 1.223
5 - val_mae: 0.6742
Epoch 626/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 18ms/step - loss: 0.1189 - mae: 0.2588 - val_loss: 1.197
7 - val_mae: 0.6742
Epoch 627/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 18ms/step - loss: 0.0933 - mae: 0.2302 - val_loss: 1.111
5 - val_mae: 0.6517
Epoch 628/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 18ms/step - loss: 0.0853 - mae: 0.2175 - val_loss: 1.145
6 - val_mae: 0.6585
Epoch 629/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 21ms/step - loss: 0.1464 - mae: 0.2983 - val_loss: 1.669
0 - val_mae: 0.8596
Epoch 630/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 17ms/step - loss: 0.2922 - mae: 0.3982 - val_loss: 1.091
4 - val_mae: 0.6664
Epoch 631/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.2679 - mae: 0.3986 - val_loss: 1.127
9 - val_mae: 0.7033
Epoch 632/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 18ms/step - loss: 0.2228 - mae: 0.3571 - val_loss: 1.487
8 - val_mae: 0.7262
Epoch 633/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 18ms/step - loss: 0.1721 - mae: 0.3208 - val_loss: 1.226
4 - val_mae: 0.6684
Epoch 634/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 18ms/step - loss: 0.1086 - mae: 0.2599 - val_loss: 1.209
0 - val_mae: 0.6570
Epoch 635/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 18ms/step - loss: 0.1163 - mae: 0.2699 - val_loss: 1.180
```

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3 - val_mae: 0.6705
Epoch 636/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 18ms/step - loss: 0.1159 - mae: 0.2659 - val_loss: 1.374
8 - val_mae: 0.7232
Epoch 637/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 17ms/step - loss: 0.1159 - mae: 0.2569 - val_loss: 1.209
3 - val_mae: 0.6737
Epoch 638/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 18ms/step - loss: 0.1240 - mae: 0.2734 - val_loss: 1.216
2 - val_mae: 0.6822
Epoch 639/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.1464 - mae: 0.2880 - val_loss: 1.141
6 - val_mae: 0.6440
Epoch 640/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 17ms/step - loss: 0.1883 - mae: 0.3170 - val_loss: 1.331
0 - val_mae: 0.7125
Epoch 641/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 17ms/step - loss: 0.1034 - mae: 0.2365 - val_loss: 1.093
4 - val_mae: 0.6376
Epoch 642/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 18ms/step - loss: 0.1006 - mae: 0.2322 - val_loss: 1.155
7 - val_mae: 0.6586
Epoch 643/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0763 - mae: 0.1982 - val_loss: 1.190
5 - val_mae: 0.6635
Epoch 644/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 17ms/step - loss: 0.1018 - mae: 0.2362 - val_loss: 1.217
3 - val_mae: 0.6759
Epoch 645/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 18ms/step - loss: 0.1028 - mae: 0.2349 - val_loss: 1.096
9 - val_mae: 0.6396
Epoch 646/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.1206 - mae: 0.2530 - val_loss: 1.171
3 - val_mae: 0.6572
Epoch 647/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0916 - mae: 0.2178 - val_loss: 1.215
3 - val_mae: 0.6647
Epoch 648/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.1628 - mae: 0.3040 - val_loss: 1.340
5 - val_mae: 0.7351
Epoch 649/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 18ms/step - loss: 0.1600 - mae: 0.3001 - val_loss: 1.177
8 - val_mae: 0.6562
Epoch 650/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 18ms/step - loss: 0.1402 - mae: 0.2965 - val_loss: 1.040
9 - val_mae: 0.6386
Epoch 651/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 18ms/step - loss: 0.1176 - mae: 0.2711 - val_loss: 1.303
2 - val_mae: 0.6942
Epoch 652/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.1089 - mae: 0.2522 - val_loss: 1.221
2 - val_mae: 0.6533
Epoch 653/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.1346 - mae: 0.2790 - val_loss: 1.103
4 - val_mae: 0.6400
Epoch 654/1400
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5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 18ms/step - loss: 0.1170 - mae: 0.2495 - val_loss: 1.588
2 - val_mae: 0.8192
Epoch 655/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 17ms/step - loss: 0.2016 - mae: 0.3405 - val_loss: 1.319
2 - val_mae: 0.7066
Epoch 656/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 18ms/step - loss: 0.1792 - mae: 0.3224 - val_loss: 1.151
8 - val_mae: 0.7263
Epoch 657/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.2518 - mae: 0.4113 - val_loss: 1.236
7 - val_mae: 0.7058
Epoch 658/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 21ms/step - loss: 0.1904 - mae: 0.3324 - val_loss: 1.316
9 - val_mae: 0.7115
Epoch 659/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 18ms/step - loss: 0.1666 - mae: 0.3082 - val_loss: 1.394
7 - val_mae: 0.7553
Epoch 660/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.2257 - mae: 0.3678 - val_loss: 1.398
0 - val_mae: 0.6994
Epoch 661/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.1544 - mae: 0.2991 - val_loss: 1.071
0 - val_mae: 0.6351
Epoch 662/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 24ms/step - loss: 0.1262 - mae: 0.2689 - val_loss: 1.156
0 - val_mae: 0.6806
Epoch 663/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 17ms/step - loss: 0.2087 - mae: 0.3598 - val_loss: 1.192
4 - val_mae: 0.6503
Epoch 664/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 18ms/step - loss: 0.1424 - mae: 0.2771 - val_loss: 1.224
7 - val_mae: 0.6729
Epoch 665/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 18ms/step - loss: 0.0825 - mae: 0.2091 - val_loss: 1.193
1 - val_mae: 0.6682
Epoch 666/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 30ms/step - loss: 0.1532 - mae: 0.3136 - val_loss: 1.284
1 - val_mae: 0.7733
Epoch 667/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 27ms/step - loss: 0.3113 - mae: 0.4445 - val_loss: 1.059
9 - val_mae: 0.6942
Epoch 668/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 36ms/step - loss: 0.2609 - mae: 0.3975 - val_loss: 1.393
0 - val_mae: 0.7563
Epoch 669/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 36ms/step - loss: 0.2246 - mae: 0.3644 - val_loss: 1.166
8 - val_mae: 0.7129
Epoch 670/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 38ms/step - loss: 0.1465 - mae: 0.3044 - val_loss: 1.300
7 - val_mae: 0.6723
Epoch 671/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 37ms/step - loss: 0.1221 - mae: 0.2721 - val_loss: 1.212
8 - val_mae: 0.6549
Epoch 672/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 25ms/step - loss: 0.1354 - mae: 0.2744 - val_loss: 1.258
2 - val_mae: 0.6924
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Epoch 673/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 27ms/step - loss: 0.1161 - mae: 0.2575 - val_loss: 1.181
4 - val_mae: 0.6567
Epoch 674/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 33ms/step - loss: 0.1861 - mae: 0.3251 - val_loss: 1.234
2 - val_mae: 0.7006
Epoch 675/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 30ms/step - loss: 0.2254 - mae: 0.3903 - val_loss: 1.554
6 - val_mae: 0.8352
Epoch 676/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.2062 - mae: 0.3511 - val_loss: 1.293
5 - val_mae: 0.6806
Epoch 677/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 28ms/step - loss: 0.2800 - mae: 0.4097 - val_loss: 1.030
0 - val_mae: 0.6618
Epoch 678/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 27ms/step - loss: 0.1573 - mae: 0.3089 - val_loss: 1.182
6 - val_mae: 0.6921
Epoch 679/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.1304 - mae: 0.2811 - val_loss: 1.155
0 - val_mae: 0.6738
Epoch 680/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 17ms/step - loss: 0.1055 - mae: 0.2472 - val_loss: 1.094
2 - val_mae: 0.6528
Epoch 681/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 17ms/step - loss: 0.1042 - mae: 0.2484 - val_loss: 1.185
3 - val_mae: 0.6635
Epoch 682/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 17ms/step - loss: 0.1158 - mae: 0.2529 - val_loss: 1.343
4 - val_mae: 0.7269
Epoch 683/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 17ms/step - loss: 0.1587 - mae: 0.3012 - val_loss: 1.077
2 - val_mae: 0.6398
Epoch 684/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 18ms/step - loss: 0.1083 - mae: 0.2458 - val_loss: 1.245
2 - val_mae: 0.6643
Epoch 685/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 18ms/step - loss: 0.1000 - mae: 0.2352 - val_loss: 1.267
6 - val_mae: 0.6973
Epoch 686/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 18ms/step - loss: 0.1245 - mae: 0.2686 - val_loss: 1.137
6 - val_mae: 0.6599
Epoch 687/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 17ms/step - loss: 0.1376 - mae: 0.2757 - val_loss: 1.171
6 - val_mae: 0.7005
Epoch 688/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 17ms/step - loss: 0.1159 - mae: 0.2616 - val_loss: 1.264
4 - val_mae: 0.6792
Epoch 689/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 18ms/step - loss: 0.1138 - mae: 0.2351 - val_loss: 1.059
2 - val_mae: 0.6526
Epoch 690/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.1849 - mae: 0.3199 - val_loss: 1.389
7 - val_mae: 0.7397
Epoch 691/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 18ms/step - loss: 0.1886 - mae: 0.3419 - val_loss: 1.428
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5 - val_mae: 0.7777
Epoch 692/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.2095 - mae: 0.3620 - val_loss: 1.162
0 - val_mae: 0.6713
Epoch 693/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 17ms/step - loss: 0.1536 - mae: 0.3071 - val_loss: 1.231
8 - val_mae: 0.6775
Epoch 694/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 18ms/step - loss: 0.1096 - mae: 0.2527 - val_loss: 1.406
4 - val_mae: 0.7198
Epoch 695/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.1324 - mae: 0.2942 - val_loss: 1.240
7 - val_mae: 0.7081
Epoch 696/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 18ms/step - loss: 0.1302 - mae: 0.2742 - val_loss: 1.252
6 - val_mae: 0.6683
Epoch 697/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 18ms/step - loss: 0.1234 - mae: 0.2737 - val_loss: 1.240
5 - val_mae: 0.7042
Epoch 698/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 16ms/step - loss: 0.2020 - mae: 0.3347 - val_loss: 1.139
3 - val_mae: 0.7043
Epoch 699/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 17ms/step - loss: 0.1900 - mae: 0.3571 - val_loss: 1.126
4 - val_mae: 0.6727
Epoch 700/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 18ms/step - loss: 0.1072 - mae: 0.2591 - val_loss: 1.282
8 - val_mae: 0.6910
Epoch 701/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 18ms/step - loss: 0.1030 - mae: 0.2513 - val_loss: 1.095
8 - val_mae: 0.6439
Epoch 702/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 18ms/step - loss: 0.0955 - mae: 0.2372 - val_loss: 1.122
4 - val_mae: 0.6491
Epoch 703/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 17ms/step - loss: 0.1122 - mae: 0.2576 - val_loss: 1.498
9 - val_mae: 0.8071
Epoch 704/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 18ms/step - loss: 0.1876 - mae: 0.3166 - val_loss: 1.113
8 - val_mae: 0.6748
Epoch 705/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 18ms/step - loss: 0.1260 - mae: 0.2693 - val_loss: 0.963
2 - val_mae: 0.6461
Epoch 706/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 18ms/step - loss: 0.1193 - mae: 0.2747 - val_loss: 1.279
8 - val_mae: 0.6931
Epoch 707/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.1297 - mae: 0.2899 - val_loss: 1.239
3 - val_mae: 0.6735
Epoch 708/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 18ms/step - loss: 0.1375 - mae: 0.2906 - val_loss: 1.034
2 - val_mae: 0.6218
Epoch 709/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 21ms/step - loss: 0.1487 - mae: 0.2947 - val_loss: 1.098
7 - val_mae: 0.6317
Epoch 710/1400
```

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5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 18ms/step - loss: 0.0963 - mae: 0.2334 - val_loss: 1.158
4 - val_mae: 0.6943
Epoch 711/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 17ms/step - loss: 0.1485 - mae: 0.2928 - val_loss: 1.144
2 - val_mae: 0.6428
Epoch 712/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 18ms/step - loss: 0.1127 - mae: 0.2515 - val_loss: 1.239
7 - val_mae: 0.6803
Epoch 713/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 18ms/step - loss: 0.0841 - mae: 0.2129 - val_loss: 0.974
8 - val_mae: 0.6220
Epoch 714/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 17ms/step - loss: 0.1144 - mae: 0.2543 - val_loss: 1.185
7 - val_mae: 0.6761
Epoch 715/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 18ms/step - loss: 0.1206 - mae: 0.2642 - val_loss: 1.513
5 - val_mae: 0.8055
Epoch 716/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 17ms/step - loss: 0.1590 - mae: 0.3058 - val_loss: 1.123
9 - val_mae: 0.6534
Epoch 717/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 17ms/step - loss: 0.1105 - mae: 0.2481 - val_loss: 1.060
8 - val_mae: 0.7020
Epoch 718/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 18ms/step - loss: 0.1723 - mae: 0.3261 - val_loss: 1.265
7 - val_mae: 0.8095
Epoch 719/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 21ms/step - loss: 0.2301 - mae: 0.3644 - val_loss: 1.206
8 - val_mae: 0.6577
Epoch 720/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 18ms/step - loss: 0.1770 - mae: 0.3301 - val_loss: 1.242
7 - val_mae: 0.6887
Epoch 721/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.2394 - mae: 0.3531 - val_loss: 1.178
7 - val_mae: 0.6868
Epoch 722/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 18ms/step - loss: 0.1938 - mae: 0.3521 - val_loss: 1.133
3 - val_mae: 0.7030
Epoch 723/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 17ms/step - loss: 0.1858 - mae: 0.3390 - val_loss: 1.109
5 - val_mae: 0.6330
Epoch 724/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.1280 - mae: 0.2839 - val_loss: 1.259
2 - val_mae: 0.6607
Epoch 725/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.1071 - mae: 0.2571 - val_loss: 1.075
7 - val_mae: 0.6581
Epoch 726/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 17ms/step - loss: 0.1147 - mae: 0.2613 - val_loss: 1.127
8 - val_mae: 0.6966
Epoch 727/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.1863 - mae: 0.3413 - val_loss: 1.412
2 - val_mae: 0.7574
Epoch 728/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 17ms/step - loss: 0.2011 - mae: 0.3448 - val_loss: 1.038
8 - val_mae: 0.6316
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Epoch 729/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.1192 - mae: 0.2596 - val_loss: 1.091
1 - val_mae: 0.6362
Epoch 730/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0784 - mae: 0.2163 - val_loss: 1.349
6 - val_mae: 0.6895
Epoch 731/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 18ms/step - loss: 0.0890 - mae: 0.2216 - val_loss: 1.118
7 - val_mae: 0.6430
Epoch 732/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 17ms/step - loss: 0.0856 - mae: 0.2212 - val_loss: 1.027
5 - val_mae: 0.6212
Epoch 733/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0760 - mae: 0.2036 - val_loss: 1.157
7 - val_mae: 0.6548
Epoch 734/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0793 - mae: 0.2027 - val_loss: 1.194
9 - val_mae: 0.6679
Epoch 735/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 17ms/step - loss: 0.0817 - mae: 0.2024 - val_loss: 1.118
5 - val_mae: 0.6544
Epoch 736/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.2209 - mae: 0.3417 - val_loss: 1.303
7 - val_mae: 0.7392
Epoch 737/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.1874 - mae: 0.3322 - val_loss: 1.476
7 - val_mae: 0.8022
Epoch 738/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.1752 - mae: 0.3220 - val_loss: 1.190
6 - val_mae: 0.6457
Epoch 739/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.1057 - mae: 0.2453 - val_loss: 1.045
8 - val_mae: 0.6503
Epoch 740/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 18ms/step - loss: 0.2337 - mae: 0.3711 - val_loss: 1.094
0 - val_mae: 0.7090
Epoch 741/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 18ms/step - loss: 0.2252 - mae: 0.3761 - val_loss: 1.202
5 - val_mae: 0.6841
Epoch 742/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 18ms/step - loss: 0.1642 - mae: 0.2977 - val_loss: 1.561
6 - val_mae: 0.8335
Epoch 743/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.2976 - mae: 0.4117 - val_loss: 1.093
4 - val_mae: 0.6837
Epoch 744/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 18ms/step - loss: 0.1480 - mae: 0.2876 - val_loss: 1.068
3 - val_mae: 0.6255
Epoch 745/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 17ms/step - loss: 0.1613 - mae: 0.3152 - val_loss: 1.160
2 - val_mae: 0.6330
Epoch 746/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 18ms/step - loss: 0.1286 - mae: 0.2744 - val_loss: 1.282
9 - val_mae: 0.6872
Epoch 747/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 18ms/step - loss: 0.2083 - mae: 0.3661 - val_loss: 1.344
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0 - val_mae: 0.7279
Epoch 748/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 21ms/step - loss: 0.2402 - mae: 0.3981 - val_loss: 1.102
0 - val_mae: 0.7000
Epoch 749/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 18ms/step - loss: 0.2144 - mae: 0.3436 - val_loss: 1.393
0 - val_mae: 0.8195
Epoch 750/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 18ms/step - loss: 0.3135 - mae: 0.4519 - val_loss: 1.117
7 - val_mae: 0.6864
Epoch 751/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 17ms/step - loss: 0.1774 - mae: 0.3188 - val_loss: 1.290
8 - val_mae: 0.6549
Epoch 752/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 17ms/step - loss: 0.1535 - mae: 0.2943 - val_loss: 1.110
2 - val_mae: 0.6663
Epoch 753/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.2342 - mae: 0.3746 - val_loss: 1.253
8 - val_mae: 0.6704
Epoch 754/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 18ms/step - loss: 0.3480 - mae: 0.4582 - val_loss: 1.569
4 - val_mae: 0.8041
Epoch 755/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 18ms/step - loss: 0.1870 - mae: 0.3468 - val_loss: 1.150
4 - val_mae: 0.6478
Epoch 756/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 18ms/step - loss: 0.1317 - mae: 0.2898 - val_loss: 1.161
5 - val_mae: 0.6593
Epoch 757/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 18ms/step - loss: 0.1050 - mae: 0.2522 - val_loss: 1.350
6 - val_mae: 0.7068
Epoch 758/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 22ms/step - loss: 0.1397 - mae: 0.2876 - val_loss: 1.151
9 - val_mae: 0.6466
Epoch 759/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0785 - mae: 0.2133 - val_loss: 1.168
3 - val_mae: 0.6551
Epoch 760/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 18ms/step - loss: 0.0823 - mae: 0.2183 - val_loss: 1.235
2 - val_mae: 0.6779
Epoch 761/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 17ms/step - loss: 0.0892 - mae: 0.2212 - val_loss: 1.150
6 - val_mae: 0.6463
Epoch 762/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 18ms/step - loss: 0.0968 - mae: 0.2201 - val_loss: 1.139
7 - val_mae: 0.6563
Epoch 763/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.1133 - mae: 0.2549 - val_loss: 1.071
0 - val_mae: 0.6656
Epoch 764/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 18ms/step - loss: 0.1685 - mae: 0.3121 - val_loss: 1.187
8 - val_mae: 0.6606
Epoch 765/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 18ms/step - loss: 0.1212 - mae: 0.2601 - val_loss: 1.222
6 - val_mae: 0.6805
Epoch 766/1400
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5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 17ms/step - loss: 0.1237 - mae: 0.2755 - val_loss: 1.083
3 - val_mae: 0.6436
Epoch 767/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 18ms/step - loss: 0.0888 - mae: 0.2297 - val_loss: 1.111
2 - val_mae: 0.6945
Epoch 768/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 30ms/step - loss: 0.1967 - mae: 0.3444 - val_loss: 1.055
2 - val_mae: 0.6461
Epoch 769/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.2402 - mae: 0.3731 - val_loss: 1.613
0 - val_mae: 0.8187
Epoch 770/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 17ms/step - loss: 0.2579 - mae: 0.3942 - val_loss: 1.087
6 - val_mae: 0.6410
Epoch 771/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 18ms/step - loss: 0.1193 - mae: 0.2608 - val_loss: 1.154
5 - val_mae: 0.6351
Epoch 772/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 18ms/step - loss: 0.0948 - mae: 0.2409 - val_loss: 1.158
3 - val_mae: 0.6437
Epoch 773/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 28ms/step - loss: 0.1038 - mae: 0.2466 - val_loss: 1.138
5 - val_mae: 0.6446
Epoch 774/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 26ms/step - loss: 0.1094 - mae: 0.2637 - val_loss: 1.110
9 - val_mae: 0.6466
Epoch 775/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 37ms/step - loss: 0.1082 - mae: 0.2484 - val_loss: 1.236
3 - val_mae: 0.6707
Epoch 776/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 27ms/step - loss: 0.1129 - mae: 0.2506 - val_loss: 1.117
0 - val_mae: 0.6429
Epoch 777/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 25ms/step - loss: 0.1042 - mae: 0.2540 - val_loss: 1.113
8 - val_mae: 0.6593
Epoch 778/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 32ms/step - loss: 0.1401 - mae: 0.2907 - val_loss: 1.142
2 - val_mae: 0.6321
Epoch 779/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 29ms/step - loss: 0.1189 - mae: 0.2643 - val_loss: 1.146
3 - val_mae: 0.7025
Epoch 780/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 24ms/step - loss: 0.1238 - mae: 0.2713 - val_loss: 1.334
6 - val_mae: 0.7070
Epoch 781/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 33ms/step - loss: 0.1515 - mae: 0.2982 - val_loss: 1.225
7 - val_mae: 0.6651
Epoch 782/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 18ms/step - loss: 0.0890 - mae: 0.2212 - val_loss: 1.244
3 - val_mae: 0.6640
Epoch 783/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 18ms/step - loss: 0.1267 - mae: 0.2784 - val_loss: 1.051
8 - val_mae: 0.6574
Epoch 784/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 18ms/step - loss: 0.0944 - mae: 0.2279 - val_loss: 1.168
0 - val_mae: 0.6439
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Epoch 785/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 17ms/step - loss: 0.1254 - mae: 0.2770 - val_loss: 1.143
6 - val_mae: 0.6574
Epoch 786/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.1091 - mae: 0.2488 - val_loss: 1.257
6 - val_mae: 0.6996
Epoch 787/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 18ms/step - loss: 0.1242 - mae: 0.2784 - val_loss: 1.045
7 - val_mae: 0.6462
Epoch 788/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 18ms/step - loss: 0.1026 - mae: 0.2377 - val_loss: 1.194
4 - val_mae: 0.6703
Epoch 789/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 17ms/step - loss: 0.0895 - mae: 0.2331 - val_loss: 1.005
0 - val_mae: 0.5979
Epoch 790/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 17ms/step - loss: 0.0865 - mae: 0.2198 - val_loss: 1.100
5 - val_mae: 0.6312
Epoch 791/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 18ms/step - loss: 0.0804 - mae: 0.2137 - val_loss: 1.148
8 - val_mae: 0.6449
Epoch 792/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 18ms/step - loss: 0.0780 - mae: 0.2034 - val_loss: 1.262
4 - val_mae: 0.6722
Epoch 793/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 18ms/step - loss: 0.0881 - mae: 0.2138 - val_loss: 1.079
7 - val_mae: 0.6616
Epoch 794/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 18ms/step - loss: 0.1784 - mae: 0.3263 - val_loss: 1.017
6 - val_mae: 0.6517
Epoch 795/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.1250 - mae: 0.2659 - val_loss: 1.239
6 - val_mae: 0.6890
Epoch 796/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 18ms/step - loss: 0.1582 - mae: 0.3102 - val_loss: 1.172
8 - val_mae: 0.6870
Epoch 797/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 18ms/step - loss: 0.1028 - mae: 0.2451 - val_loss: 1.095
6 - val_mae: 0.6237
Epoch 798/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.1644 - mae: 0.2977 - val_loss: 1.113
3 - val_mae: 0.7632
Epoch 799/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 18ms/step - loss: 0.3547 - mae: 0.4807 - val_loss: 1.353
6 - val_mae: 0.7489
Epoch 800/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.4216 - mae: 0.4834 - val_loss: 1.443
0 - val_mae: 0.8012
Epoch 801/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 18ms/step - loss: 0.1530 - mae: 0.3220 - val_loss: 1.133
5 - val_mae: 0.6126
Epoch 802/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 17ms/step - loss: 0.1158 - mae: 0.2616 - val_loss: 1.198
2 - val_mae: 0.6333
Epoch 803/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 17ms/step - loss: 0.1736 - mae: 0.3170 - val_loss: 1.186
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7 - val_mae: 0.6393
Epoch 804/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 18ms/step - loss: 0.1142 - mae: 0.2575 - val_loss: 1.086
5 - val_mae: 0.6620
Epoch 805/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 17ms/step - loss: 0.1264 - mae: 0.2886 - val_loss: 1.309
8 - val_mae: 0.6784
Epoch 806/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 17ms/step - loss: 0.1543 - mae: 0.3064 - val_loss: 1.153
5 - val_mae: 0.7159
Epoch 807/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 18ms/step - loss: 0.2063 - mae: 0.3750 - val_loss: 1.050
2 - val_mae: 0.6904
Epoch 808/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 18ms/step - loss: 0.3753 - mae: 0.4929 - val_loss: 1.807
4 - val_mae: 0.9887
Epoch 809/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 18ms/step - loss: 0.6542 - mae: 0.6774 - val_loss: 2.209
6 - val_mae: 1.0235
Epoch 810/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 17ms/step - loss: 0.6038 - mae: 0.6321 - val_loss: 1.221
1 - val_mae: 0.6976
Epoch 811/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 18ms/step - loss: 0.3738 - mae: 0.4731 - val_loss: 1.393
7 - val_mae: 0.8552
Epoch 812/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 18ms/step - loss: 0.2963 - mae: 0.4334 - val_loss: 1.308
8 - val_mae: 0.7281
Epoch 813/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.1976 - mae: 0.3496 - val_loss: 1.098
9 - val_mae: 0.6195
Epoch 814/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 18ms/step - loss: 0.2354 - mae: 0.3699 - val_loss: 1.051
2 - val_mae: 0.6357
Epoch 815/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.1909 - mae: 0.3472 - val_loss: 1.307
7 - val_mae: 0.6768
Epoch 816/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 18ms/step - loss: 0.1087 - mae: 0.2623 - val_loss: 1.212
9 - val_mae: 0.7064
Epoch 817/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.1432 - mae: 0.2871 - val_loss: 1.198
4 - val_mae: 0.6755
Epoch 818/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 18ms/step - loss: 0.0997 - mae: 0.2445 - val_loss: 1.149
1 - val_mae: 0.6425
Epoch 819/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.1070 - mae: 0.2573 - val_loss: 1.131
9 - val_mae: 0.6127
Epoch 820/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 18ms/step - loss: 0.0971 - mae: 0.2398 - val_loss: 1.034
5 - val_mae: 0.6119
Epoch 821/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 18ms/step - loss: 0.0823 - mae: 0.2213 - val_loss: 1.182
1 - val_mae: 0.6423
Epoch 822/1400
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5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.1158 - mae: 0.2536 - val_loss: 1.159
3 - val_mae: 0.6612
Epoch 823/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.1304 - mae: 0.2772 - val_loss: 1.077
3 - val_mae: 0.6180
Epoch 824/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 18ms/step - loss: 0.1188 - mae: 0.2552 - val_loss: 1.086
3 - val_mae: 0.6129
Epoch 825/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 21ms/step - loss: 0.0770 - mae: 0.2119 - val_loss: 1.131
2 - val_mae: 0.6108
Epoch 826/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 18ms/step - loss: 0.0731 - mae: 0.1941 - val_loss: 1.052
5 - val_mae: 0.6073
Epoch 827/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.1008 - mae: 0.2417 - val_loss: 1.113
0 - val_mae: 0.6522
Epoch 828/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.1518 - mae: 0.2966 - val_loss: 1.035
2 - val_mae: 0.6491
Epoch 829/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 17ms/step - loss: 0.1754 - mae: 0.3195 - val_loss: 1.173
8 - val_mae: 0.6516
Epoch 830/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 18ms/step - loss: 0.2606 - mae: 0.3861 - val_loss: 1.137
3 - val_mae: 0.7397
Epoch 831/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 18ms/step - loss: 0.3425 - mae: 0.4615 - val_loss: 1.144
7 - val_mae: 0.7398
Epoch 832/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.4707 - mae: 0.5465 - val_loss: 1.395
8 - val_mae: 0.8820
Epoch 833/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 18ms/step - loss: 0.3866 - mae: 0.4955 - val_loss: 1.448
4 - val_mae: 0.6891
Epoch 834/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.3166 - mae: 0.4457 - val_loss: 1.195
6 - val_mae: 0.7131
Epoch 835/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.2043 - mae: 0.3579 - val_loss: 1.147
0 - val_mae: 0.6568
Epoch 836/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 18ms/step - loss: 0.1816 - mae: 0.3340 - val_loss: 1.080
3 - val_mae: 0.6162
Epoch 837/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.1463 - mae: 0.2989 - val_loss: 1.086
3 - val_mae: 0.6197
Epoch 838/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 18ms/step - loss: 0.1560 - mae: 0.3203 - val_loss: 1.240
4 - val_mae: 0.7150
Epoch 839/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.1339 - mae: 0.2847 - val_loss: 1.113
5 - val_mae: 0.6142
Epoch 840/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.1234 - mae: 0.2851 - val_loss: 1.217
5 - val_mae: 0.7425
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Epoch 841/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 18ms/step - loss: 0.2433 - mae: 0.3853 - val_loss: 1.051
4 - val_mae: 0.6228
Epoch 842/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 17ms/step - loss: 0.1608 - mae: 0.3010 - val_loss: 1.131
2 - val_mae: 0.6197
Epoch 843/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 18ms/step - loss: 0.0977 - mae: 0.2509 - val_loss: 1.078
9 - val_mae: 0.5881
Epoch 844/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 18ms/step - loss: 0.0907 - mae: 0.2245 - val_loss: 1.130
1 - val_mae: 0.6300
Epoch 845/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0737 - mae: 0.2042 - val_loss: 1.116
7 - val_mae: 0.6179
Epoch 846/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 18ms/step - loss: 0.1155 - mae: 0.2717 - val_loss: 1.114
3 - val_mae: 0.6342
Epoch 847/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 17ms/step - loss: 0.1329 - mae: 0.2774 - val_loss: 1.088
7 - val_mae: 0.6141
Epoch 848/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 18ms/step - loss: 0.1114 - mae: 0.2432 - val_loss: 1.153
2 - val_mae: 0.6436
Epoch 849/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 18ms/step - loss: 0.1195 - mae: 0.2638 - val_loss: 1.044
7 - val_mae: 0.6345
Epoch 850/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.1193 - mae: 0.2677 - val_loss: 1.244
8 - val_mae: 0.6721
Epoch 851/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 18ms/step - loss: 0.0866 - mae: 0.2206 - val_loss: 1.100
9 - val_mae: 0.6354
Epoch 852/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0896 - mae: 0.2278 - val_loss: 1.054
4 - val_mae: 0.6231
Epoch 853/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0836 - mae: 0.2276 - val_loss: 1.034
2 - val_mae: 0.6116
Epoch 854/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 22ms/step - loss: 0.0701 - mae: 0.2020 - val_loss: 1.133
6 - val_mae: 0.6345
Epoch 855/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.1216 - mae: 0.2835 - val_loss: 1.093
5 - val_mae: 0.6923
Epoch 856/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 17ms/step - loss: 0.1897 - mae: 0.3429 - val_loss: 1.302
3 - val_mae: 0.8227
Epoch 857/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 17ms/step - loss: 0.3058 - mae: 0.4422 - val_loss: 1.172
8 - val_mae: 0.6565
Epoch 858/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 18ms/step - loss: 0.2182 - mae: 0.3730 - val_loss: 1.689
6 - val_mae: 0.8038
Epoch 859/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 18ms/step - loss: 0.2279 - mae: 0.3398 - val_loss: 0.927
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9 - val_mae: 0.6233
Epoch 860/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.1619 - mae: 0.3100 - val_loss: 1.022
9 - val_mae: 0.6606
Epoch 861/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.1617 - mae: 0.3163 - val_loss: 1.205
6 - val_mae: 0.6433
Epoch 862/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0939 - mae: 0.2322 - val_loss: 1.110
4 - val_mae: 0.6389
Epoch 863/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 22ms/step - loss: 0.0932 - mae: 0.2284 - val_loss: 0.960
3 - val_mae: 0.5836
Epoch 864/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 18ms/step - loss: 0.0745 - mae: 0.2058 - val_loss: 1.051
7 - val_mae: 0.6035
Epoch 865/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 18ms/step - loss: 0.0822 - mae: 0.2178 - val_loss: 1.177
1 - val_mae: 0.6444
Epoch 866/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0737 - mae: 0.1989 - val_loss: 1.084
9 - val_mae: 0.5987
Epoch 867/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0696 - mae: 0.1921 - val_loss: 1.075
7 - val_mae: 0.6056
Epoch 868/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0777 - mae: 0.2048 - val_loss: 1.099
9 - val_mae: 0.6070
Epoch 869/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.1030 - mae: 0.2321 - val_loss: 1.348
4 - val_mae: 0.7319
Epoch 870/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.1385 - mae: 0.2753 - val_loss: 1.049
7 - val_mae: 0.6375
Epoch 871/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 21ms/step - loss: 0.1153 - mae: 0.2554 - val_loss: 1.047
5 - val_mae: 0.5957
Epoch 872/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 25ms/step - loss: 0.1246 - mae: 0.2622 - val_loss: 1.222
5 - val_mae: 0.6544
Epoch 873/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 18ms/step - loss: 0.0950 - mae: 0.2287 - val_loss: 1.152
8 - val_mae: 0.6457
Epoch 874/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 27ms/step - loss: 0.0970 - mae: 0.2296 - val_loss: 1.065
6 - val_mae: 0.6194
Epoch 875/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 34ms/step - loss: 0.0839 - mae: 0.2147 - val_loss: 1.040
0 - val_mae: 0.6183
Epoch 876/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 34ms/step - loss: 0.0726 - mae: 0.1899 - val_loss: 1.024
4 - val_mae: 0.5988
Epoch 877/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 32ms/step - loss: 0.0579 - mae: 0.1727 - val_loss: 1.096
1 - val_mae: 0.6065
Epoch 878/1400
```

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5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 33ms/step - loss: 0.0663 - mae: 0.1919 - val_loss: 1.122
9 - val_mae: 0.6351
Epoch 879/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 34ms/step - loss: 0.0763 - mae: 0.2113 - val_loss: 1.095
2 - val_mae: 0.6082
Epoch 880/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 25ms/step - loss: 0.0756 - mae: 0.2020 - val_loss: 0.995
7 - val_mae: 0.5942
Epoch 881/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 34ms/step - loss: 0.0905 - mae: 0.2335 - val_loss: 1.077
4 - val_mae: 0.6010
Epoch 882/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 34ms/step - loss: 0.0771 - mae: 0.2048 - val_loss: 1.115
7 - val_mae: 0.6418
Epoch 883/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 29ms/step - loss: 0.0816 - mae: 0.2148 - val_loss: 1.054
3 - val_mae: 0.6069
Epoch 884/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 40ms/step - loss: 0.0971 - mae: 0.2308 - val_loss: 1.282
7 - val_mae: 0.7168
Epoch 885/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 29ms/step - loss: 0.1128 - mae: 0.2612 - val_loss: 1.044
7 - val_mae: 0.6125
Epoch 886/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.1200 - mae: 0.2675 - val_loss: 1.054
7 - val_mae: 0.6221
Epoch 887/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.1143 - mae: 0.2577 - val_loss: 1.229
7 - val_mae: 0.6629
Epoch 888/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.1032 - mae: 0.2599 - val_loss: 1.061
1 - val_mae: 0.5906
Epoch 889/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 18ms/step - loss: 0.0747 - mae: 0.2052 - val_loss: 1.007
0 - val_mae: 0.6207
Epoch 890/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 21ms/step - loss: 0.1046 - mae: 0.2546 - val_loss: 1.034
7 - val_mae: 0.6022
Epoch 891/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 23ms/step - loss: 0.1136 - mae: 0.2533 - val_loss: 1.273
4 - val_mae: 0.6940
Epoch 892/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.1081 - mae: 0.2440 - val_loss: 0.876
9 - val_mae: 0.5914
Epoch 893/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.1016 - mae: 0.2379 - val_loss: 1.126
6 - val_mae: 0.6144
Epoch 894/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.1355 - mae: 0.2721 - val_loss: 1.408
1 - val_mae: 0.7786
Epoch 895/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.2649 - mae: 0.4027 - val_loss: 0.933
0 - val_mae: 0.6329
Epoch 896/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 18ms/step - loss: 0.2413 - mae: 0.3959 - val_loss: 1.102
8 - val_mae: 0.7614
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Epoch 897/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.4131 - mae: 0.5396 - val_loss: 1.383
4 - val_mae: 0.7058
Epoch 898/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.2498 - mae: 0.3909 - val_loss: 1.704
5 - val_mae: 0.9556
Epoch 899/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 18ms/step - loss: 0.4112 - mae: 0.4905 - val_loss: 1.258
7 - val_mae: 0.6497
Epoch 900/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 25ms/step - loss: 0.2373 - mae: 0.3791 - val_loss: 1.090
6 - val_mae: 0.6034
Epoch 901/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 18ms/step - loss: 0.1204 - mae: 0.2715 - val_loss: 1.086
0 - val_mae: 0.6128
Epoch 902/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0924 - mae: 0.2402 - val_loss: 1.135
0 - val_mae: 0.6168
Epoch 903/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0755 - mae: 0.2055 - val_loss: 0.981
4 - val_mae: 0.5863
Epoch 904/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 18ms/step - loss: 0.0769 - mae: 0.2032 - val_loss: 0.970
6 - val_mae: 0.6279
Epoch 905/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 18ms/step - loss: 0.1174 - mae: 0.2573 - val_loss: 0.973
0 - val_mae: 0.5709
Epoch 906/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 18ms/step - loss: 0.0914 - mae: 0.2284 - val_loss: 1.238
9 - val_mae: 0.6599
Epoch 907/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0948 - mae: 0.2330 - val_loss: 1.052
2 - val_mae: 0.5930
Epoch 908/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 22ms/step - loss: 0.0845 - mae: 0.2191 - val_loss: 1.110
0 - val_mae: 0.6216
Epoch 909/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0749 - mae: 0.2002 - val_loss: 1.065
5 - val_mae: 0.5990
Epoch 910/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0746 - mae: 0.2031 - val_loss: 0.917
0 - val_mae: 0.5655
Epoch 911/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0743 - mae: 0.2042 - val_loss: 1.042
6 - val_mae: 0.5932
Epoch 912/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0647 - mae: 0.1784 - val_loss: 0.973
0 - val_mae: 0.5855
Epoch 913/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 18ms/step - loss: 0.0783 - mae: 0.2101 - val_loss: 0.922
0 - val_mae: 0.5820
Epoch 914/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 18ms/step - loss: 0.0740 - mae: 0.2031 - val_loss: 1.134
5 - val_mae: 0.6367
Epoch 915/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 32ms/step - loss: 0.0878 - mae: 0.2305 - val_loss: 1.080
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9 - val_mae: 0.6644
Epoch 916/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0867 - mae: 0.2232 - val_loss: 1.077
5 - val_mae: 0.6091
Epoch 917/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 22ms/step - loss: 0.0891 - mae: 0.2178 - val_loss: 1.046
3 - val_mae: 0.5965
Epoch 918/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 18ms/step - loss: 0.0940 - mae: 0.2416 - val_loss: 1.012
0 - val_mae: 0.5910
Epoch 919/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0647 - mae: 0.1944 - val_loss: 0.997
1 - val_mae: 0.5942
Epoch 920/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0978 - mae: 0.2319 - val_loss: 1.056
9 - val_mae: 0.5941
Epoch 921/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 18ms/step - loss: 0.0702 - mae: 0.1921 - val_loss: 1.099
5 - val_mae: 0.6189
Epoch 922/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.1029 - mae: 0.2447 - val_loss: 0.900
6 - val_mae: 0.5913
Epoch 923/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0877 - mae: 0.2206 - val_loss: 1.123
6 - val_mae: 0.6207
Epoch 924/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 18ms/step - loss: 0.0753 - mae: 0.2050 - val_loss: 0.955
7 - val_mae: 0.6243
Epoch 925/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 18ms/step - loss: 0.0909 - mae: 0.2280 - val_loss: 0.948
1 - val_mae: 0.5919
Epoch 926/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 17ms/step - loss: 0.0690 - mae: 0.1985 - val_loss: 1.146
8 - val_mae: 0.6479
Epoch 927/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 22ms/step - loss: 0.1278 - mae: 0.2758 - val_loss: 0.975
9 - val_mae: 0.5935
Epoch 928/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 18ms/step - loss: 0.1201 - mae: 0.2674 - val_loss: 0.989
4 - val_mae: 0.6535
Epoch 929/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.2458 - mae: 0.3955 - val_loss: 1.462
1 - val_mae: 0.8279
Epoch 930/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.3046 - mae: 0.4459 - val_loss: 1.019
8 - val_mae: 0.6623
Epoch 931/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.2063 - mae: 0.3658 - val_loss: 1.081
5 - val_mae: 0.7013
Epoch 932/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.2182 - mae: 0.3677 - val_loss: 1.844
5 - val_mae: 0.9521
Epoch 933/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 18ms/step - loss: 0.2330 - mae: 0.3761 - val_loss: 1.343
9 - val_mae: 0.7276
Epoch 934/1400
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5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.1591 - mae: 0.3141 - val_loss: 0.881
3 - val_mae: 0.6504
Epoch 935/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 18ms/step - loss: 0.1962 - mae: 0.3356 - val_loss: 1.214
3 - val_mae: 0.6612
Epoch 936/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 18ms/step - loss: 0.1727 - mae: 0.3292 - val_loss: 1.277
8 - val_mae: 0.7576
Epoch 937/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 21ms/step - loss: 0.5570 - mae: 0.5871 - val_loss: 1.636
4 - val_mae: 1.0814
Epoch 938/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 21ms/step - loss: 0.7911 - mae: 0.7292 - val_loss: 1.374
7 - val_mae: 0.9975
Epoch 939/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.6065 - mae: 0.6125 - val_loss: 1.306
9 - val_mae: 0.7169
Epoch 940/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 18ms/step - loss: 0.4196 - mae: 0.5168 - val_loss: 1.750
8 - val_mae: 0.8690
Epoch 941/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.3552 - mae: 0.4697 - val_loss: 1.287
9 - val_mae: 0.6286
Epoch 942/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 18ms/step - loss: 0.2460 - mae: 0.3715 - val_loss: 1.115
6 - val_mae: 0.6850
Epoch 943/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.1492 - mae: 0.3007 - val_loss: 1.007
4 - val_mae: 0.6260
Epoch 944/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.1289 - mae: 0.2889 - val_loss: 1.002
4 - val_mae: 0.6166
Epoch 945/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.1099 - mae: 0.2646 - val_loss: 0.911
1 - val_mae: 0.5687
Epoch 946/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 24ms/step - loss: 0.0882 - mae: 0.2166 - val_loss: 0.995
5 - val_mae: 0.6064
Epoch 947/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0827 - mae: 0.2140 - val_loss: 0.963
4 - val_mae: 0.5803
Epoch 948/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 21ms/step - loss: 0.0823 - mae: 0.2165 - val_loss: 1.076
3 - val_mae: 0.6209
Epoch 949/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0928 - mae: 0.2305 - val_loss: 0.969
7 - val_mae: 0.5753
Epoch 950/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0635 - mae: 0.1828 - val_loss: 0.912
7 - val_mae: 0.5659
Epoch 951/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0771 - mae: 0.2079 - val_loss: 0.965
3 - val_mae: 0.5743
Epoch 952/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0731 - mae: 0.1970 - val_loss: 1.160
4 - val_mae: 0.6727
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Epoch 953/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 18ms/step - loss: 0.1176 - mae: 0.2739 - val_loss: 1.037
5 - val_mae: 0.6072
Epoch 954/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 21ms/step - loss: 0.1206 - mae: 0.2643 - val_loss: 1.038
6 - val_mae: 0.6025
Epoch 955/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 23ms/step - loss: 0.0832 - mae: 0.2047 - val_loss: 1.100
0 - val_mae: 0.6568
Epoch 956/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.1119 - mae: 0.2513 - val_loss: 0.975
0 - val_mae: 0.5857
Epoch 957/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0880 - mae: 0.2257 - val_loss: 0.884
6 - val_mae: 0.5681
Epoch 958/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 18ms/step - loss: 0.1039 - mae: 0.2368 - val_loss: 1.094
9 - val_mae: 0.6238
Epoch 959/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0736 - mae: 0.1989 - val_loss: 1.019
0 - val_mae: 0.6048
Epoch 960/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 18ms/step - loss: 0.0703 - mae: 0.1993 - val_loss: 1.023
3 - val_mae: 0.5859
Epoch 961/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0725 - mae: 0.2001 - val_loss: 1.025
2 - val_mae: 0.5873
Epoch 962/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0679 - mae: 0.1937 - val_loss: 0.945
3 - val_mae: 0.6068
Epoch 963/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.1106 - mae: 0.2603 - val_loss: 0.900
7 - val_mae: 0.5757
Epoch 964/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.1188 - mae: 0.2633 - val_loss: 1.280
0 - val_mae: 0.7018
Epoch 965/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 22ms/step - loss: 0.1795 - mae: 0.3318 - val_loss: 1.087
4 - val_mae: 0.6662
Epoch 966/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.2847 - mae: 0.4319 - val_loss: 0.911
5 - val_mae: 0.7230
Epoch 967/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 18ms/step - loss: 0.2892 - mae: 0.4213 - val_loss: 1.061
2 - val_mae: 0.7259
Epoch 968/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 18ms/step - loss: 0.2924 - mae: 0.4445 - val_loss: 1.343
2 - val_mae: 0.8329
Epoch 969/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.4751 - mae: 0.5529 - val_loss: 1.359
9 - val_mae: 0.8056
Epoch 970/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.5136 - mae: 0.5492 - val_loss: 2.323
1 - val_mae: 1.0839
Epoch 971/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.4444 - mae: 0.5080 - val_loss: 1.194
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2 - val_mae: 0.7093
Epoch 972/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.2529 - mae: 0.4120 - val_loss: 1.010
1 - val_mae: 0.6243
Epoch 973/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.2192 - mae: 0.3868 - val_loss: 1.202
3 - val_mae: 0.7095
Epoch 974/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.2207 - mae: 0.3735 - val_loss: 1.059
5 - val_mae: 0.6521
Epoch 975/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 30ms/step - loss: 0.1667 - mae: 0.3163 - val_loss: 1.046
2 - val_mae: 0.6038
Epoch 976/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 26ms/step - loss: 0.1244 - mae: 0.2707 - val_loss: 1.073
4 - val_mae: 0.5934
Epoch 977/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 36ms/step - loss: 0.0865 - mae: 0.2252 - val_loss: 1.015
9 - val_mae: 0.5823
Epoch 978/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 25ms/step - loss: 0.0953 - mae: 0.2315 - val_loss: 1.002
6 - val_mae: 0.5876
Epoch 979/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 37ms/step - loss: 0.0800 - mae: 0.2210 - val_loss: 0.967
9 - val_mae: 0.5914
Epoch 980/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 42ms/step - loss: 0.0824 - mae: 0.2115 - val_loss: 0.902
5 - val_mae: 0.5580
Epoch 981/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 26ms/step - loss: 0.0561 - mae: 0.1754 - val_loss: 0.932
1 - val_mae: 0.5619
Epoch 982/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 27ms/step - loss: 0.0619 - mae: 0.1838 - val_loss: 0.888
7 - val_mae: 0.5630
Epoch 983/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 45ms/step - loss: 0.0751 - mae: 0.2048 - val_loss: 1.063
3 - val_mae: 0.6051
Epoch 984/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 41ms/step - loss: 0.0729 - mae: 0.2079 - val_loss: 0.990
7 - val_mae: 0.6151
Epoch 985/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 42ms/step - loss: 0.1108 - mae: 0.2546 - val_loss: 0.979
6 - val_mae: 0.5886
Epoch 986/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 30ms/step - loss: 0.0995 - mae: 0.2323 - val_loss: 0.936
8 - val_mae: 0.6540
Epoch 987/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.1301 - mae: 0.2696 - val_loss: 1.037
3 - val_mae: 0.5782
Epoch 988/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 18ms/step - loss: 0.1091 - mae: 0.2450 - val_loss: 1.082
2 - val_mae: 0.6228
Epoch 989/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.1144 - mae: 0.2572 - val_loss: 1.112
4 - val_mae: 0.6833
Epoch 990/1400
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5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.1221 - mae: 0.2753 - val_loss: 1.061
9 - val_mae: 0.5915
Epoch 991/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0681 - mae: 0.1921 - val_loss: 0.858
2 - val_mae: 0.5555
Epoch 992/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0798 - mae: 0.2172 - val_loss: 0.880
2 - val_mae: 0.5427
Epoch 993/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0568 - mae: 0.1783 - val_loss: 0.991
6 - val_mae: 0.5671
Epoch 994/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 18ms/step - loss: 0.0607 - mae: 0.1821 - val_loss: 0.911
6 - val_mae: 0.5753
Epoch 995/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 21ms/step - loss: 0.0672 - mae: 0.1959 - val_loss: 0.915
9 - val_mae: 0.5497
Epoch 996/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0641 - mae: 0.1848 - val_loss: 0.992
0 - val_mae: 0.5835
Epoch 997/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0669 - mae: 0.1856 - val_loss: 0.984
7 - val_mae: 0.5772
Epoch 998/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 18ms/step - loss: 0.0838 - mae: 0.2090 - val_loss: 1.024
6 - val_mae: 0.5841
Epoch 999/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.1220 - mae: 0.2618 - val_loss: 0.997
5 - val_mae: 0.5985
Epoch 1000/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 18ms/step - loss: 0.1159 - mae: 0.2537 - val_loss: 0.902
0 - val_mae: 0.5748
Epoch 1001/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 17ms/step - loss: 0.1259 - mae: 0.2629 - val_loss: 0.962
8 - val_mae: 0.5754
Epoch 1002/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0829 - mae: 0.2161 - val_loss: 0.888
6 - val_mae: 0.5845
Epoch 1003/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.1393 - mae: 0.2858 - val_loss: 1.502
2 - val_mae: 0.7732
Epoch 1004/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.2855 - mae: 0.4155 - val_loss: 0.858
0 - val_mae: 0.6396
Epoch 1005/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.2262 - mae: 0.3561 - val_loss: 1.135
5 - val_mae: 0.5990
Epoch 1006/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 18ms/step - loss: 0.1198 - mae: 0.2632 - val_loss: 1.032
5 - val_mae: 0.5929
Epoch 1007/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0886 - mae: 0.2220 - val_loss: 1.043
2 - val_mae: 0.6035
Epoch 1008/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.1172 - mae: 0.2573 - val_loss: 0.966
5 - val_mae: 0.5757
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Epoch 1009/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 18ms/step - loss: 0.1560 - mae: 0.3049 - val_loss: 1.041
7 - val_mae: 0.6363
Epoch 1010/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.1505 - mae: 0.3006 - val_loss: 1.160
3 - val_mae: 0.6505
Epoch 1011/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.1173 - mae: 0.2651 - val_loss: 1.116
2 - val_mae: 0.6387
Epoch 1012/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 21ms/step - loss: 0.1098 - mae: 0.2524 - val_loss: 0.828
6 - val_mae: 0.5849
Epoch 1013/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0896 - mae: 0.2206 - val_loss: 1.091
5 - val_mae: 0.6119
Epoch 1014/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 18ms/step - loss: 0.0701 - mae: 0.1987 - val_loss: 1.058
8 - val_mae: 0.6016
Epoch 1015/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 17ms/step - loss: 0.0893 - mae: 0.2256 - val_loss: 1.267
1 - val_mae: 0.7256
Epoch 1016/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 18ms/step - loss: 0.2352 - mae: 0.3971 - val_loss: 0.852
2 - val_mae: 0.5481
Epoch 1017/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.1257 - mae: 0.2694 - val_loss: 0.949
2 - val_mae: 0.5951
Epoch 1018/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.1027 - mae: 0.2513 - val_loss: 1.124
6 - val_mae: 0.5979
Epoch 1019/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0869 - mae: 0.2286 - val_loss: 1.052
8 - val_mae: 0.6117
Epoch 1020/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0897 - mae: 0.2226 - val_loss: 1.225
4 - val_mae: 0.7095
Epoch 1021/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.1888 - mae: 0.3419 - val_loss: 0.999
0 - val_mae: 0.5684
Epoch 1022/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 22ms/step - loss: 0.1442 - mae: 0.3024 - val_loss: 1.045
4 - val_mae: 0.6183
Epoch 1023/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.1250 - mae: 0.2662 - val_loss: 0.963
7 - val_mae: 0.5789
Epoch 1024/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.1132 - mae: 0.2615 - val_loss: 1.211
6 - val_mae: 0.6607
Epoch 1025/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 18ms/step - loss: 0.0928 - mae: 0.2365 - val_loss: 1.037
7 - val_mae: 0.6153
Epoch 1026/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 18ms/step - loss: 0.1462 - mae: 0.2996 - val_loss: 1.265
7 - val_mae: 0.6623
Epoch 1027/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 18ms/step - loss: 0.1437 - mae: 0.2726 - val_loss: 0.882
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8 - val_mae: 0.6317
Epoch 1028/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.1921 - mae: 0.3443 - val_loss: 1.141
5 - val_mae: 0.7107
Epoch 1029/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 17ms/step - loss: 0.2374 - mae: 0.3919 - val_loss: 1.016
4 - val_mae: 0.6653
Epoch 1030/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 18ms/step - loss: 0.1669 - mae: 0.3197 - val_loss: 1.525
8 - val_mae: 0.7879
Epoch 1031/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.1362 - mae: 0.2853 - val_loss: 1.204
5 - val_mae: 0.6724
Epoch 1032/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 21ms/step - loss: 0.1192 - mae: 0.2638 - val_loss: 1.072
4 - val_mae: 0.6262
Epoch 1033/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.2237 - mae: 0.3681 - val_loss: 1.004
8 - val_mae: 0.6726
Epoch 1034/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 18ms/step - loss: 0.2416 - mae: 0.3998 - val_loss: 1.198
8 - val_mae: 0.6988
Epoch 1035/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.1686 - mae: 0.3282 - val_loss: 0.996
4 - val_mae: 0.6419
Epoch 1036/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 18ms/step - loss: 0.1253 - mae: 0.2629 - val_loss: 1.190
0 - val_mae: 0.6146
Epoch 1037/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 18ms/step - loss: 0.0886 - mae: 0.2318 - val_loss: 1.246
7 - val_mae: 0.6877
Epoch 1038/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 18ms/step - loss: 0.1452 - mae: 0.3012 - val_loss: 1.403
4 - val_mae: 0.7758
Epoch 1039/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 18ms/step - loss: 0.1969 - mae: 0.3438 - val_loss: 0.983
2 - val_mae: 0.6261
Epoch 1040/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.1836 - mae: 0.3063 - val_loss: 1.146
5 - val_mae: 0.8124
Epoch 1041/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.3079 - mae: 0.4510 - val_loss: 1.096
7 - val_mae: 0.7118
Epoch 1042/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.2155 - mae: 0.3597 - val_loss: 1.149
0 - val_mae: 0.6397
Epoch 1043/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.1111 - mae: 0.2495 - val_loss: 1.054
9 - val_mae: 0.6340
Epoch 1044/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0992 - mae: 0.2393 - val_loss: 1.029
6 - val_mae: 0.6191
Epoch 1045/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.1194 - mae: 0.2626 - val_loss: 1.043
5 - val_mae: 0.5868
Epoch 1046/1400
```

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5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 17ms/step - loss: 0.0969 - mae: 0.2301 - val_loss: 0.983
8 - val_mae: 0.5762
Epoch 1047/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 21ms/step - loss: 0.0693 - mae: 0.1951 - val_loss: 1.039
8 - val_mae: 0.5923
Epoch 1048/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0671 - mae: 0.1938 - val_loss: 1.043
6 - val_mae: 0.5874
Epoch 1049/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 18ms/step - loss: 0.0679 - mae: 0.1867 - val_loss: 1.051
1 - val_mae: 0.5876
Epoch 1050/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0870 - mae: 0.2199 - val_loss: 0.936
8 - val_mae: 0.5878
Epoch 1051/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 23ms/step - loss: 0.1057 - mae: 0.2492 - val_loss: 1.135
4 - val_mae: 0.7439
Epoch 1052/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.2001 - mae: 0.3566 - val_loss: 0.908
3 - val_mae: 0.6504
Epoch 1053/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.1895 - mae: 0.3320 - val_loss: 0.948
1 - val_mae: 0.6062
Epoch 1054/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.1594 - mae: 0.3253 - val_loss: 1.340
3 - val_mae: 0.7731
Epoch 1055/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.2221 - mae: 0.3816 - val_loss: 1.435
5 - val_mae: 0.7935
Epoch 1056/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.1413 - mae: 0.2916 - val_loss: 0.998
7 - val_mae: 0.5825
Epoch 1057/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 18ms/step - loss: 0.0739 - mae: 0.2057 - val_loss: 0.878
9 - val_mae: 0.5805
Epoch 1058/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0573 - mae: 0.1778 - val_loss: 1.107
8 - val_mae: 0.6064
Epoch 1059/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 18ms/step - loss: 0.0887 - mae: 0.2244 - val_loss: 0.955
3 - val_mae: 0.5841
Epoch 1060/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0991 - mae: 0.2321 - val_loss: 0.914
9 - val_mae: 0.5769
Epoch 1061/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 21ms/step - loss: 0.1304 - mae: 0.2705 - val_loss: 0.857
7 - val_mae: 0.6203
Epoch 1062/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 18ms/step - loss: 0.2073 - mae: 0.3453 - val_loss: 1.094
1 - val_mae: 0.6439
Epoch 1063/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.1339 - mae: 0.2824 - val_loss: 1.096
9 - val_mae: 0.6304
Epoch 1064/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 18ms/step - loss: 0.2475 - mae: 0.4027 - val_loss: 0.938
7 - val_mae: 0.5821
```

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Epoch 1065/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.2414 - mae: 0.3617 - val_loss: 1.625
4 - val_mae: 0.8511
Epoch 1066/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 18ms/step - loss: 0.2232 - mae: 0.3623 - val_loss: 1.083
6 - val_mae: 0.6456
Epoch 1067/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.1411 - mae: 0.3030 - val_loss: 1.099
6 - val_mae: 0.6229
Epoch 1068/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 18ms/step - loss: 0.0867 - mae: 0.2235 - val_loss: 1.102
1 - val_mae: 0.5953
Epoch 1069/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0933 - mae: 0.2278 - val_loss: 1.051
4 - val_mae: 0.6991
Epoch 1070/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 21ms/step - loss: 0.1791 - mae: 0.3192 - val_loss: 1.121
4 - val_mae: 0.6267
Epoch 1071/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.1037 - mae: 0.2545 - val_loss: 1.292
0 - val_mae: 0.7330
Epoch 1072/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 18ms/step - loss: 0.2328 - mae: 0.3604 - val_loss: 0.932
5 - val_mae: 0.5839
Epoch 1073/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 18ms/step - loss: 0.1859 - mae: 0.3421 - val_loss: 0.984
7 - val_mae: 0.6483
Epoch 1074/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.2383 - mae: 0.3823 - val_loss: 1.041
6 - val_mae: 0.6118
Epoch 1075/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 18ms/step - loss: 0.1891 - mae: 0.3150 - val_loss: 1.112
1 - val_mae: 0.6385
Epoch 1076/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 18ms/step - loss: 0.1189 - mae: 0.2629 - val_loss: 1.040
6 - val_mae: 0.6124
Epoch 1077/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.1374 - mae: 0.2973 - val_loss: 0.970
0 - val_mae: 0.6333
Epoch 1078/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 22ms/step - loss: 0.2332 - mae: 0.3996 - val_loss: 0.949
1 - val_mae: 0.5972
Epoch 1079/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.2180 - mae: 0.3358 - val_loss: 1.219
6 - val_mae: 0.7084
Epoch 1080/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 27ms/step - loss: 0.1426 - mae: 0.2858 - val_loss: 1.306
9 - val_mae: 0.6721
Epoch 1081/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 28ms/step - loss: 0.1069 - mae: 0.2567 - val_loss: 1.030
2 - val_mae: 0.6302
Epoch 1082/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 23ms/step - loss: 0.0678 - mae: 0.1890 - val_loss: 0.830
8 - val_mae: 0.5890
Epoch 1083/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 27ms/step - loss: 0.0850 - mae: 0.2192 - val_loss: 1.044
```

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5 - val_mae: 0.6386
Epoch 1084/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 42ms/step - loss: 0.0864 - mae: 0.2350 - val_loss: 0.968
6 - val_mae: 0.5974
Epoch 1085/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 37ms/step - loss: 0.0613 - mae: 0.1851 - val_loss: 0.961
7 - val_mae: 0.5979
Epoch 1086/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 33ms/step - loss: 0.0506 - mae: 0.1673 - val_loss: 0.937
3 - val_mae: 0.5838
Epoch 1087/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 34ms/step - loss: 0.0569 - mae: 0.1763 - val_loss: 0.956
7 - val_mae: 0.5899
Epoch 1088/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 33ms/step - loss: 0.0585 - mae: 0.1853 - val_loss: 0.933
4 - val_mae: 0.5834
Epoch 1089/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 32ms/step - loss: 0.0636 - mae: 0.1853 - val_loss: 0.984
1 - val_mae: 0.5918
Epoch 1090/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 33ms/step - loss: 0.0820 - mae: 0.2225 - val_loss: 0.950
0 - val_mae: 0.5859
Epoch 1091/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 22ms/step - loss: 0.0546 - mae: 0.1689 - val_loss: 1.012
7 - val_mae: 0.5905
Epoch 1092/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0472 - mae: 0.1552 - val_loss: 1.120
7 - val_mae: 0.6603
Epoch 1093/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 21ms/step - loss: 0.0687 - mae: 0.1857 - val_loss: 0.963
5 - val_mae: 0.5827
Epoch 1094/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 21ms/step - loss: 0.0731 - mae: 0.2015 - val_loss: 1.016
1 - val_mae: 0.5898
Epoch 1095/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0637 - mae: 0.1832 - val_loss: 1.079
0 - val_mae: 0.6071
Epoch 1096/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0839 - mae: 0.2189 - val_loss: 0.953
1 - val_mae: 0.6014
Epoch 1097/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.1121 - mae: 0.2536 - val_loss: 0.900
3 - val_mae: 0.6145
Epoch 1098/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.1103 - mae: 0.2542 - val_loss: 0.938
9 - val_mae: 0.5832
Epoch 1099/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.1442 - mae: 0.2900 - val_loss: 1.305
3 - val_mae: 0.7779
Epoch 1100/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.1680 - mae: 0.3212 - val_loss: 1.180
2 - val_mae: 0.6518
Epoch 1101/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 18ms/step - loss: 0.0970 - mae: 0.2434 - val_loss: 0.919
3 - val_mae: 0.5802
Epoch 1102/1400
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5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0894 - mae: 0.2339 - val_loss: 1.188
1 - val_mae: 0.6636
Epoch 1103/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.1157 - mae: 0.2493 - val_loss: 1.052
1 - val_mae: 0.5884
Epoch 1104/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0987 - mae: 0.2385 - val_loss: 1.002
9 - val_mae: 0.6455
Epoch 1105/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.1332 - mae: 0.2746 - val_loss: 1.018
0 - val_mae: 0.6046
Epoch 1106/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.1191 - mae: 0.2514 - val_loss: 1.231
2 - val_mae: 0.6684
Epoch 1107/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.1288 - mae: 0.2895 - val_loss: 0.900
4 - val_mae: 0.6079
Epoch 1108/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 22ms/step - loss: 0.1062 - mae: 0.2590 - val_loss: 1.171
5 - val_mae: 0.6183
Epoch 1109/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0977 - mae: 0.2423 - val_loss: 1.362
9 - val_mae: 0.7429
Epoch 1110/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 18ms/step - loss: 0.1410 - mae: 0.2990 - val_loss: 1.167
6 - val_mae: 0.6750
Epoch 1111/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 18ms/step - loss: 0.1203 - mae: 0.2764 - val_loss: 1.036
5 - val_mae: 0.5876
Epoch 1112/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 18ms/step - loss: 0.1049 - mae: 0.2489 - val_loss: 0.966
1 - val_mae: 0.5788
Epoch 1113/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0850 - mae: 0.2131 - val_loss: 1.184
7 - val_mae: 0.6503
Epoch 1114/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0885 - mae: 0.2246 - val_loss: 0.978
4 - val_mae: 0.5966
Epoch 1115/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.1312 - mae: 0.2975 - val_loss: 0.932
5 - val_mae: 0.5918
Epoch 1116/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0817 - mae: 0.2206 - val_loss: 1.051
2 - val_mae: 0.5799
Epoch 1117/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 30ms/step - loss: 0.0572 - mae: 0.1789 - val_loss: 1.000
6 - val_mae: 0.5780
Epoch 1118/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0521 - mae: 0.1692 - val_loss: 1.025
0 - val_mae: 0.6021
Epoch 1119/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0551 - mae: 0.1793 - val_loss: 0.988
2 - val_mae: 0.5830
Epoch 1120/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 21ms/step - loss: 0.0596 - mae: 0.1875 - val_loss: 1.069
5 - val_mae: 0.6149
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Epoch 1121/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0704 - mae: 0.1934 - val_loss: 1.006
5 - val_mae: 0.5799
Epoch 1122/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0444 - mae: 0.1533 - val_loss: 0.956
6 - val_mae: 0.5855
Epoch 1123/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0509 - mae: 0.1691 - val_loss: 0.962
4 - val_mae: 0.5943
Epoch 1124/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0650 - mae: 0.1926 - val_loss: 0.994
7 - val_mae: 0.5931
Epoch 1125/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 23ms/step - loss: 0.0572 - mae: 0.1800 - val_loss: 1.169
4 - val_mae: 0.6536
Epoch 1126/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0685 - mae: 0.1903 - val_loss: 1.034
0 - val_mae: 0.5876
Epoch 1127/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0612 - mae: 0.1820 - val_loss: 0.946
7 - val_mae: 0.5826
Epoch 1128/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 21ms/step - loss: 0.0612 - mae: 0.1909 - val_loss: 1.090
4 - val_mae: 0.6138
Epoch 1129/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0440 - mae: 0.1575 - val_loss: 0.993
1 - val_mae: 0.5821
Epoch 1130/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0590 - mae: 0.1806 - val_loss: 1.077
5 - val_mae: 0.6106
Epoch 1131/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0451 - mae: 0.1501 - val_loss: 0.918
5 - val_mae: 0.5646
Epoch 1132/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0852 - mae: 0.2221 - val_loss: 0.967
0 - val_mae: 0.6233
Epoch 1133/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0934 - mae: 0.2371 - val_loss: 0.997
2 - val_mae: 0.5850
Epoch 1134/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 18ms/step - loss: 0.0765 - mae: 0.2192 - val_loss: 1.060
4 - val_mae: 0.6043
Epoch 1135/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 21ms/step - loss: 0.0661 - mae: 0.1876 - val_loss: 1.272
5 - val_mae: 0.6832
Epoch 1136/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0889 - mae: 0.2355 - val_loss: 1.085
5 - val_mae: 0.5919
Epoch 1137/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 18ms/step - loss: 0.0519 - mae: 0.1729 - val_loss: 0.962
0 - val_mae: 0.5756
Epoch 1138/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0642 - mae: 0.1916 - val_loss: 1.134
8 - val_mae: 0.6224
Epoch 1139/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 18ms/step - loss: 0.0695 - mae: 0.2089 - val_loss: 1.203
```

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6 - val_mae: 0.6856
Epoch 1140/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.1004 - mae: 0.2478 - val_loss: 1.021
3 - val_mae: 0.5800
Epoch 1141/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0766 - mae: 0.2059 - val_loss: 1.085
4 - val_mae: 0.6064
Epoch 1142/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0555 - mae: 0.1729 - val_loss: 1.103
4 - val_mae: 0.6234
Epoch 1143/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0801 - mae: 0.2149 - val_loss: 0.995
0 - val_mae: 0.5763
Epoch 1144/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 21ms/step - loss: 0.0781 - mae: 0.2030 - val_loss: 1.051
4 - val_mae: 0.6172
Epoch 1145/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0822 - mae: 0.2290 - val_loss: 0.976
2 - val_mae: 0.5858
Epoch 1146/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0963 - mae: 0.2434 - val_loss: 1.427
8 - val_mae: 0.7594
Epoch 1147/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 18ms/step - loss: 0.1119 - mae: 0.2593 - val_loss: 1.253
3 - val_mae: 0.6932
Epoch 1148/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.1023 - mae: 0.2460 - val_loss: 1.109
4 - val_mae: 0.6185
Epoch 1149/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0960 - mae: 0.2262 - val_loss: 1.154
8 - val_mae: 0.6454
Epoch 1150/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 18ms/step - loss: 0.0844 - mae: 0.2262 - val_loss: 1.085
5 - val_mae: 0.5908
Epoch 1151/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 17ms/step - loss: 0.0858 - mae: 0.2243 - val_loss: 1.125
0 - val_mae: 0.6560
Epoch 1152/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 18ms/step - loss: 0.1140 - mae: 0.2654 - val_loss: 0.958
8 - val_mae: 0.5924
Epoch 1153/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 17ms/step - loss: 0.0778 - mae: 0.2122 - val_loss: 1.108
9 - val_mae: 0.6271
Epoch 1154/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 23ms/step - loss: 0.0635 - mae: 0.1978 - val_loss: 0.974
0 - val_mae: 0.5793
Epoch 1155/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0901 - mae: 0.2274 - val_loss: 1.097
7 - val_mae: 0.6441
Epoch 1156/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.1108 - mae: 0.2608 - val_loss: 1.101
8 - val_mae: 0.6620
Epoch 1157/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.1081 - mae: 0.2458 - val_loss: 1.298
8 - val_mae: 0.6790
Epoch 1158/1400
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5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.1418 - mae: 0.2856 - val_loss: 0.936
8 - val_mae: 0.6257
Epoch 1159/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0997 - mae: 0.2308 - val_loss: 1.002
7 - val_mae: 0.6103
Epoch 1160/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 25ms/step - loss: 0.0767 - mae: 0.2091 - val_loss: 1.014
0 - val_mae: 0.5895
Epoch 1161/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0769 - mae: 0.2137 - val_loss: 1.045
7 - val_mae: 0.6006
Epoch 1162/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 17ms/step - loss: 0.0696 - mae: 0.1972 - val_loss: 1.118
7 - val_mae: 0.6171
Epoch 1163/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 21ms/step - loss: 0.0978 - mae: 0.2305 - val_loss: 0.904
4 - val_mae: 0.6300
Epoch 1164/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.1290 - mae: 0.2767 - val_loss: 1.067
8 - val_mae: 0.6805
Epoch 1165/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.1062 - mae: 0.2428 - val_loss: 1.024
8 - val_mae: 0.6044
Epoch 1166/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 18ms/step - loss: 0.1066 - mae: 0.2378 - val_loss: 1.140
2 - val_mae: 0.6607
Epoch 1167/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 18ms/step - loss: 0.1403 - mae: 0.2843 - val_loss: 1.040
4 - val_mae: 0.5955
Epoch 1168/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 21ms/step - loss: 0.1043 - mae: 0.2550 - val_loss: 0.920
1 - val_mae: 0.6689
Epoch 1169/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.1264 - mae: 0.2789 - val_loss: 1.152
6 - val_mae: 0.6585
Epoch 1170/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 26ms/step - loss: 0.1303 - mae: 0.2675 - val_loss: 0.858
2 - val_mae: 0.6360
Epoch 1171/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 18ms/step - loss: 0.2012 - mae: 0.3513 - val_loss: 1.087
9 - val_mae: 0.6112
Epoch 1172/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.2903 - mae: 0.4126 - val_loss: 1.805
6 - val_mae: 0.9359
Epoch 1173/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 18ms/step - loss: 0.5808 - mae: 0.6337 - val_loss: 1.130
4 - val_mae: 0.6812
Epoch 1174/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 18ms/step - loss: 0.3222 - mae: 0.4489 - val_loss: 1.236
2 - val_mae: 0.8261
Epoch 1175/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 18ms/step - loss: 0.2312 - mae: 0.3697 - val_loss: 1.184
3 - val_mae: 0.7380
Epoch 1176/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 18ms/step - loss: 0.2411 - mae: 0.3848 - val_loss: 1.363
0 - val_mae: 0.7056
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Epoch 1177/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 21ms/step - loss: 0.2027 - mae: 0.3625 - val_loss: 1.108
3 - val_mae: 0.5765
Epoch 1178/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.1611 - mae: 0.3102 - val_loss: 0.919
3 - val_mae: 0.6752
Epoch 1179/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.1909 - mae: 0.3525 - val_loss: 1.257
6 - val_mae: 0.6648
Epoch 1180/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 25ms/step - loss: 0.1388 - mae: 0.2825 - val_loss: 0.807
2 - val_mae: 0.5935
Epoch 1181/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 36ms/step - loss: 0.1302 - mae: 0.2869 - val_loss: 1.101
5 - val_mae: 0.6097
Epoch 1182/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 25ms/step - loss: 0.1424 - mae: 0.2908 - val_loss: 0.969
8 - val_mae: 0.5894
Epoch 1183/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 36ms/step - loss: 0.1050 - mae: 0.2542 - val_loss: 0.911
8 - val_mae: 0.5964
Epoch 1184/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 31ms/step - loss: 0.0959 - mae: 0.2344 - val_loss: 0.945
4 - val_mae: 0.6082
Epoch 1185/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1179 - mae: 0.2649 - val_loss: 1.039
2 - val_mae: 0.6084
Epoch 1186/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 42ms/step - loss: 0.0814 - mae: 0.2197 - val_loss: 1.071
9 - val_mae: 0.5983
Epoch 1187/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 25ms/step - loss: 0.1187 - mae: 0.2538 - val_loss: 1.044
8 - val_mae: 0.6688
Epoch 1188/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 34ms/step - loss: 0.2394 - mae: 0.3746 - val_loss: 0.914
5 - val_mae: 0.5865
Epoch 1189/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 28ms/step - loss: 0.2354 - mae: 0.3729 - val_loss: 1.354
9 - val_mae: 0.7813
Epoch 1190/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 26ms/step - loss: 0.2931 - mae: 0.4413 - val_loss: 1.063
0 - val_mae: 0.6271
Epoch 1191/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.2220 - mae: 0.3678 - val_loss: 0.962
1 - val_mae: 0.6476
Epoch 1192/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.2027 - mae: 0.3664 - val_loss: 0.831
5 - val_mae: 0.5592
Epoch 1193/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 18ms/step - loss: 0.1348 - mae: 0.2836 - val_loss: 1.240
2 - val_mae: 0.7301
Epoch 1194/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.2706 - mae: 0.4021 - val_loss: 1.128
2 - val_mae: 0.6664
Epoch 1195/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.1721 - mae: 0.3302 - val_loss: 0.991
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6 - val_mae: 0.7293
Epoch 1196/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 18ms/step - loss: 0.2450 - mae: 0.3880 - val_loss: 0.885
7 - val_mae: 0.5561
Epoch 1197/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.1593 - mae: 0.3103 - val_loss: 1.331
9 - val_mae: 0.7771
Epoch 1198/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.1952 - mae: 0.3538 - val_loss: 1.586
1 - val_mae: 0.8670
Epoch 1199/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 18ms/step - loss: 0.3317 - mae: 0.4475 - val_loss: 0.824
4 - val_mae: 0.5555
Epoch 1200/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.2095 - mae: 0.3570 - val_loss: 0.932
0 - val_mae: 0.5590
Epoch 1201/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 22ms/step - loss: 0.1712 - mae: 0.3327 - val_loss: 1.176
8 - val_mae: 0.6624
Epoch 1202/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.1616 - mae: 0.2908 - val_loss: 1.085
5 - val_mae: 0.5848
Epoch 1203/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 17ms/step - loss: 0.1489 - mae: 0.2978 - val_loss: 1.096
0 - val_mae: 0.6787
Epoch 1204/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 17ms/step - loss: 0.1597 - mae: 0.3107 - val_loss: 1.105
2 - val_mae: 0.5848
Epoch 1205/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.1127 - mae: 0.2642 - val_loss: 0.955
6 - val_mae: 0.5968
Epoch 1206/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.1075 - mae: 0.2628 - val_loss: 0.760
1 - val_mae: 0.5774
Epoch 1207/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0984 - mae: 0.2435 - val_loss: 0.936
4 - val_mae: 0.5988
Epoch 1208/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 18ms/step - loss: 0.1090 - mae: 0.2508 - val_loss: 0.905
6 - val_mae: 0.5761
Epoch 1209/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0753 - mae: 0.2120 - val_loss: 1.131
9 - val_mae: 0.6295
Epoch 1210/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 22ms/step - loss: 0.0751 - mae: 0.2169 - val_loss: 0.861
2 - val_mae: 0.5980
Epoch 1211/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 21ms/step - loss: 0.1161 - mae: 0.2670 - val_loss: 0.946
8 - val_mae: 0.6027
Epoch 1212/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.1272 - mae: 0.2684 - val_loss: 1.057
7 - val_mae: 0.6201
Epoch 1213/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.1295 - mae: 0.2844 - val_loss: 1.270
5 - val_mae: 0.6904
Epoch 1214/1400
```

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5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 22ms/step - loss: 0.1198 - mae: 0.2615 - val_loss: 0.929
5 - val_mae: 0.5760
Epoch 1215/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.1051 - mae: 0.2414 - val_loss: 1.002
1 - val_mae: 0.5806
Epoch 1216/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0806 - mae: 0.2190 - val_loss: 0.965
1 - val_mae: 0.6442
Epoch 1217/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 18ms/step - loss: 0.1495 - mae: 0.3167 - val_loss: 0.922
1 - val_mae: 0.6298
Epoch 1218/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.1335 - mae: 0.2851 - val_loss: 1.144
6 - val_mae: 0.6454
Epoch 1219/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 21ms/step - loss: 0.1204 - mae: 0.2705 - val_loss: 0.825
7 - val_mae: 0.5749
Epoch 1220/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 22ms/step - loss: 0.0864 - mae: 0.2370 - val_loss: 0.988
8 - val_mae: 0.5986
Epoch 1221/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 18ms/step - loss: 0.1441 - mae: 0.3005 - val_loss: 1.064
4 - val_mae: 0.7086
Epoch 1222/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 18ms/step - loss: 0.1760 - mae: 0.3397 - val_loss: 1.009
9 - val_mae: 0.5953
Epoch 1223/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.1366 - mae: 0.2943 - val_loss: 1.018
1 - val_mae: 0.5831
Epoch 1224/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0995 - mae: 0.2385 - val_loss: 1.210
7 - val_mae: 0.7121
Epoch 1225/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 18ms/step - loss: 0.1545 - mae: 0.3145 - val_loss: 1.019
9 - val_mae: 0.6586
Epoch 1226/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 18ms/step - loss: 0.1432 - mae: 0.2866 - val_loss: 0.998
1 - val_mae: 0.6021
Epoch 1227/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 18ms/step - loss: 0.0726 - mae: 0.2027 - val_loss: 0.871
4 - val_mae: 0.5999
Epoch 1228/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0847 - mae: 0.2118 - val_loss: 1.033
5 - val_mae: 0.6094
Epoch 1229/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0739 - mae: 0.2079 - val_loss: 1.027
1 - val_mae: 0.6334
Epoch 1230/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0783 - mae: 0.2002 - val_loss: 1.136
8 - val_mae: 0.6671
Epoch 1231/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.1090 - mae: 0.2428 - val_loss: 0.924
6 - val_mae: 0.6213
Epoch 1232/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 21ms/step - loss: 0.0676 - mae: 0.1941 - val_loss: 0.970
1 - val_mae: 0.5852
```

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Epoch 1233/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0686 - mae: 0.1984 - val_loss: 0.926
5 - val_mae: 0.5704
Epoch 1234/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0622 - mae: 0.1883 - val_loss: 0.865
9 - val_mae: 0.5606
Epoch 1235/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 18ms/step - loss: 0.0622 - mae: 0.1837 - val_loss: 0.956
6 - val_mae: 0.5957
Epoch 1236/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 18ms/step - loss: 0.1086 - mae: 0.2485 - val_loss: 0.854
4 - val_mae: 0.5632
Epoch 1237/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 21ms/step - loss: 0.1091 - mae: 0.2483 - val_loss: 0.994
0 - val_mae: 0.6051
Epoch 1238/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0631 - mae: 0.1831 - val_loss: 0.920
6 - val_mae: 0.5588
Epoch 1239/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 24ms/step - loss: 0.0509 - mae: 0.1727 - val_loss: 1.026
1 - val_mae: 0.5905
Epoch 1240/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0424 - mae: 0.1464 - val_loss: 0.885
5 - val_mae: 0.5700
Epoch 1241/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 18ms/step - loss: 0.0454 - mae: 0.1605 - val_loss: 0.950
9 - val_mae: 0.5761
Epoch 1242/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 17ms/step - loss: 0.0471 - mae: 0.1572 - val_loss: 0.931
5 - val_mae: 0.5731
Epoch 1243/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 17ms/step - loss: 0.0487 - mae: 0.1620 - val_loss: 0.918
8 - val_mae: 0.5777
Epoch 1244/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0508 - mae: 0.1662 - val_loss: 0.994
1 - val_mae: 0.5691
Epoch 1245/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0542 - mae: 0.1696 - val_loss: 1.060
7 - val_mae: 0.6101
Epoch 1246/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0712 - mae: 0.1957 - val_loss: 0.905
7 - val_mae: 0.5710
Epoch 1247/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0519 - mae: 0.1701 - val_loss: 1.018
6 - val_mae: 0.6046
Epoch 1248/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 21ms/step - loss: 0.0716 - mae: 0.2034 - val_loss: 0.911
6 - val_mae: 0.5658
Epoch 1249/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0587 - mae: 0.1712 - val_loss: 0.957
6 - val_mae: 0.5772
Epoch 1250/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 22ms/step - loss: 0.0471 - mae: 0.1582 - val_loss: 1.055
8 - val_mae: 0.6015
Epoch 1251/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0762 - mae: 0.2034 - val_loss: 0.952
```

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4 - val_mae: 0.5831
Epoch 1252/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 18ms/step - loss: 0.0564 - mae: 0.1803 - val_loss: 1.007
7 - val_mae: 0.5828
Epoch 1253/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 18ms/step - loss: 0.0487 - mae: 0.1698 - val_loss: 1.053
5 - val_mae: 0.6199
Epoch 1254/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0555 - mae: 0.1775 - val_loss: 0.945
3 - val_mae: 0.5846
Epoch 1255/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0536 - mae: 0.1683 - val_loss: 0.969
4 - val_mae: 0.5805
Epoch 1256/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0500 - mae: 0.1682 - val_loss: 1.076
0 - val_mae: 0.6273
Epoch 1257/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0557 - mae: 0.1746 - val_loss: 0.992
4 - val_mae: 0.5916
Epoch 1258/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 22ms/step - loss: 0.0412 - mae: 0.1454 - val_loss: 0.964
4 - val_mae: 0.5697
Epoch 1259/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 21ms/step - loss: 0.0450 - mae: 0.1570 - val_loss: 0.987
8 - val_mae: 0.5803
Epoch 1260/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 18ms/step - loss: 0.0576 - mae: 0.1790 - val_loss: 1.040
0 - val_mae: 0.5880
Epoch 1261/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0576 - mae: 0.1750 - val_loss: 0.910
2 - val_mae: 0.5943
Epoch 1262/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 18ms/step - loss: 0.1010 - mae: 0.2375 - val_loss: 1.022
2 - val_mae: 0.5884
Epoch 1263/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 18ms/step - loss: 0.0806 - mae: 0.2166 - val_loss: 1.074
6 - val_mae: 0.6355
Epoch 1264/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.1376 - mae: 0.2888 - val_loss: 0.999
4 - val_mae: 0.5952
Epoch 1265/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.1043 - mae: 0.2547 - val_loss: 0.961
4 - val_mae: 0.6563
Epoch 1266/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.3008 - mae: 0.4584 - val_loss: 1.015
5 - val_mae: 0.6898
Epoch 1267/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.1683 - mae: 0.3176 - val_loss: 0.968
0 - val_mae: 0.6259
Epoch 1268/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 21ms/step - loss: 0.3884 - mae: 0.4912 - val_loss: 1.220
4 - val_mae: 0.6188
Epoch 1269/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.2791 - mae: 0.3991 - val_loss: 1.179
0 - val_mae: 0.7409
Epoch 1270/1400
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5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.1681 - mae: 0.3136 - val_loss: 1.132
0 - val_mae: 0.6549
Epoch 1271/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.1509 - mae: 0.3020 - val_loss: 0.945
0 - val_mae: 0.5907
Epoch 1272/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.1752 - mae: 0.3347 - val_loss: 1.070
5 - val_mae: 0.6095
Epoch 1273/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.2199 - mae: 0.3621 - val_loss: 1.121
9 - val_mae: 0.8157
Epoch 1274/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.3138 - mae: 0.4369 - val_loss: 1.069
8 - val_mae: 0.6396
Epoch 1275/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.1503 - mae: 0.3010 - val_loss: 1.281
6 - val_mae: 0.7379
Epoch 1276/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.1814 - mae: 0.3150 - val_loss: 0.816
8 - val_mae: 0.5968
Epoch 1277/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 23ms/step - loss: 0.0758 - mae: 0.2125 - val_loss: 0.971
2 - val_mae: 0.6120
Epoch 1278/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0960 - mae: 0.2385 - val_loss: 1.021
0 - val_mae: 0.6150
Epoch 1279/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.1421 - mae: 0.2807 - val_loss: 1.470
2 - val_mae: 0.8371
Epoch 1280/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.2790 - mae: 0.3997 - val_loss: 0.866
5 - val_mae: 0.5799
Epoch 1281/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.1450 - mae: 0.2942 - val_loss: 0.878
1 - val_mae: 0.6183
Epoch 1282/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 38ms/step - loss: 0.1327 - mae: 0.2937 - val_loss: 0.944
5 - val_mae: 0.5896
Epoch 1283/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 39ms/step - loss: 0.0812 - mae: 0.2207 - val_loss: 1.052
3 - val_mae: 0.5923
Epoch 1284/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 28ms/step - loss: 0.0692 - mae: 0.2008 - val_loss: 1.018
7 - val_mae: 0.6339
Epoch 1285/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.0735 - mae: 0.2083 - val_loss: 0.837
1 - val_mae: 0.5709
Epoch 1286/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 26ms/step - loss: 0.0795 - mae: 0.2072 - val_loss: 1.029
1 - val_mae: 0.6394
Epoch 1287/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 40ms/step - loss: 0.1002 - mae: 0.2422 - val_loss: 0.830
3 - val_mae: 0.5647
Epoch 1288/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 25ms/step - loss: 0.0836 - mae: 0.2150 - val_loss: 0.991
0 - val_mae: 0.6052
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Epoch 1289/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 25ms/step - loss: 0.0609 - mae: 0.1876 - val_loss: 0.848
3 - val_mae: 0.5722
Epoch 1290/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 38ms/step - loss: 0.0555 - mae: 0.1767 - val_loss: 1.005
0 - val_mae: 0.5924
Epoch 1291/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 33ms/step - loss: 0.0496 - mae: 0.1637 - val_loss: 0.886
1 - val_mae: 0.5602
Epoch 1292/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 29ms/step - loss: 0.0454 - mae: 0.1527 - val_loss: 1.004
9 - val_mae: 0.5845
Epoch 1293/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 18ms/step - loss: 0.1066 - mae: 0.2461 - val_loss: 0.992
7 - val_mae: 0.7288
Epoch 1294/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.2073 - mae: 0.3708 - val_loss: 0.965
9 - val_mae: 0.6749
Epoch 1295/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 30ms/step - loss: 0.1597 - mae: 0.3077 - val_loss: 1.127
9 - val_mae: 0.6535
Epoch 1296/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.1329 - mae: 0.2843 - val_loss: 1.201
3 - val_mae: 0.7101
Epoch 1297/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.1394 - mae: 0.2872 - val_loss: 1.001
5 - val_mae: 0.5795
Epoch 1298/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0845 - mae: 0.2181 - val_loss: 0.910
3 - val_mae: 0.5915
Epoch 1299/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.1086 - mae: 0.2578 - val_loss: 0.989
7 - val_mae: 0.5820
Epoch 1300/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.1083 - mae: 0.2464 - val_loss: 1.132
0 - val_mae: 0.6656
Epoch 1301/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.1043 - mae: 0.2457 - val_loss: 0.879
7 - val_mae: 0.6111
Epoch 1302/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 18ms/step - loss: 0.1022 - mae: 0.2303 - val_loss: 0.951
5 - val_mae: 0.6050
Epoch 1303/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0838 - mae: 0.2152 - val_loss: 0.785
2 - val_mae: 0.5501
Epoch 1304/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 21ms/step - loss: 0.0660 - mae: 0.1979 - val_loss: 0.931
3 - val_mae: 0.5877
Epoch 1305/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0668 - mae: 0.1941 - val_loss: 0.868
7 - val_mae: 0.5656
Epoch 1306/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 18ms/step - loss: 0.0655 - mae: 0.1910 - val_loss: 0.944
4 - val_mae: 0.5908
Epoch 1307/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0955 - mae: 0.2294 - val_loss: 1.136
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8 - val_mae: 0.6857
Epoch 1308/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0934 - mae: 0.2402 - val_loss: 1.044
6 - val_mae: 0.6361
Epoch 1309/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0749 - mae: 0.2128 - val_loss: 0.892
4 - val_mae: 0.5642
Epoch 1310/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 18ms/step - loss: 0.0568 - mae: 0.1853 - val_loss: 1.110
9 - val_mae: 0.6362
Epoch 1311/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0979 - mae: 0.2465 - val_loss: 0.982
7 - val_mae: 0.5864
Epoch 1312/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0792 - mae: 0.2113 - val_loss: 0.907
5 - val_mae: 0.6014
Epoch 1313/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0660 - mae: 0.1942 - val_loss: 1.040
2 - val_mae: 0.6059
Epoch 1314/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 23ms/step - loss: 0.0615 - mae: 0.1873 - val_loss: 0.872
5 - val_mae: 0.5772
Epoch 1315/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0820 - mae: 0.2141 - val_loss: 0.932
3 - val_mae: 0.5878
Epoch 1316/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0936 - mae: 0.2329 - val_loss: 0.870
2 - val_mae: 0.5726
Epoch 1317/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0682 - mae: 0.1954 - val_loss: 1.009
2 - val_mae: 0.6257
Epoch 1318/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0906 - mae: 0.2427 - val_loss: 1.199
8 - val_mae: 0.6631
Epoch 1319/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0635 - mae: 0.2027 - val_loss: 0.971
0 - val_mae: 0.5828
Epoch 1320/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0752 - mae: 0.2084 - val_loss: 0.815
9 - val_mae: 0.5869
Epoch 1321/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 21ms/step - loss: 0.1121 - mae: 0.2665 - val_loss: 0.868
5 - val_mae: 0.6187
Epoch 1322/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 22ms/step - loss: 0.1034 - mae: 0.2498 - val_loss: 0.831
0 - val_mae: 0.6027
Epoch 1323/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 24ms/step - loss: 0.1038 - mae: 0.2353 - val_loss: 1.107
9 - val_mae: 0.6141
Epoch 1324/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.1389 - mae: 0.2855 - val_loss: 1.098
8 - val_mae: 0.7092
Epoch 1325/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.1977 - mae: 0.3526 - val_loss: 1.289
0 - val_mae: 0.7498
Epoch 1326/1400
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5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.2298 - mae: 0.4047 - val_loss: 0.970
0 - val_mae: 0.6071
Epoch 1327/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 21ms/step - loss: 0.1794 - mae: 0.3343 - val_loss: 1.049
4 - val_mae: 0.7547
Epoch 1328/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.2723 - mae: 0.4157 - val_loss: 0.820
1 - val_mae: 0.5667
Epoch 1329/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 21ms/step - loss: 0.1162 - mae: 0.2694 - val_loss: 0.999
5 - val_mae: 0.6069
Epoch 1330/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.1417 - mae: 0.2878 - val_loss: 1.042
1 - val_mae: 0.5948
Epoch 1331/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 21ms/step - loss: 0.0966 - mae: 0.2425 - val_loss: 0.924
1 - val_mae: 0.5861
Epoch 1332/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 22ms/step - loss: 0.0651 - mae: 0.2025 - val_loss: 0.902
7 - val_mae: 0.5981
Epoch 1333/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0579 - mae: 0.1820 - val_loss: 0.934
9 - val_mae: 0.5996
Epoch 1334/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 21ms/step - loss: 0.0490 - mae: 0.1697 - val_loss: 0.854
6 - val_mae: 0.5953
Epoch 1335/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0444 - mae: 0.1595 - val_loss: 0.895
4 - val_mae: 0.5759
Epoch 1336/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0426 - mae: 0.1500 - val_loss: 0.907
6 - val_mae: 0.5904
Epoch 1337/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0511 - mae: 0.1691 - val_loss: 0.927
2 - val_mae: 0.5942
Epoch 1338/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 21ms/step - loss: 0.0618 - mae: 0.1909 - val_loss: 0.885
4 - val_mae: 0.5887
Epoch 1339/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 21ms/step - loss: 0.0636 - mae: 0.1928 - val_loss: 1.067
4 - val_mae: 0.6497
Epoch 1340/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0675 - mae: 0.1950 - val_loss: 0.865
6 - val_mae: 0.5712
Epoch 1341/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 28ms/step - loss: 0.0397 - mae: 0.1509 - val_loss: 0.882
4 - val_mae: 0.5866
Epoch 1342/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 21ms/step - loss: 0.0394 - mae: 0.1448 - val_loss: 0.834
4 - val_mae: 0.5836
Epoch 1343/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 21ms/step - loss: 0.0700 - mae: 0.1969 - val_loss: 0.875
9 - val_mae: 0.6083
Epoch 1344/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0454 - mae: 0.1603 - val_loss: 0.906
8 - val_mae: 0.5794
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Epoch 1345/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 21ms/step - loss: 0.0399 - mae: 0.1437 - val_loss: 1.037
1 - val_mae: 0.6281
Epoch 1346/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 21ms/step - loss: 0.0484 - mae: 0.1606 - val_loss: 0.783
7 - val_mae: 0.5777
Epoch 1347/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 21ms/step - loss: 0.0417 - mae: 0.1499 - val_loss: 0.921
0 - val_mae: 0.5946
Epoch 1348/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 22ms/step - loss: 0.0408 - mae: 0.1489 - val_loss: 0.833
1 - val_mae: 0.5742
Epoch 1349/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0370 - mae: 0.1421 - val_loss: 0.878
7 - val_mae: 0.5838
Epoch 1350/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 23ms/step - loss: 0.0443 - mae: 0.1498 - val_loss: 1.019
4 - val_mae: 0.6331
Epoch 1351/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0478 - mae: 0.1678 - val_loss: 0.908
2 - val_mae: 0.5875
Epoch 1352/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0540 - mae: 0.1723 - val_loss: 0.842
6 - val_mae: 0.5770
Epoch 1353/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 21ms/step - loss: 0.0421 - mae: 0.1512 - val_loss: 0.999
5 - val_mae: 0.6047
Epoch 1354/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 21ms/step - loss: 0.0506 - mae: 0.1733 - val_loss: 0.835
0 - val_mae: 0.5628
Epoch 1355/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0449 - mae: 0.1550 - val_loss: 0.994
4 - val_mae: 0.6060
Epoch 1356/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 22ms/step - loss: 0.0442 - mae: 0.1484 - val_loss: 0.776
6 - val_mae: 0.5610
Epoch 1357/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 21ms/step - loss: 0.0488 - mae: 0.1658 - val_loss: 0.878
5 - val_mae: 0.5636
Epoch 1358/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0523 - mae: 0.1719 - val_loss: 1.032
3 - val_mae: 0.6204
Epoch 1359/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 22ms/step - loss: 0.0539 - mae: 0.1705 - val_loss: 0.820
8 - val_mae: 0.5669
Epoch 1360/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0678 - mae: 0.1972 - val_loss: 0.872
6 - val_mae: 0.5817
Epoch 1361/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0518 - mae: 0.1731 - val_loss: 0.863
6 - val_mae: 0.5628
Epoch 1362/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0590 - mae: 0.1785 - val_loss: 0.967
5 - val_mae: 0.5803
Epoch 1363/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 22ms/step - loss: 0.0435 - mae: 0.1501 - val_loss: 0.968
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4 - val_mae: 0.6014
Epoch 1364/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0530 - mae: 0.1784 - val_loss: 0.815
7 - val_mae: 0.5697
Epoch 1365/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0614 - mae: 0.1816 - val_loss: 0.926
2 - val_mae: 0.6067
Epoch 1366/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 21ms/step - loss: 0.0772 - mae: 0.2079 - val_loss: 0.946
4 - val_mae: 0.5994
Epoch 1367/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.1012 - mae: 0.2400 - val_loss: 0.783
7 - val_mae: 0.5784
Epoch 1368/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 26ms/step - loss: 0.1297 - mae: 0.2679 - val_loss: 1.177
4 - val_mae: 0.6990
Epoch 1369/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.1605 - mae: 0.3079 - val_loss: 0.904
0 - val_mae: 0.6227
Epoch 1370/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.1380 - mae: 0.2947 - val_loss: 1.014
7 - val_mae: 0.6435
Epoch 1371/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.1358 - mae: 0.2812 - val_loss: 0.793
5 - val_mae: 0.5887
Epoch 1372/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0966 - mae: 0.2325 - val_loss: 1.089
3 - val_mae: 0.6605
Epoch 1373/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.1287 - mae: 0.2850 - val_loss: 1.110
1 - val_mae: 0.7533
Epoch 1374/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.1979 - mae: 0.3545 - val_loss: 1.263
8 - val_mae: 0.6847
Epoch 1375/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.2997 - mae: 0.4324 - val_loss: 1.084
0 - val_mae: 0.7829
Epoch 1376/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.3641 - mae: 0.4964 - val_loss: 1.064
0 - val_mae: 0.7242
Epoch 1377/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 23ms/step - loss: 0.1814 - mae: 0.3377 - val_loss: 0.888
1 - val_mae: 0.5821
Epoch 1378/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 28ms/step - loss: 0.1322 - mae: 0.2707 - val_loss: 1.177
9 - val_mae: 0.7134
Epoch 1379/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 38ms/step - loss: 0.2422 - mae: 0.3861 - val_loss: 0.975
4 - val_mae: 0.5799
Epoch 1380/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 43ms/step - loss: 0.1432 - mae: 0.2905 - val_loss: 1.017
8 - val_mae: 0.6756
Epoch 1381/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 47ms/step - loss: 0.1278 - mae: 0.2853 - val_loss: 0.950
8 - val_mae: 0.5979
Epoch 1382/1400
```

```
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 33ms/step - loss: 0.0750 - mae: 0.2192 - val_loss: 1.102
9 - val_mae: 0.6704
Epoch 1383/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 39ms/step - loss: 0.0927 - mae: 0.2353 - val_loss: 1.052
2 - val_mae: 0.6558
Epoch 1384/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 39ms/step - loss: 0.1354 - mae: 0.2877 - val_loss: 1.194
3 - val_mae: 0.6798
Epoch 1385/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 27ms/step - loss: 0.2663 - mae: 0.4087 - val_loss: 0.987
0 - val_mae: 0.7052
Epoch 1386/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 27ms/step - loss: 0.2688 - mae: 0.4313 - val_loss: 0.969
6 - val_mae: 0.7647
Epoch 1387/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 34ms/step - loss: 0.2321 - mae: 0.3811 - val_loss: 0.932
7 - val_mae: 0.6170
Epoch 1388/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 30ms/step - loss: 0.1458 - mae: 0.2926 - val_loss: 0.871
7 - val_mae: 0.6166
Epoch 1389/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 43ms/step - loss: 0.0983 - mae: 0.2413 - val_loss: 0.971
1 - val_mae: 0.6298
Epoch 1390/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0779 - mae: 0.2196 - val_loss: 0.929
8 - val_mae: 0.6006
Epoch 1391/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0569 - mae: 0.1880 - val_loss: 0.908
9 - val_mae: 0.5975
Epoch 1392/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 22ms/step - loss: 0.0609 - mae: 0.1886 - val_loss: 1.029
4 - val_mae: 0.6189
Epoch 1393/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 24ms/step - loss: 0.0946 - mae: 0.2332 - val_loss: 0.863
4 - val_mae: 0.6149
Epoch 1394/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 21ms/step - loss: 0.1537 - mae: 0.3228 - val_loss: 1.006
9 - val_mae: 0.6019
Epoch 1395/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.1001 - mae: 0.2440 - val_loss: 0.892
4 - val_mae: 0.6227
Epoch 1396/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 21ms/step - loss: 0.0806 - mae: 0.2205 - val_loss: 1.039
4 - val_mae: 0.6132
Epoch 1397/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0608 - mae: 0.1939 - val_loss: 0.899
4 - val_mae: 0.5914
Epoch 1398/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 21ms/step - loss: 0.0618 - mae: 0.1849 - val_loss: 0.923
7 - val_mae: 0.6004
Epoch 1399/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 22ms/step - loss: 0.0743 - mae: 0.2147 - val_loss: 0.895
9 - val_mae: 0.5868
Epoch 1400/1400
5/5 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0731 - mae: 0.1988 - val_loss: 0.856
5 - val_mae: 0.5805
```

model.fit() menjalankan training menggunakan x_train dan y_train selama maksimal 1400 epoch. satu epoch berarti model melewati seluruh data training satu kali. validation_data meminta keras menghitung loss dan mae pada data validasi setelah setiap epoch yg sdah berjalan.

1. forward propagation: data input diteruskan melalui layer untuk menghasilkan prediksi.
2. menghitung loss: prediksi dibandingkan dengan target menggunakan mse.
3. backpropagation: kesalahan diteruskan mundur dari output menuju layer sebelumnya untuk menghitung gradien setiap bobot.
4. optimizer adam memperbarui bobot dan bias berdasarkan gradien agar loss semakin kecil.
5. proses diulang pada epoch berikutnya. output epoch menampilkan loss dan mae untuk training serta val_loss dan val_mae untuk validasi. nilai yang semakin kecil umumnya menunjukkan prediksi semakin dekat dengan target. early_stop sudah dibuat dengan memantau val_loss, tetapi callback tersebut belum dimasukkan ke argumen model.fit(). akibatnya, early stopping belum aktif dan training tetap mengikuti batas epoch yang ditentukan.

In [34]:
```
test_loss, test_mae = model.evaluate(X_test, y_test)
print(f"Mean Absolute Error pada data pengujian: {test_mae}")
```

```
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 18ms/step - loss: 0.8565 - mae: 0.5805
Mean Absolute Error pada data pengujian: 0.5804809331893921
```

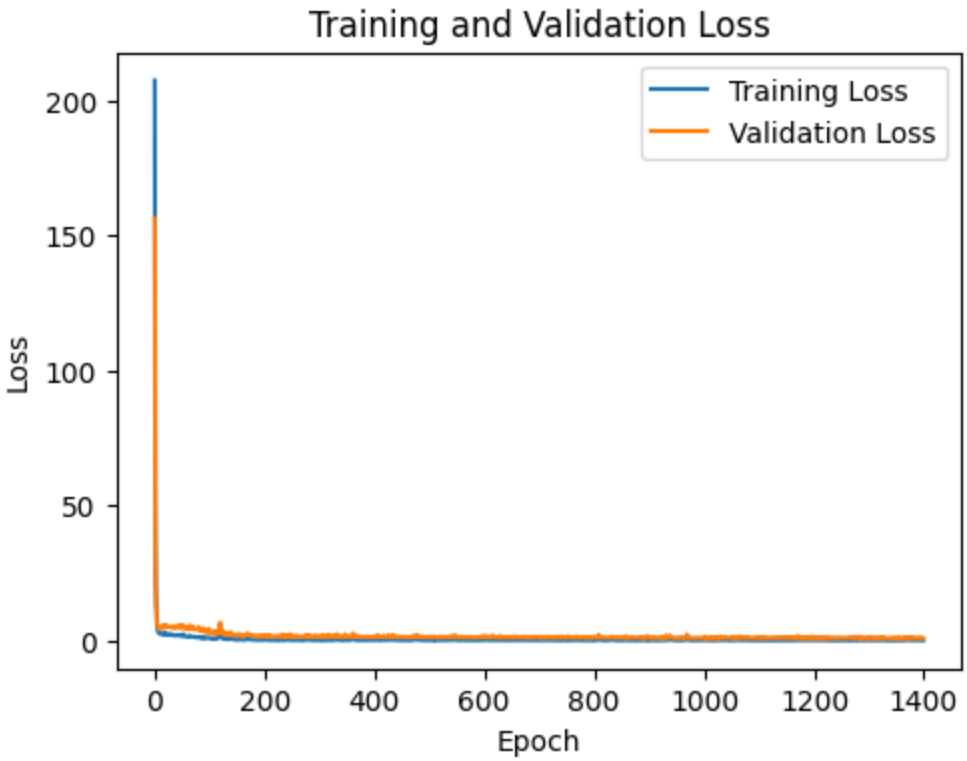
model.evaluate(x_test, y_test) menghitung performa model pada data testing. test_loss berisi nilai loss mse, sedangkan test_mae berisi rata-rata selisih absolut antara prediksi dan nilai sales sebenarnya.

output menunjukkan mae sekitar 1.2908. artinya, pada data uji ini, rata-rata selisih absolut prediksi sales terhadap nilai sebenarnya sekitar 1.29 satuan sales. semakin kecil mae, semakin dekat hasil prediksi dengan nilai sebenarnya. nilai ini perlu dibaca bersama grafik dan kondisi pembagian data, karena data testing juga dipakai sebagai validation saat training, hasil pengujian belum sepenuhnya independen.

In [35]:
```
# Plot Loss dan MAE selama pelatihan
plt.figure(figsize=(12, 4))

# Plot Loss
plt.subplot(1, 2, 1)
plt.plot(history.history['loss'], label='Training Loss')
plt.plot(history.history['val_loss'], label='Validation Loss')
plt.title('Training and Validation Loss')
plt.xlabel('Epoch')
plt.ylabel('Loss')
plt.legend()
```

Out[35]:
```
<matplotlib.legend.Legend at 0x7f94bc643c50>
```



history menyimpan catatan metrik dari setiap epoch. grafik pertama membandingkan training loss dan validation loss, sedangkan grafik kedua membandingkan training mae dan validation mae.

loss pada grafik ini menggunakan mse. jika kedua kurva menurun, model sedang memperbaiki prediksinya. jika training loss terus turun tetapi validation loss mulai naik, hal tersebut dapat menjadi tanda overfitting. jika keduanya masih tinggi atau tidak membaik, model mungkin belum mempelajari pola dengan baik.

In [36]:
```
y_pred_test = model.predict(X_test)

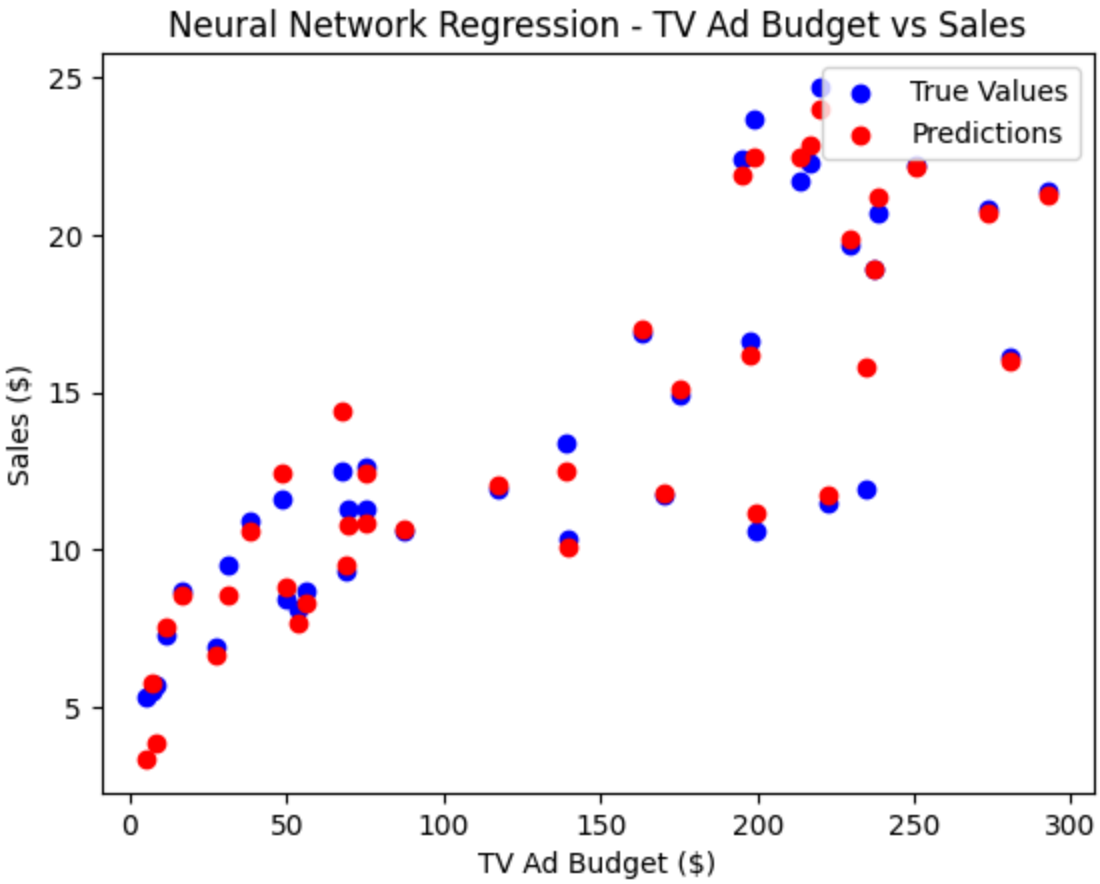
plt.scatter(X_test['TV'], y_test, color='blue')

# Scatter plot of predictions (red)
plt.scatter(X_test['TV'], y_pred_test, color='red')

# Set the axis labels and title
plt.xlabel('TV Ad Budget ($)')
plt.ylabel('Sales ($)')
plt.title('Neural Network Regression - TV Ad Budget vs Sales')

# Add a legend to the plot
plt.legend(['True Values', 'Predictions'])
plt.show()
```

```
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 45ms/step
```



model.predict(x_test) meminta model menghasilkan prediksi untuk seluruh baris data testing. hasilnya disimpan pada y_pred_test. scatter plot kemudian membandingkan nilai sales sebenarnya dengan prediksi, dengan anggaran tv sebagai sumbu x.

titik biru menunjukkan nilai sales aktual, sedangkan titik merah menunjukkan hasil prediksi mlp. jika titik merah berdekatan dengan titik biru pada rentang anggaran yang sama, prediksi model relatif mendekati nilai aktual. jarak yang jauh menunjukkan salah dalam memprresdiksi yang lebih besar.

grafik ini hanya memilih tv sebagai sumbu x untuk membantu membaca hubungan secara visual. prediksi model sebenarnya menggunakan ketiga fitur sekaligus, yaitu tv, radio, dan newspaper.

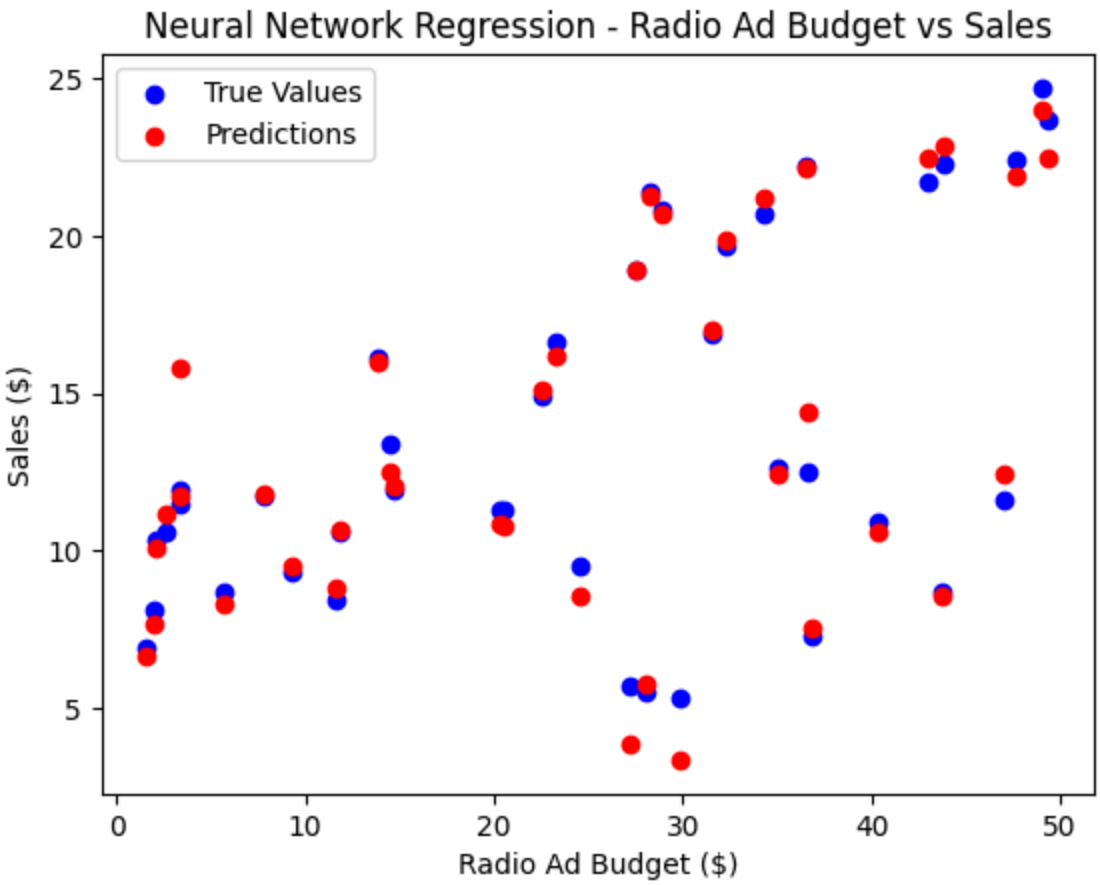
In [37]:
```
plt.scatter(X_test['Radio'], y_test, color='blue')

# Scatter plot of predictions (red)
plt.scatter(X_test['Radio'], y_pred_test, color='red')

# Set the axis labels and title
plt.xlabel('Radio Ad Budget ($)')
plt.ylabel('Sales ($)')
plt.title('Neural Network Regression - Radio Ad Budget vs Sales')

# Add a legend to the plot
```

```
plt.legend(['True Values', 'Predictions'])
plt.show()
```



cell ini menggunakan y_pred_test yang sama, tetapi menampilkan anggaran radio pada sumbu x. titik biru adalah sales aktual dan titik merah adalah sales hasil prediksi.

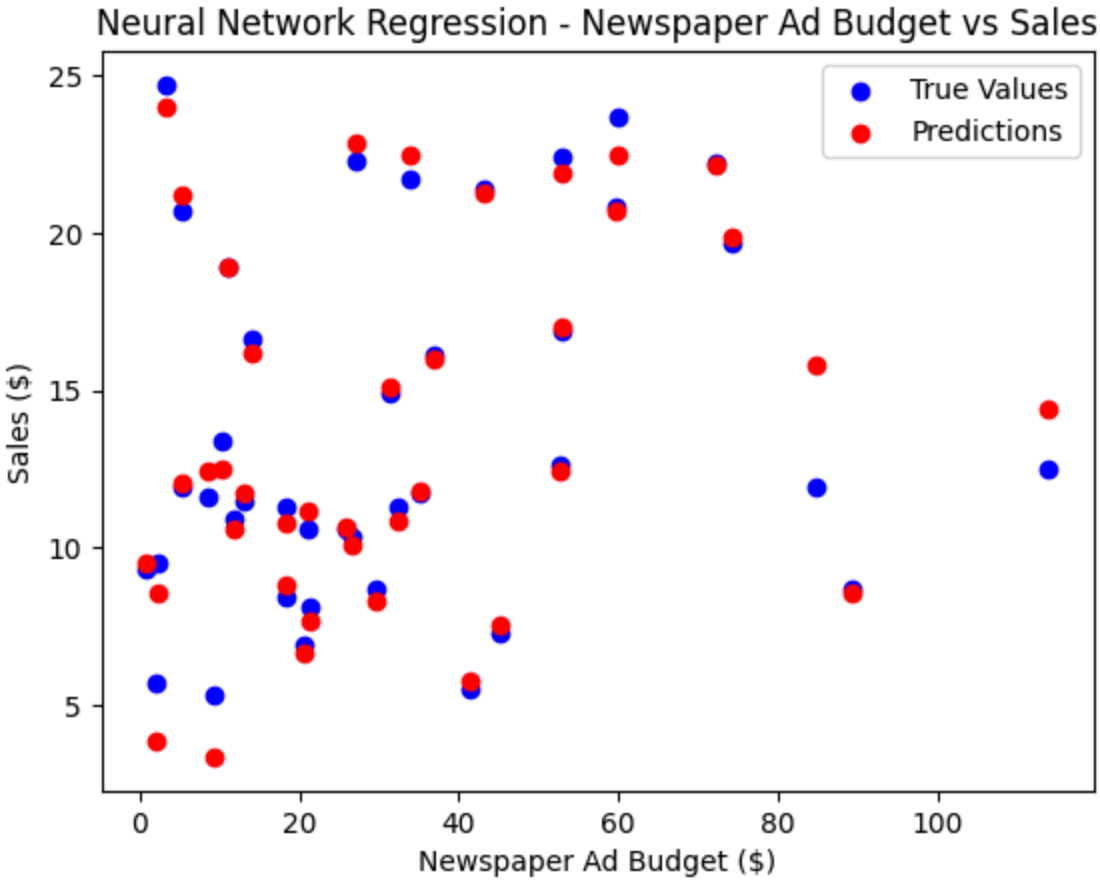
tujuannya membandingkan kedekatan prediksi dan nilai aktual ketika data dilihat berdasarkan anggaran radio. perlu diingat bahwa model tidak memprediksi sales hanya dari radio; model menerima tiga fitur iklan secara bersamaan. karena itu, grafik ini merupakan cara visualisasi, bukan model regresi satu fitur.

In [38]:
```
plt.scatter(X_test['Newspaper'], y_test, color='blue')

# Scatter plot of predictions (red)
plt.scatter(X_test['Newspaper'], y_pred_test, color='red')

# Set the axis labels and title
plt.xlabel('Newspaper Ad Budget ($)')
plt.ylabel('Sales ($)')
plt.title('Neural Network Regression - Newspaper Ad Budget vs Sales')

# Add a legend to the plot
plt.legend(['True Values', 'Predictions'])
plt.show()
```



cell ini membuat perbandingan yang sama dengan dua grafik sebelumnya, tetapi menggunakan anggaran newspaper sebagai sumbu x. titik biru menunjukkan nilai sales sebenarnya dan titik merah menunjukkan hasil prediksi.

grafik membantu melihat apakah prediksi model berada dekat dengan nilai aktual pada berbagai tingkat anggaran newspaper. kesimpulan performa tetap perlu mempertimbangkan ketiga fitur dan metrik mae, karena model mlp menggunakan tv, radio, dan newspaper secara bersama-sama.